GE Digital

Browser

4.4

Browser User Guide

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About This Document

This document is supplied as a part of GE's Browser product.

Purpose of This Document

The *Browser User Guide* has two main purposes:

- To provide a general overview of the features and components of Browser.
- To provide step-by-step instructions for using the features and components of Browser.

Who Should Use This Document

This guide is for all users of Browser, and all those who are designing or building applications that will utilize Browser.

This guide will be most useful to those who are familiar with the Microsoft Windows operating system.

For More Information

For more information about Browser, refer to the following:

- *Browser Software Installation and Maintenance Guide*
- *Browser Reference Manual*
- *Display Builder User Guide*

For more information about Habitat, refer to the following:

- *Habitat User Guide*
- *Habitat Software Installation and Maintenance Guide*

For more information about Platform, refer to the following:

- *Platform System Overview*: An overall view of Platform and its components, their functions and use, and how they are related functionally and structurally.

1. Overview

This chapter provides an overview of Browser and its subsystems, and describes the relationships among Browser and Habitat.

The Browser user interface software gives operators complete and rapid access to Habitat or Control. The user interface software employs web-based technology to display interactive views of system data via a local area network, a corporate intranet, or the Internet.

Users of Browser can interact with Habitat applications or with databases on the GE Scada management platform, Control. Displays may be viewed using industry-standard web browsers, or by using GE proprietary Display Viewer software. Display Viewer provides a fully managed desktop environment, including powerful navigation aids and customization features.

Browser supports real-time and near-real-time operating modes, and read-only vs. read-write configuration options. Browser displays support a wide variety of interactive mechanisms for viewing and controlling the operation of the power network and computer system. Finally, the Browser Display Builder program provides powerful tools for creating, customizing, and deploying full-graphic displays.

1.1 Browser and Habitat

Browser is the Windows technology-based user interface for Habitat[1], a real-time application development tool. Browser performs the following functions:

- Provides a graphics-based interface application between the operator and Habitat.
- Receives and interprets commands issued by the operator and operator processes.
- Processes, paints, and refreshes specially developed graphic displays that show data from application databases developed in Habitat.

Refer to the Habitat documentation set for more information about the Habitat development and real-time control environment.

1.2 Browser Subsystems

Browser is made up of three components: the Browser server, the Browser client, and Display Builder.

> **Note:** It is important to note that, even though there may appear to be two separate applications, server and client, there is only one user interface.

Refer to the *Browser Software Installation and Maintenance Guide* for more information about the procedures and requirements for installing and maintaining Browser and its components.

1.2.1 Browser Server

Browser operates in a distributed environment with an EMS-side data server process and one or more network-resident or intranet/Internet-based client processes.

[1] Browser is also the user interface for Control.

The Browser server is a Habitat application running under Energy Management System (EMS) control. Its primary purpose is to supply application data to Browser client applications running on remote workstations.

The benefits of distributing the display processing include:

- Reduced EMS-side CPU loading.
- Reduced network traffic.
- Flexible configuration: hardware choices can be tailored to actual usage.
- Re-locatability: workstations in multiple locations can work with databases on the same EMS platform.

1.2.2 Browser Client

The Browser client is the software component installed on workstations in the control room or on corporate desktops, to provide operators and administrators with a user interface to Habitat, Platform, and Control applications. The installation requires a framework (APF) and a component (Browser).

1.2.2.1 Framework (APF)

The APF provides an intuitive interface for controlling the Browser application, allowing operators to configure the work environment to best meet their needs, to open viewports (see section 2.2.1.2 The Viewport and section 5.3 Viewports), and to create rooms (see section 5.4.1 Creating Rooms and section 3.6 Calling Up Rooms).

1.2.2.2 Display Viewer (Browser)

Display Viewer is the proprietary software component providing sophisticated access to displays. It provides viewport windows.

Viewport windows (also called “viewports”) are an extremely flexible way to view displays. Multiple displays can be viewed in a single viewport by using the “frames” feature, and multiple viewports can be open at any given time on one or more screens, giving the operator access to a wide array of information.

1.2.3 Display Builder

The Display Builder is used to create displays and display components. It is unique, in that it is not required that Browser be installed on the Display Builder workstation in order to build displays.

The Display Builder is installed independently on standalone workstations, and it utilizes Microsoft Windows technology. This technology supports special handling of graphical objects and dynamically linked libraries.

Additional information about the Display Builder can be found in the *Display Builder User Guide*.

1.3 Browser Features

Browser supports real-time and near-real-time operating modes, and read-only vs. read-write configuration options. Browser displays support a wide variety of interactive mechanisms for viewing and controlling the operation of the power network and computer system. In addition, the Browser Display Builder software provides powerful tools for creating, customizing, and deploying displays.

The Browser user interface provides the following:

- A user-friendly interactive style.
- A consistent look and feel across GE applications.
- The ability to easily customize and extend the user interface.
- A graphics-oriented interface, where you interact with Habitat and issue commands by interacting with graphical items on the screen.
- The ability to design and develop graphic interfaces for user applications developed with Habitat.
- The ability to view multiple Habitat displays concurrently, each display accessing data either from different, or the same, applications.
- Flexible workstation operations and permissions.
- System security at multiple levels.
- Rapid notification of state and device changes.
- The ability to distribute application processing throughout a network.
- The option to insulate operators and application programmers from low-level details, so they can concentrate on the application-specific factors of the user interface.

1.4 Browser Workstation Requirements

The Browser client requires a suitably configured workstation. The following operating systems are supported:

- Windows 10 64-bit – Build version 21H2 (certified)
- Windows 11 (supported)
- Windows Server 2019 64-bit (supported)

Another requirement is:

- Network or Internet access to the data server.

See the *Browser Software Installation and Maintenance Guide* for additional hardware and software requirements.

2. Concepts

To make effective use of Browser's features, an understanding of the Windows environment, as well as basic Browser concepts, is needed.

2.1 Windows Concepts

Applications designed for the Windows environment, including Browser, share numerous similar features. These features, and how they relate to Browser, are discussed below.

Refer to your operating system documentation for more-detailed information.

2.1.1 Windows

The term "windows" means areas on the workstation screen in which an application is accessed, viewed, and controlled. Browser provides windows for controlling the Browser process, viewing displays, and creating displays, among other purposes.

Browser windows have the features standard to all Microsoft Windows applications. They can be moved, re-sized, stacked, maximized, and minimized.

Only one window on a workstation can have "input focus" — the ability to accept text or commands — at any given time. That window is called the "active window."

Figure 1 and Figure 2 show typical Browser windows: the Advanced Presentation Framework (APF) window and a typical viewport.

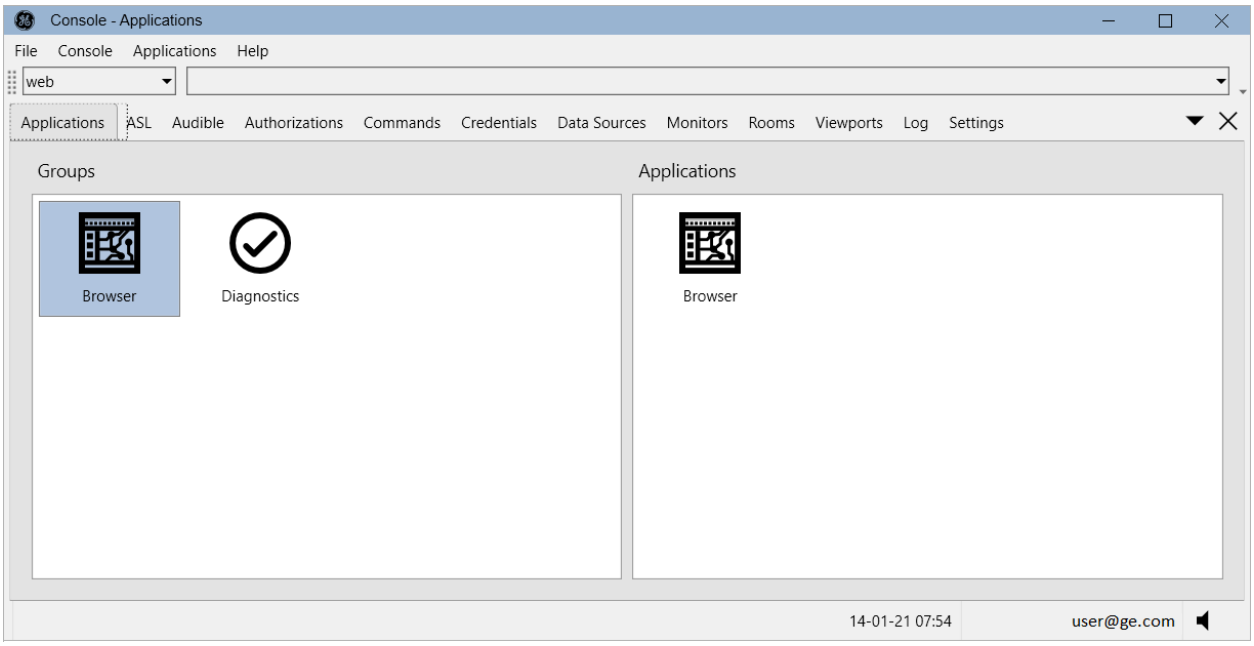


Figure 1. Advanced Presentation Framework (APF) Window

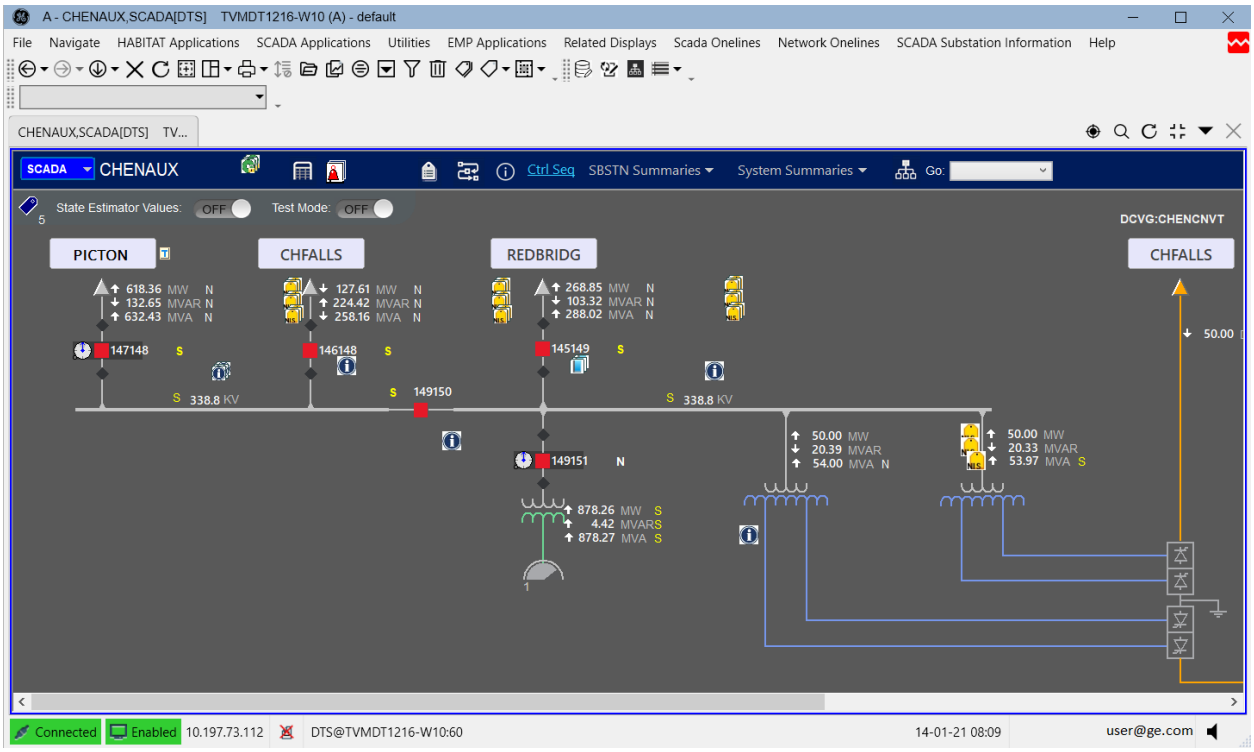


Figure 2. Typical Viewport

2.1.2 The Application Title Bar

The title bar (Figure 3) is an important feature of a Browser window. The title bar includes several buttons that are useful for controlling the application, and that are common to all Windows applications. It also conveys the viewport Name and Title. The Title normally contains information specific to the display loaded in the viewport.

Figure 3. Application Title Bar

The three buttons that will be useful to operators are, *from right to left*:

- **Close:** Clicking this button closes the application.
- **Maximize:** Clicking this button toggles between displaying the window in full-screen mode or at a smaller size, depending on the current state of the window.
- **Minimize:** Clicking this button causes the window to disappear from view and be stored as an icon on the Windows task bar (see section 2.1.3 The Windows Task Bar). Minimizing the window is not the same as closing the application; the application is still running, and all displays are still open and available for immediate viewing.

2.1.3 The Windows Task Bar

The Windows task bar normally appears at the bottom of the workstation display. However, it can be repositioned at the discretion of the operator.

The task bar includes the Start button, and it may include other options depending on how the workstation is configured. All open applications will appear as icons on the task bar, whether they are maximized or minimized. This is useful as a navigation tool if numerous applications are open concurrently. When an application or window is minimized (see section 2.1.2 The Application Title Bar), it is temporarily removed from view, but its icon is still accessible on the task bar.

Your operating system documentation will have additional information about the Windows task bar.

2.1.4 Menus

Menus provide a fast system navigation tool, and an easy way to issue commands without having to type them (see section 2.2.9.4 Menus). Menus are lists of one or more labels. The labels are text strings that describe the choices available from the menu. Each label is associated with a command that will be executed when the label is selected.

Menus are accessed by clicking the menu name. Menu items are selected by clicking the desired label.

Browser includes two default main menus, the File and Help menus (see section 3.3.3 Using Menus). Custom menus can be built into the display at design time, and can be organized to best meet the needs of the operator. See the Display Builder documentation for more information about designing menus.

2.1.4.1 Drop-Down Menus

Drop-down menus are those menus that appear in the main menu bar of Browser windows. These menus operate exactly like those found in any other Windows application.

2.1.4.2 Pop-Up Menus

Pop-up menus can be included in Browser displays. These menus must be defined in Display Builder, and they must be attached to an object in a display.

2.1.5 Toolbars

Toolbars are groups of icons grouped together in a row immediately below the application's menu bar. Toolbars provide a fast, easy way to access commonly used commands (see section 2.2.9.5 Toolbars).

2.1.6 Dialog Boxes

Dialog boxes (Figure 4) solicit additional information from operators. Information is supplied by any of the following means:

- Typing into text boxes, areas in a dialog box where text can be typed.
- Selecting check boxes and radio buttons. Check boxes let you turn an attribute on or off. Radio buttons allow you to make one choice from an option group.
- Selecting a value from a list box.
- Activating commands using command buttons. Command buttons (such as OK, Apply, Cancel, etc.) let you tell Browser what to do with the information in the dialog box.

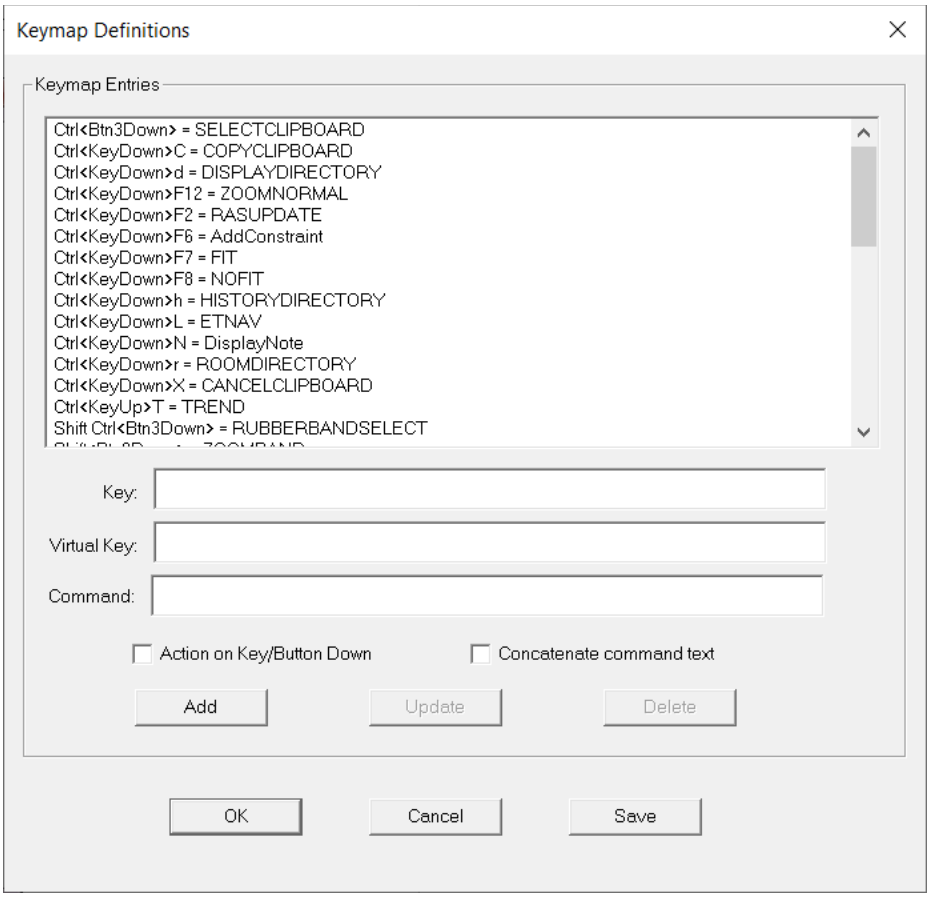


Figure 4. Typical Dialog Box

2.1.7 Message Boxes

During routine operations, or in the event of an operator error, message boxes appear on the screen. Message boxes contain a text message and one or more push buttons. The type and number of push buttons will depend on the type of message and its context.

Browser message boxes are of two types: informative and decision.

2.1.7.1 Informative Message Boxes

Informative message boxes (Figure 5) provide you with information about the current process or executing command. They have a single push button, OK, for simultaneously acknowledging the message and closing the message box.

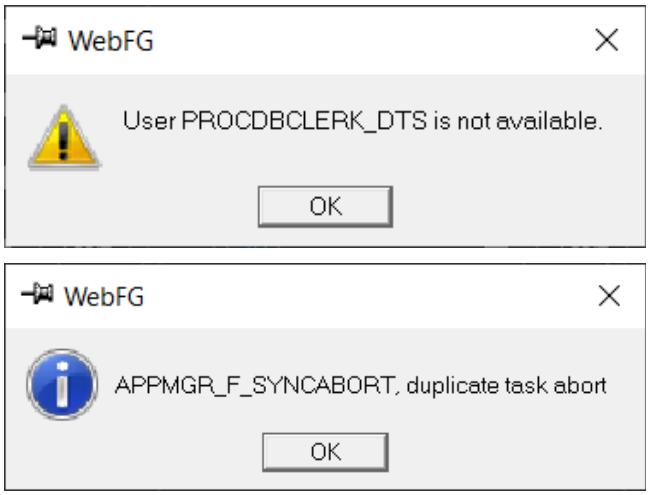


Figure 5. Typical Informative Message Boxes

2.1.7.2 Decision Message Boxes

Decision message boxes (Figure 6) provide you with information regarding a current activity, but they also ask you to make a decision about how the process or command should proceed. You inform Browser how the process/command should continue using the available push buttons.

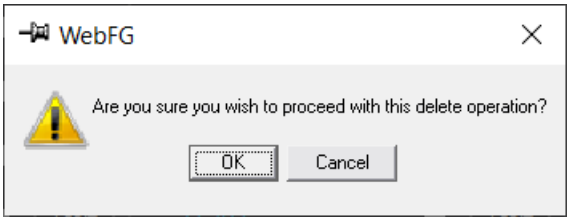


Figure 6. Typical Decision Message Box

Decision message boxes always contain OK and Cancel command buttons. They may also include additional elements, depending on the process or command that produced the message and the nature of the message itself.

- Selecting OK will always mean: "Continue the process/command, as indicated by the message." The message indicates the action that occurs when you select OK.
- Selecting Cancel will always mean: "Do not continue the process/command, and stop its execution without completion." The process or command exits or aborts and does not complete when you select Cancel.

2.2 Browser Concepts

In addition to Windows concepts, several Browser concepts must be understood in order to use the Browser software efficiently.

2.2.1 Display Viewer

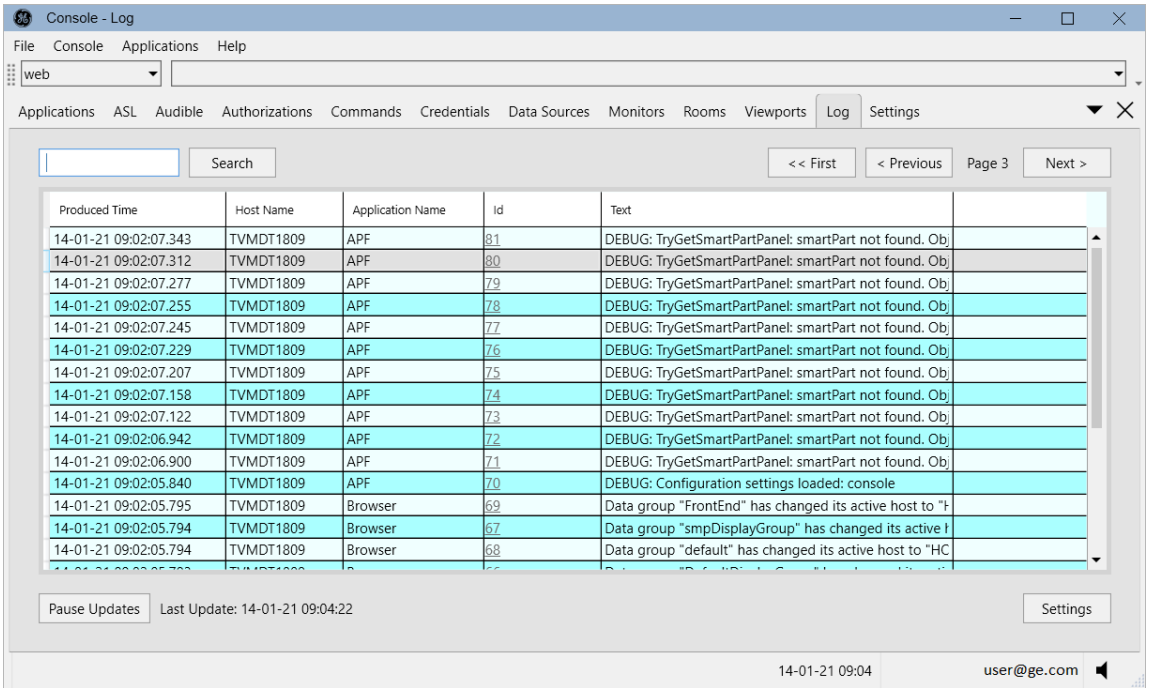
Display Viewer is proprietary software that has been designed with special features for viewing Browser displays. It allows for easy customization by the user. Display Viewer has been designed with specialized features that make it the best choice for viewing displays and interacting with other applications in the control room.

2.2.1.1 The Manager (APF Console)

The Manager is a tabbed dialog box from which the operator can control and configure Display Viewer attributes such as start-up options, alarm tones, and scripts.

In a multi-monitor configuration (more than one physical screen attached to a single CPU or workstation), one APF console manages and controls the viewports, rooms, displays, and commands for all the screens.

Refer to the *Habitat Software Installation and Maintenance Guide* for details about multi-headed workstation setup.



Produced Time	Host Name	Application Name	Id	Text
14-01-21 09:02:07.343	TVMDT1809	APF	81	DEBUG: TryGetSmartPartPanel: smartPart not found. Obj
14-01-21 09:02:07.312	TVMDT1809	APF	80	DEBUG: TryGetSmartPartPanel: smartPart not found. Obj
14-01-21 09:02:07.277	TVMDT1809	APF	79	DEBUG: TryGetSmartPartPanel: smartPart not found. Obj
14-01-21 09:02:07.255	TVMDT1809	APF	78	DEBUG: TryGetSmartPartPanel: smartPart not found. Obj
14-01-21 09:02:07.245	TVMDT1809	APF	77	DEBUG: TryGetSmartPartPanel: smartPart not found. Obj
14-01-21 09:02:07.229	TVMDT1809	APF	76	DEBUG: TryGetSmartPartPanel: smartPart not found. Obj
14-01-21 09:02:07.207	TVMDT1809	APF	75	DEBUG: TryGetSmartPartPanel: smartPart not found. Obj
14-01-21 09:02:07.158	TVMDT1809	APF	74	DEBUG: TryGetSmartPartPanel: smartPart not found. Obj
14-01-21 09:02:07.122	TVMDT1809	APF	73	DEBUG: TryGetSmartPartPanel: smartPart not found. Obj
14-01-21 09:02:06.942	TVMDT1809	APF	72	DEBUG: TryGetSmartPartPanel: smartPart not found. Obj
14-01-21 09:02:06.900	TVMDT1809	APF	71	DEBUG: TryGetSmartPartPanel: smartPart not found. Obj
14-01-21 09:02:05.840	TVMDT1809	APF	70	DEBUG: Configuration settings loaded: console
14-01-21 09:02:05.795	TVMDT1809	Browser	69	Data group "FrontEnd" has changed its active host to "H
14-01-21 09:02:05.794	TVMDT1809	Browser	67	Data group "smpDisplayGroup" has changed its active h
14-01-21 09:02:05.794	TVMDT1809	Browser	68	Data group "default" has changed its active host to "HC

Figure 7. APF Console, Log Tab

2.2.1.2 The Viewport

The Browser viewport is a specialized window for viewing displays or any web page. As with other windows, each viewport includes a title bar with Maximize, Minimize, and Close buttons; a status bar; a menu bar; and a standard toolbar. Viewports may also include customized menus and/or toolbars.

2.2.1.2.1 Viewport Features

Viewports share several features common to all Windows applications, as well as some specific to Browser, including:

- Vertical and horizontal scrollbars.
- The ability to size the viewport, either by using the Maximize button or by dragging the viewport boundaries.
- The ability to store the viewport on the Windows taskbar using the Minimize button.
- The ability to reposition the viewport as desired by dragging.
- A standard toolbar, including:
  - Back and Forward buttons.
  - A command line.
  - History lists.
- Menus.
- The ability to display web pages, spreadsheets, and other types of documents as well as Browser displays.
- Frame-making tools.

2.2.2 Displays

In Browser, displays are user-created interfaces to application databases. They provide the means to access, display, and interact with this data.

Browser displays have a common set of features and capabilities (see section 2.2.2.1 Display Features and Capabilities), and a common set of definable attributes.

Refer to the *Display Builder User Guide* for more information about displays.

2.2.2.1 Display Features and Capabilities

Browser displays have the following features and capabilities:

- Composed of elements, which are items that are created separately from the display and then combined in various fashions to form displays.
- Accessed using Browser commands and methods.
- Viewed in special Browser windows called viewports (see section 2.2.5 Viewports).
- Defined and viewed using the World Coordinate system, a typical X,Y coordinate system where X is the horizontal position and Y is the vertical position. Position 0,0 is the upper-left corner of the entire display.
- The ability to navigate through the display by:
  - Using viewport scroll bars (see section 4.1.1 Using Scroll Bars)
  - Panning through the display (see section 4.1.2 Panning Through Displays and section 4.1.3 Scrolling vs. Panning)

    - Positioning to specified record occurrences (see section 4.1.5 Positioning to Specific Record Occurrences), with predefined placeholders (see section 4.1.7 Positioning to Placeholders), via specific pages (see section 4.1.8 Positioning to Display Pages), and via the navigation window.
- The ability to view and interact with application data from one or more application databases.
- The ability to incorporate and use the many Windows and Browser methods of user interaction.
- The ability to have windows pop up from display elements by using the pop-up picture (see section 2.2.3 Pop-Up Pictures).
- The ability to define and view the display using simple and tabular layers (see section 2.2.4 Layers). Layers can be viewed separately or as a composite, as in multiple overhead transparencies.
- Include a set of definable attributes, some of which can affect the attributes of the viewport into which they are called.
- The ability to zoom the display — reduce or enlarge the size of the display in the viewport — by providing zoom factors (see section 4.2 Zooming the Viewport).
- Scalable text, which allows text to be scaled as the zoom factor is changed. The default is to not scale the text.
- "Declutter," a feature that allows items to be removed (decluttered) or added to the display at specified zoom factors. An item is defined to declutter by providing a low and a high declutter zoom factor. The item will appear when the zoom factor of the display is greater than or equal to the low declutter zoom factor and less than or equal to the high declutter zoom factor.
- The ability to customize the display using background images. Bitmap (.bmp), gif (.gif), jpeg (.jpg), and icon (.ico) images are supported.

Refer to the *Display Builder User Guide* for details about how these features work and for instructions on defining displays using these features.

2.2.3 Pop-Up Pictures

A useful method of viewing related information is to use pop-up pictures (Figure 8). Pop-ups provide a means for building custom interactive controls or simply for offering more information (see section 4.4 Displaying Pop-up Pictures).

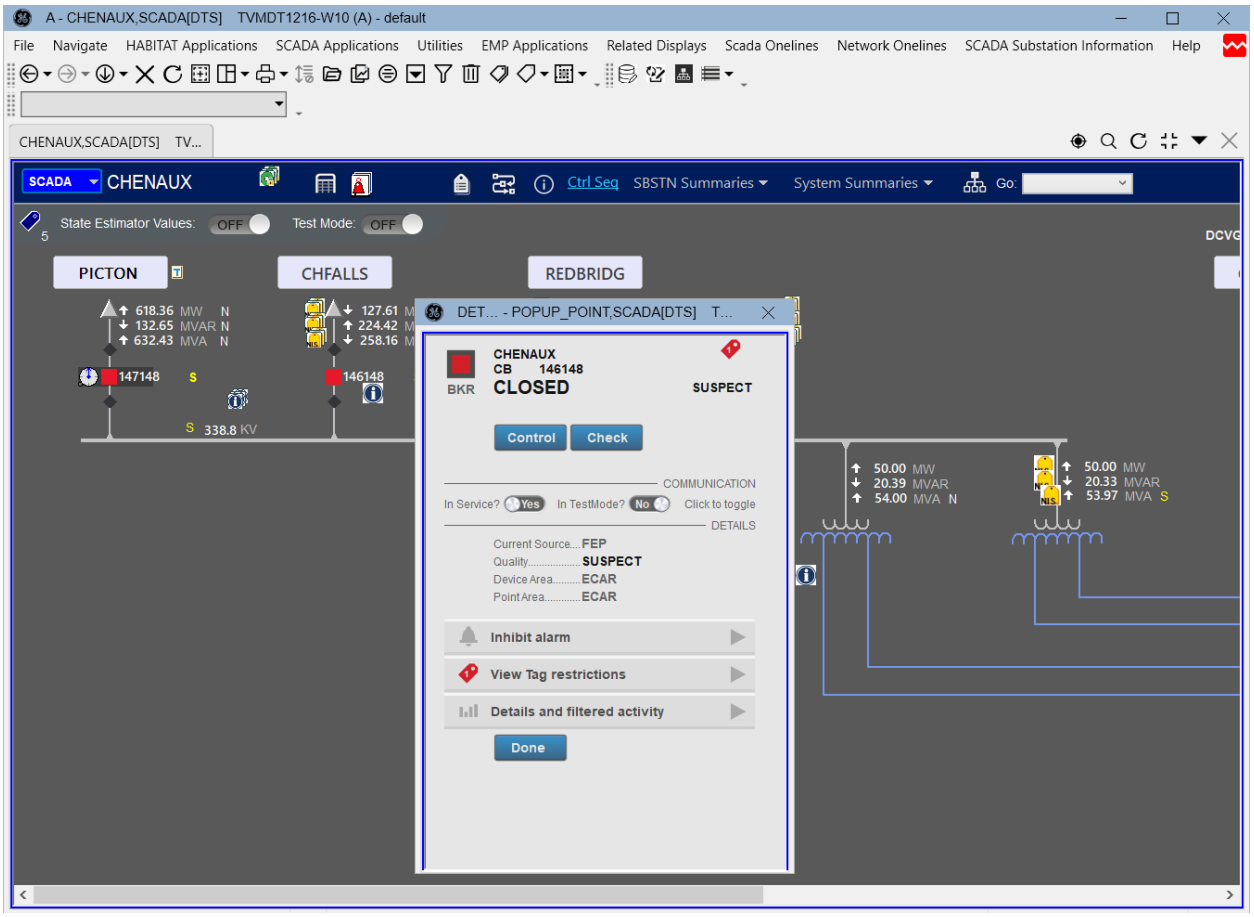


Figure 8. One-Line Display with Pop-up

2.2.4 Layers

Display layers give users the ability to selectively show one or more cross-sections of display data in a manner similar to overhead transparencies. Layers provide an additional decluttering mechanism, and they can simplify the display design process. They also provide an easy way to toggle between alternate views of the same or related data.

On a simple layer (Figure 9), display elements appear at a fixed location. When a simple layer is defined, elements are placed at specific coordinate locations. These elements will appear at their coordinate locations when the display is called to the workstation.

In addition, items placed on a simple layer are associated with a specific occurrence of a record type in the application database. The element will represent or directly display data from that occurrence only. Other placements can represent other occurrences of that or other record types in the database.

Simple layers have a feature called “placeholders” (see section 4.1.7 Positioning to Placeholders). These are named coordinate locations in the layer. The user can navigate directly to these specified locations in the online display.

Simple layers are generally used to display physical or geographic information such as system one-line displays. A one-line display consists of elements, each representing some component in the real-time system. These elements have been drawn or placed on a coordinate system to represent objects in the physical system.

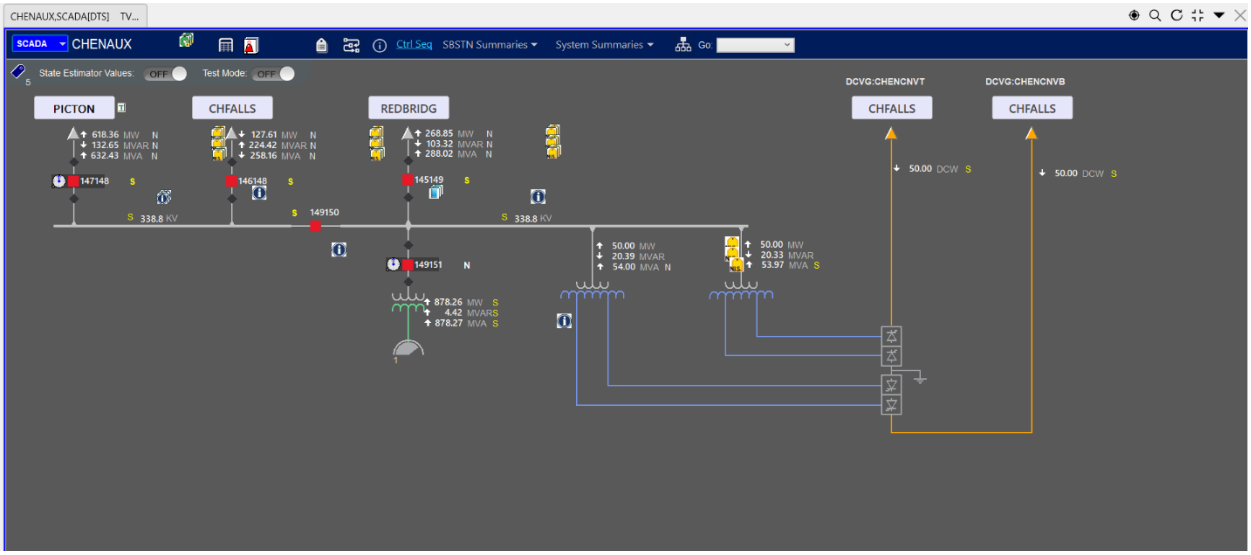


Figure 9. Simple Layer - System One-Line Display

2.2.4.1 Features of Tabular Layers

Tabular layers (Figure 10) are defined to automatically display a list of record occurrences of specified record types. Linkage to all record occurrences of the specified types is done automatically at display call-up, causing the data content to vary as the database values vary.

Tabular layers are constructed by making placements of groups of elements, each associated with a specific record type in the database. These groups of elements are called picgroups. Picgroups are the only components of a tabular layer.

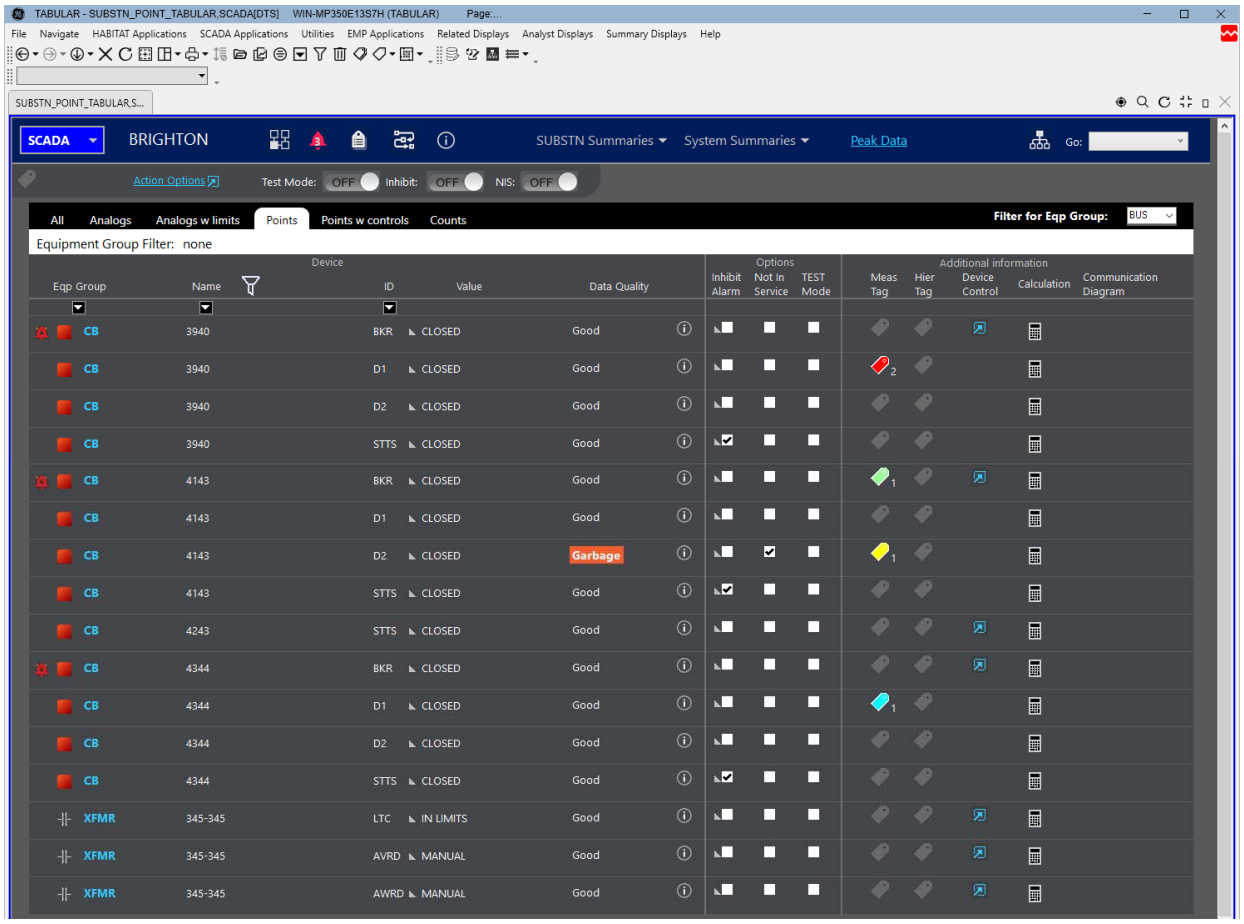


Figure 10. Tabular Layer

When a display containing a tabular layer is called up, Browser automatically displays all (or some subset of all) occurrences of the record type(s) of the picgroup(s) placed in the display. The sequence and occurrence of the picgroups are determined by the database hierarchy, the record list, and the display definition.

Tabular layers have the following features and definable attributes:

- Constructed solely of picgroups.
- The ability to automatically list all occurrences of a given record type in a database.
- The ability to automatically list all occurrences of all (or some subset of all) record types in a given hierarchy of a database.
- The ability to create a custom list of record occurrences, in any order, in any grouping, etc.
- The ability to automatically display a multi-dimensional array.
- The ability to divide the display into pages, and to position within the display page by page or by specifying the exact page number.
- The ability to create column and row headers, which will appear on every page of the display as the user scrolls up or down or right or left.

- The ability to specify the distance between, the grouping of, and the direction of repeat of the picgroups.
- The ability to sort data on a given field.

2.2.5 Viewports

Viewports are windows on a workstation screen in which users retrieve and view documents (Figure 2). Display Viewer viewports provide many navigation and configuration tools that may be familiar to users of standard web browsers, including Forward and Back buttons, “go to” lists, drop-down menus, and zoom-in/zoom-out buttons. Viewports also support many other functions specific to the GE system, such as user-definable toolbars, extended display history lists and other navigation aids, and special frame-making tools.

Another powerful feature of Browser viewports is the ability to call virtually any type of document into the viewport, provided that the necessary executable program and file-type associations have been registered with the workstation’s operating system. (Refer to your operating system documentation for more information about file-type associations.) Displays can be called into an existing viewport, or the viewport can be created when the display is called using the “dynamic viewport” feature. Display Viewer viewports may be used to show standard web pages, spreadsheets, and so forth.

Only one viewport can have input focus at a given time. The viewport that has input focus is called the “active” viewport, and it will receive any Browser commands by default. A viewport will remain the active viewport until another viewport is given input focus. Other windows, such as the APF console, can be given input focus, but the active viewport will not change until a different viewport is selected.

Each viewport has a separate set of definable attributes that can be set and modified from the APF console, from the viewport itself, and/or by the display called into it (see section 5.3.1 Viewport Attributes). Refer to the *Display Builder User Guide* for additional information about viewport attributes that can be overridden by displays.

2.2.6 Frames

Frames provide a means by which multiple displays can be viewed in a single viewport (Figure 11). Frames are controlled using a series of five toolbar buttons (see section 3.3.4.1 The Display Viewer Default Toolbar and section 4.9 Viewing Displays in Frames). Like viewports, frames can be sized by dragging the frame boundaries.

Frames include special features that increase their functionality and usability:

- Individual control for navigation and zoom features.
- Display control buttons for refreshing data, exchanging displays with other frames, etc.

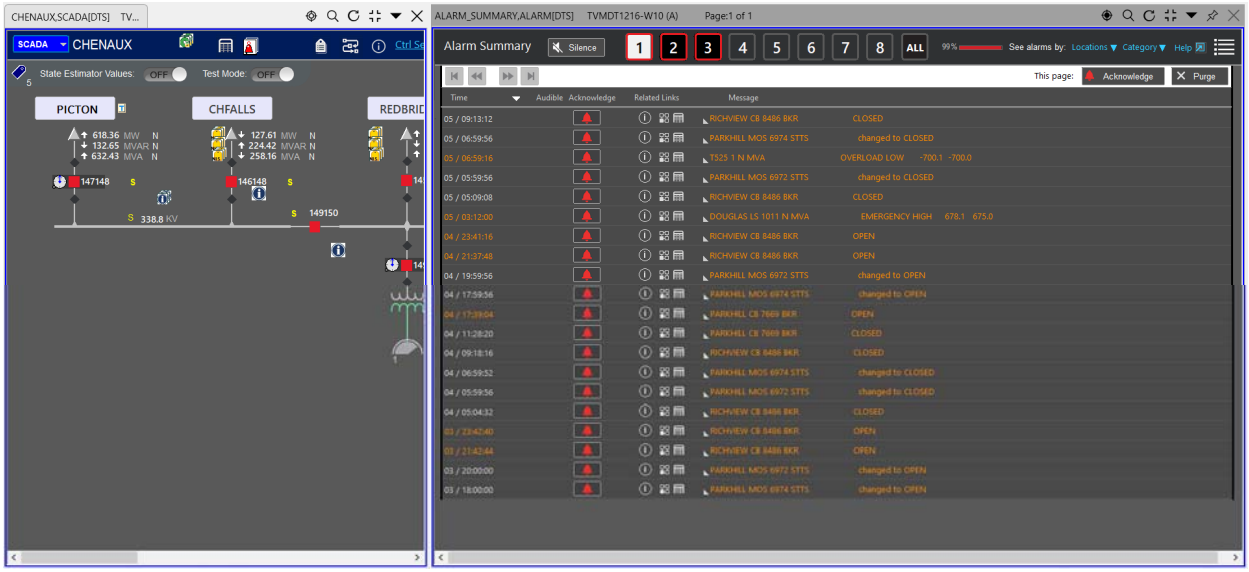


Figure 11. Viewport Utilizing Frames

2.2.7 Rooms

Each operator can also create a pre-defined screen layout called a “room.” Rooms allow users to create and position one or more viewports on the screen or screens, call displays or other documents in those viewports, and then save the arrangement for later retrieval. When the room is retrieved, all of the viewports will appear at their saved locations, with the same displays or other documents loaded and updated with current system data.

Individuals may create rooms for their own use on the local workstation, or a system analyst may create a room for shared use by placing it in a configurable location on the web server computer.

Figure 12 shows a typical room, containing (clockwise from upper left) a Scada one-line display, a Scada Tabular display, the Generation Area Status display, and the Alarm Summary display.

Note: Refer to the *Advanced Presentation Framework User Guide* for more information.

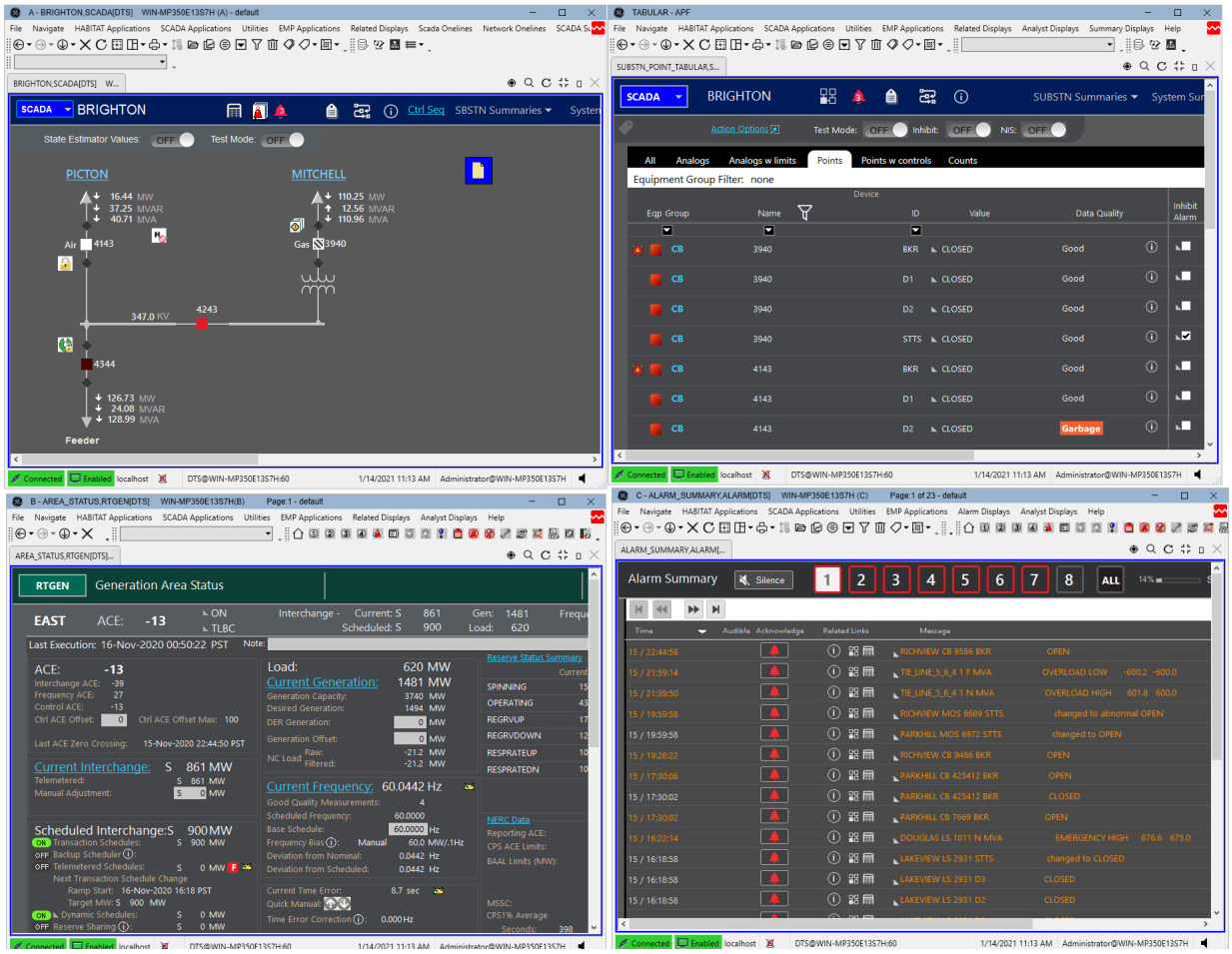


Figure 12. Typical Room with Four Viewports

2.2.8 HTML Help

Browser HTML Help is a context-sensitive help tool for accessing standard and customized help information from Browser displays (see section 3.9 Using Help). With context-sensitive help, the help information (the help context) that appears on the screen is specific to the display document in the viewport (the application context).

Help is an integrated part of Browser. It provides access to the help documents delivered with all products with a Browser user interface. Depending on the display design, help may be accessed through menus or by virtual keys.

2.2.8.1 Standard Features

Browser help provides the following features:

- Easy-to-use web-style interface.
- Three ways to search for help topics, allowing you to use the way most natural for you.
- Navigation buttons for moving easily and quickly through help documents.
- Help on the messages displayed on Browser log displays accessed through a virtual key.

2.2.9 Issuing Commands

One of the advantages of a graphical user interface is the flexibility operators have to work in the way that is most natural for them. Accordingly, commands in Browser can be issued using any of six methods:

- Command lines
- Virtual keys
- Command buttons
- Menus
- Toolbars
- Directories

Regardless of how commands are issued, their execution will be affected by two factors:

- The current context
- Command synchronization

The methods for issuing commands, and the factors affecting them, are discussed in the following sections.

2.2.9.1 Command Lines

Command lines are text boxes where Browser commands can be directly typed and executed. Each viewport created during a Browser session has a command line (Figure 13). The viewport command line appears in the standard toolbar.

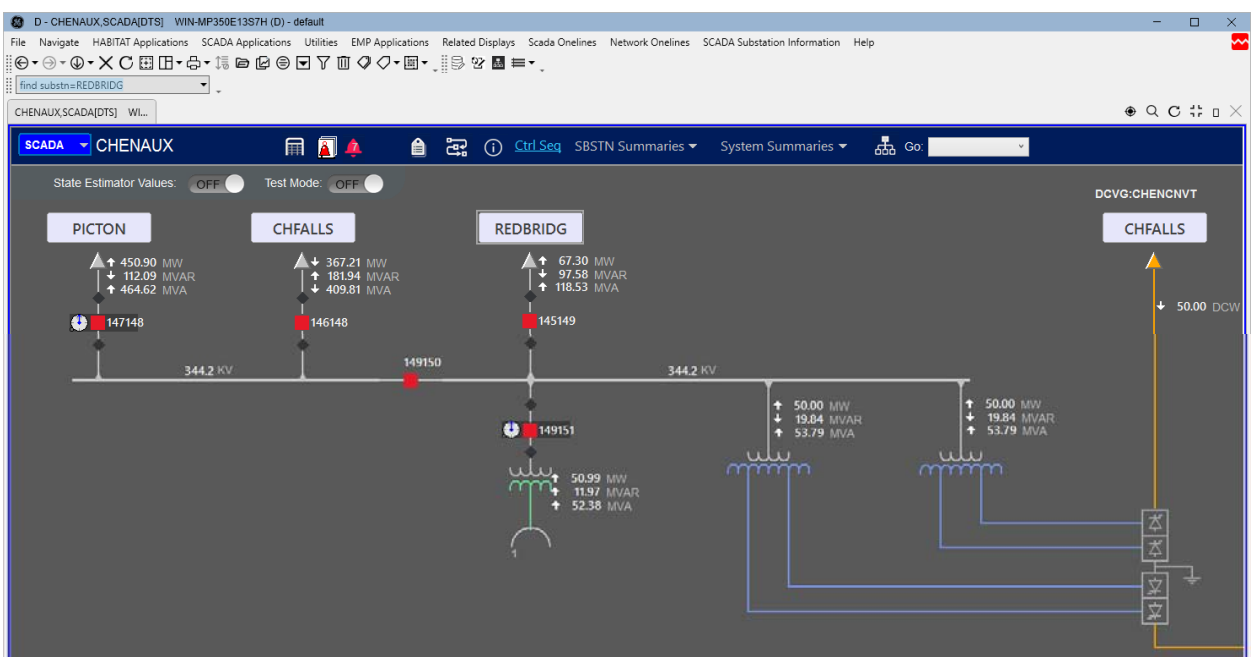


Figure 13. Command Line Box

Commands are executed in a command window by typing the command and pressing Enter. The command will execute based on the current context and any qualifiers to the command.

When more than one viewport is open, the active viewport receives commands from the APF console, unless the /VIEWPORT qualifier is used. Commands issued from viewport command windows are applied to the issuing viewport by default.

Refer to the *Browser Reference Manual* for more information about using commands and qualifiers.

2.2.9.2 Virtual Keys

Another method of issuing commands is by using a virtual key (see section 5.2.1 Defining a Virtual Key). A “virtual key” is a keyboard or mouse action that has been assigned an arbitrary name (called the “virtual key name”) and a command.

As an example, the action of pressing the F6 key could be assigned the virtual key name Zoom, with the associated command ZOOM .5. When the F6 key is pressed, the ZOOM .5 command would then execute.

Virtual keys are defined through the key map (see section 5.2 The Key Map). A key map can be defined for each Browser client, or one key map can be shared by multiple clients.

Virtual keys can be defined to accept or require additional information from the command line or pointer context. The command line or context will provide Browser with any additional information it needs to execute the command.

2.2.9.3 Command Buttons

Command buttons are display elements that, when clicked, cause a command to execute. Buttons are defined at the time the display is designed by way of the keyset.

Refer to the *Display Builder User Guide* for more information about defining buttons.

2.2.9.4 Menus

Menus are another means of issuing commands in Browser. Drop-down menus can be defined and added to the viewport menu bar for a display, and pop-up menus can be attached to displays or display elements. Each type of menu can also be defined with submenus.

2.2.9.5 Toolbars

Toolbars are used to issue frequently used commands in Browser. Toolbars can be defined and added to the viewport for a display. Toolbars are fully customizable using a proprietary configuration utility.

2.2.9.6 Directories

Directories are used to issue call-up commands for displays and rooms in Browser.

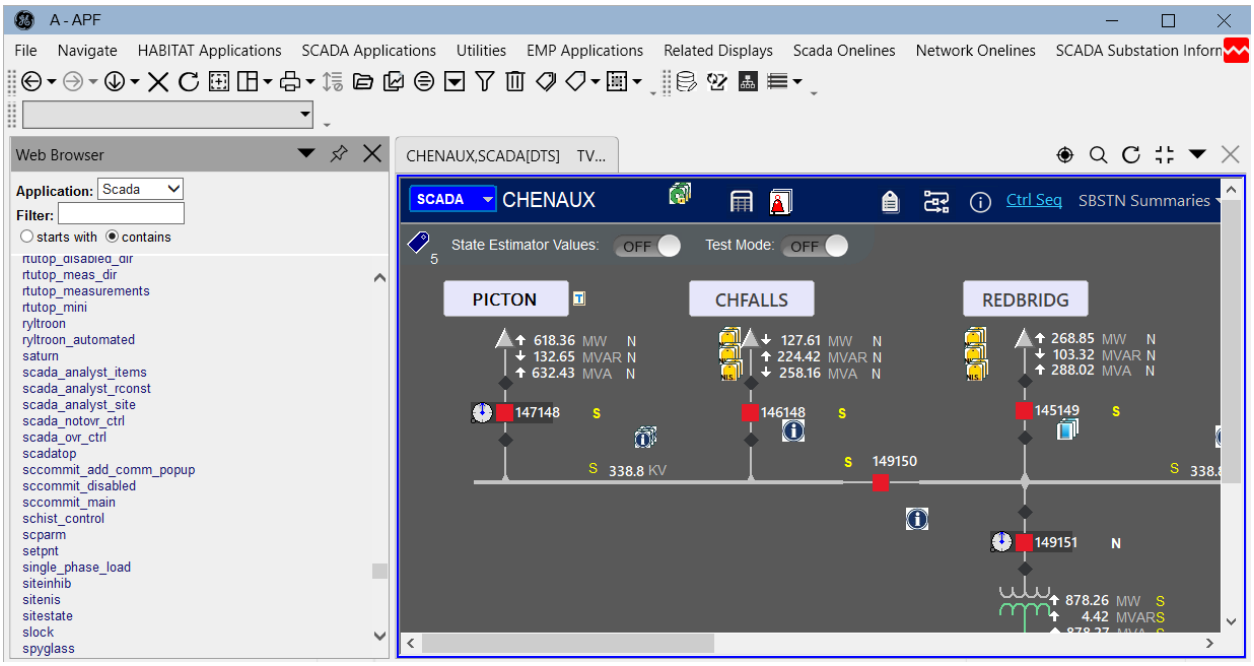


Figure 14. Viewport with Display Directory in Left Frame

2.2.9.7 The Current Context

The "current context" (see section 2.2.9.7.1 The Context Attributes) is a set of contextual information regarding four areas of the Browser session:

- **The Session Context:** This is contextual information about the Browser session itself, and it includes information about the Browser client and server, the workstation name, and the active viewport name.
- **The Viewport Context:** This is contextual information about the display in the active viewport, such as its application and family, its zoom level, its layers, etc.
- **The Pointer Context:** This is contextual information about the current position of the pointer in the active viewport. It includes information regarding the database, the current record type, the record occurrence, and the coordinates associated with the current pointer position.
- **The Mark Context:** A record occurrence in a display can be marked using the Mark command, establishing a separate set of context attributes.

The context of a display element can be viewed by positioning the mouse pointer over the object and pressing F11 to open the Context dialog box.

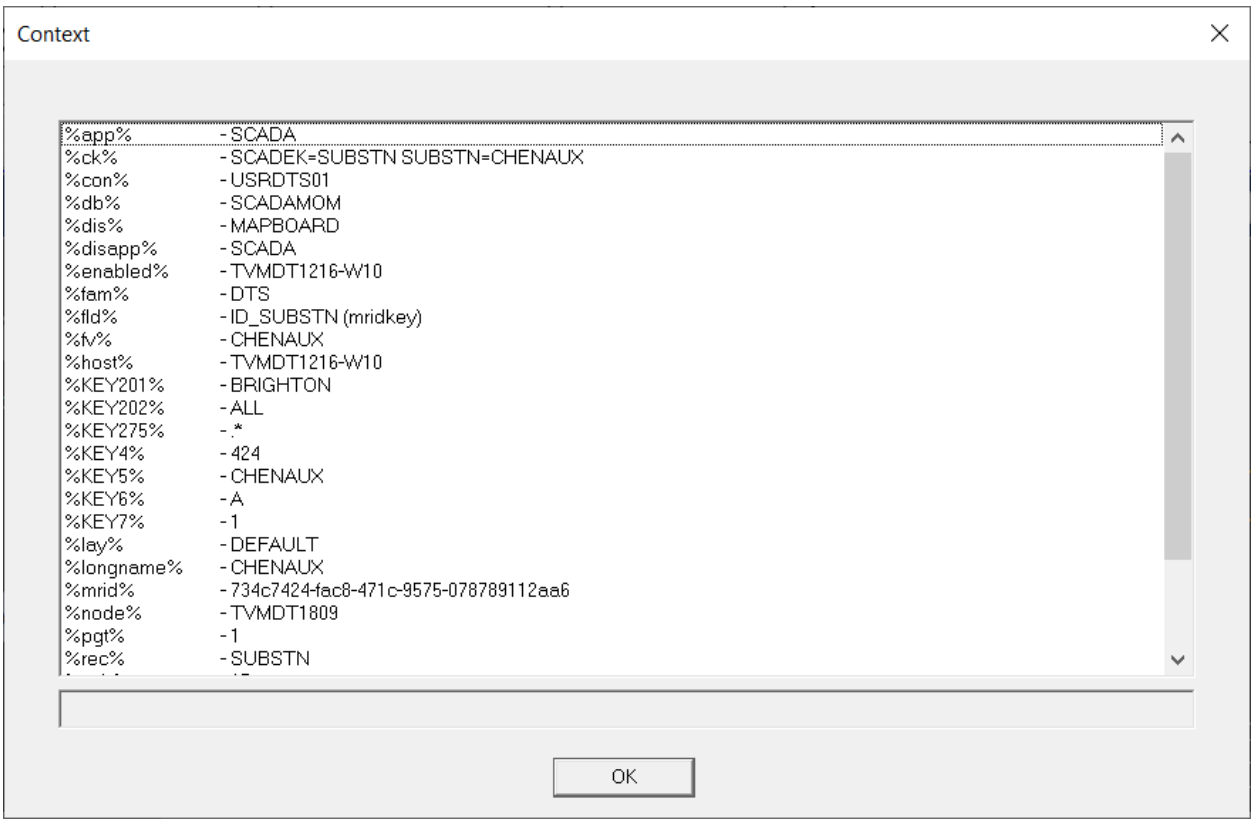


Figure 15. Context Dialog Box

Browser uses the context information as defaults to determine where and how to issue a command and to resolve "proxy strings." Proxy strings allow certain contextual information to be included automatically in a command string.

Refer to the *Browser Reference Manual* for more information about using proxy strings.

2.2.9.7.1 The Context Attributes

The following is a list of all the attributes that make up the current context:

- **APPLICATION:** The application of the display in the active viewport and the application of the item at the current pointer position.
- **DATABASE:** The database of the item at the current pointer position.
- **DISPLAY:** The name of the display in the active viewport.
- **FAMILY:** The application family name of the display in the active viewport and the application family name of the item at the current pointer position.
- **FIELD:** The name of the field at the current pointer position.
- **FIELD VALUE:** The value in the field at the current pointer position and the value in any specified field in the record at the current pointer location.
- **HOST:** The server for the display.

- **LAYER:** The name of the layer(s) currently in view in the display in the active viewport.
- **MARK Attributes:** This is a duplicate set of all the other context attributes listed here, but which applies only to the marked record occurrence, if any.
- **NODE:** The node name of the workstation.
- **PAGE NUMBER:** The page number of the tabular layer in the active viewport.
- **PAGE NUMBER MAXIMUM:** The maximum (last) page number of the tabular layer in the active viewport.
- **RECORD:** The record type associated with the item at the current pointer position.
- **REFRESH Text String:** A user-definable text string that can be used when the refresh attribute is set OFF for a viewport.
- **STATION:** The name of the STATION clone associated with the Browser session.
- **SUBSCRIPT:** The subscript value of the occurrence of the record type at the current pointer position.
- **VIEWPORT:** The active viewport, as indicated by the viewport with input focus.
- **WORKSTATION:** The name of the workstation.
- **ZOOM:** The zoom factor in the active viewport.

Context attributes are set by default based on the current Browser session, the active viewport, and the current pointer position.

2.2.9.8 Key Proxy String Values

Each Browser viewport provides a set of global fields that can be used to supply values in commands. These fields are called “key proxy strings.”

The key proxy string fields are named KEY0, KEY1, KEY2… through KEY65535, and their values are set using the SET command. Their values can be set either as character or as numeric, and the values can be set for a specific viewport or globally, affecting all viewports.

Once a key proxy string value is set, it will be substituted in any command that uses the %KEYn% proxy string, where *n* is 0 through 65535.

To set the value of a key proxy string, do <u>one</u> of the following:

- To set the value for a specific viewport, enter the following in the viewport command line:

  SET/VIEWPORT=viewport KEYx=value

  where *viewport* is the name of the viewport whose key proxy string value you want to modify, *x* is any integer between 0 and 65535, and *value* is the value you want for that field.

- To set the value for the active viewport, enter the following in the active viewport command line:

  SET KEYx=value

  where *x* is an integer between 0 and 65535, and *value* is the value you want for that field.

- To set a global value for all viewports, enter the following in the Viewer Manager command line:

  SET/ALL KEYx=value

where *x* is the integer indicating the chosen key proxy, and *value* is the value you want in that field.

The value will be substituted in all commands issued to the viewport(s) where the modified %KEYx% proxy string is used.

Refer to the *Browser Reference Manual* for additional information about the SET command and for a general discussion of proxy string resolution.

2.2.9.9 Command Synchronization

Using the /WAIT qualifier on selected commands causes a viewport to be “busy” until the command is completed before executing any other user or user-process–issued command.

The mail and data entry (Delete, Insert, Copy, Enter, and Mask) commands can busy the viewport. A busy viewport can be “un-busied” using a pre-defined set of keystrokes, such as Ctrl+Y, which is the default.

2.2.10 Automated Testing of DICE and MAIL Commands

To increase the amount of automated software testing at GE, Browser has been enhanced to capture DICE and MAIL commands initiated from Browser.

Currently, when a MAIL command is executed by Browser, implicit contextual information about the database and record is gathered and sent along with the MAIL command text to WEBFGSERVER, which will in turn forward the message to RUSERSERVER for further routing.

During capture mode, the outgoing messages (in XML format) are saved as they are being serialized and sent over the network. These saved messages are referred to as Actions.

When a capture session starts, the client creates a session file in XML format that records the commands issued as well as any Actions performed throughout the capture session.

To start or stop a capture session, use the following command:

RECORD/FILE=<filename> [ON | OFF]

To indicate that a capture session is in progress, an asterisk character (*) prepends the title of all viewports that share the same data source as the viewport that issued the RECORD command. Commands issued to this group of viewports will be recorded in the same session file.

The captured session file can be automatically transformed by the GE Automated Testing Suite into a starting point for building automated tests. For more information, refer to the GE Automated Testing Suite Installation Kit (released separately).

2.2.11 Data Entry Operations

There are four major operations that can be performed from displays to modify the contents of an application database:

- **Entering** values into database fields, including setting mask and logical field values.
- **Inserting** one or more instances of a record type into a database.
- **Copying** one or more records from one database to another or from one position to another in the same database.

- **Deleting** one or more records from a database.

Entry of data into specific fields can be performed from both simple and tabular layers for any field that is defined as enterable. Inserting, copying, and deleting can only be performed using tabular layers.

The display designer chooses which fields will be enabled for data entry, and which will support insert, copy, and delete operations.

Operational data entry is typically controlled by the applications themselves. An example is overriding the current value of a telemetered point in Scada. In this case, the point must first be removed from service. Once removed from service, the point will no longer be updated by the Scada application, and manual entry is permitted.

In all cases of data entry into the database, a full complement of error-checking procedures is provided by the system support software and is used by application developers to ensure reliable operation. Basic checks, such as valid values or ranges, are performed directly by the user interface portion of the system support software. More-complicated checks are carried out within extensions provided by individual applications.

2.2.12 Printing Displays

Displays are printed via Windows print queues. Any printer that supports Windows printing may be used to print Browser displays.

2.2.13 Audible Alarms

The Advanced Presentation Framework (APF) Audible Management & Audible Configuration applications replace the Browser AudioAlarm feature. In response to an AudioAlarm command, Browser associates the sound file with an APF Audible Profile and uses the APF API to queue a sound item, such that the sound file will be played based on its priority and in a moderated fashion among other APF applications that also play sound files. The two implementations are functionally equivalent with some minor differences; transitioning will require some configuration changes. Refer to the *Advanced Presentation Framework User Guide* for more information.

3. User Basics

Users of other Windows software will find that Browser has many features in common with other applications. Previous users of Rapport-FG can continue to use many of the methods with which they have become familiar, while learning new methods that will make their tasks faster and easier.

3.1 Starting Browser

Once the Advanced Presentation Framework (APF) and Browser client (Browser) installation is complete, a shortcut is available to launch APF:

Start > Programs > GE > APF 4.4

Once the APF console is displayed, Browser can be launched from the “Applications” tab.

Browser requires a configuration file to connect to the server. If any problems are encountered upon startup, see your system administrator for assistance.

The Browser client requires support from the Browser server component and Habitat; if this support is not present, the Browser Viewport will open, but the startup display will not appear.

3.2 Logging In

The Browser client processes (other than standalone configurations) are configured to connect to the host machine. When the Browser client initiates a connection with the host, the Connect dialog box appears (Figure 16).

Enter a valid user ID and password in the text boxes and click OK. These credentials are written to the system registry after the first successful login. Whether this dialog box is launched each time a Browser session is started depends on server configuration or authentication.

If, at any time during a Browser session, it becomes necessary to change the password, select the Change Password menu item from the standard toolbar (see section 3.3.4.1 The Display Viewer Default Toolbar).

To log out off the session, select the Logout menu item shown in Figure 17. Before logging out the section, the client queries the PERMIT database if the current section is the very last section that monitors the Critical Area; if so, the logout is cancelled and a message is shown to describe the reason. Browser removes the credential entry of the default data source from the credential cache before logging out of the section.

> **Note**: The Change Password and Logout menu items are not available by default. These buttons will appear in the toolbar if you configure them in the standard toolbar xml file.

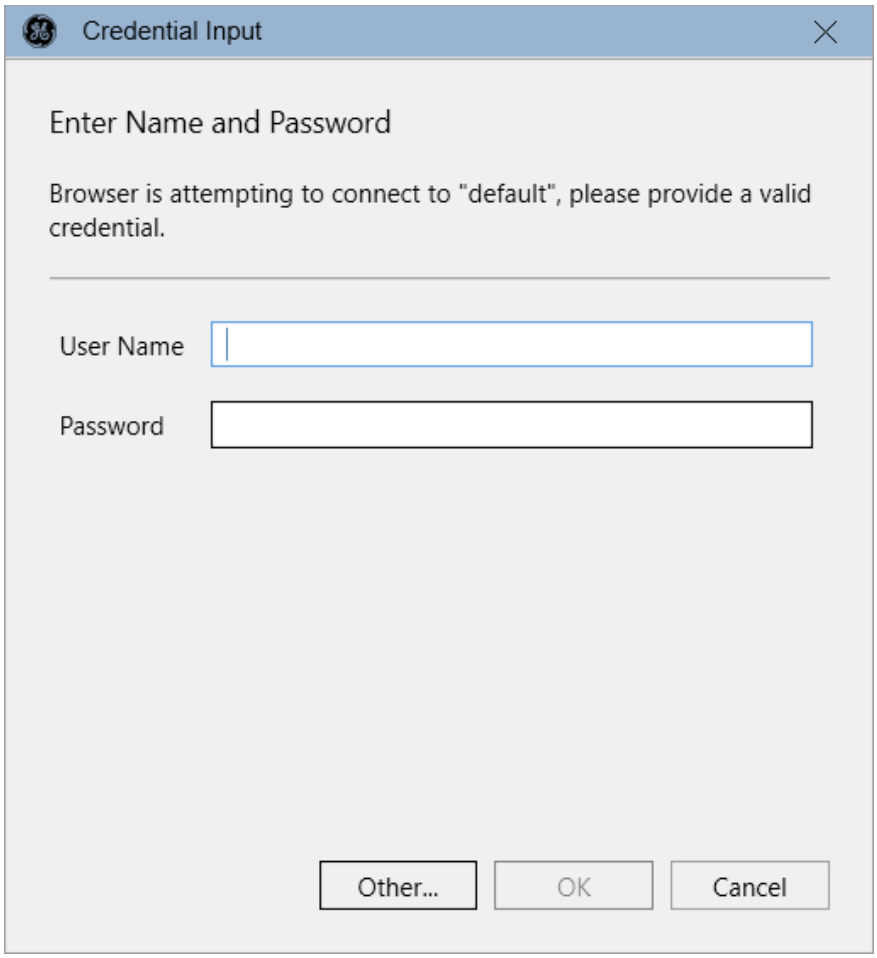


Figure 16. Credential Input Dialog Box

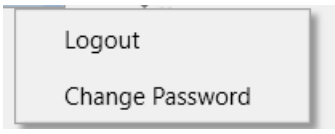


Figure 17. Change Password and Logout Menu Items

3.3 Issuing Commands

Browser provides several methods for issuing commands (see section 2.2.9 Issuing Commands), and the ability to cancel synchronized commands (see section 3.3.7 Canceling Synchronized Commands).

3.3.1 Using Virtual Keys

To execute commands using virtual keys (see section 5.2 The Key Map), use the following procedure:

1. (Optional) Type any legal command qualifiers or parameters to the virtual key command in the Active Viewport command window.
2. Position the pointer on the display or display item in the active viewport to provide context attributes for the virtual key command.

3. Press the virtual key, which will execute the desired command.

Refer to the *Browser Reference Manual* for more information about the legal command qualifiers and parameters for Browser commands.

Consult your system administrator for a list of the virtual keys and their associated Browser commands for your system.

3.3.2 Using Command Buttons

To execute commands using buttons, use the following procedure:

1. (Optional) Some button commands may require additional information to execute. In this case, type command qualifiers or parameters in the Viewport command box prior to proceeding.
2. Click the command button.

Refer to the *Browser Reference Manual* for more information about the legal command qualifiers and parameters to Browser commands.

Consult your system administrator for information about the commands associated with buttons.

3.3.3 Using Menus

Commands can be issued from both drop-down and pop-up menus. By default, all menus are activated by clicking the left mouse button.

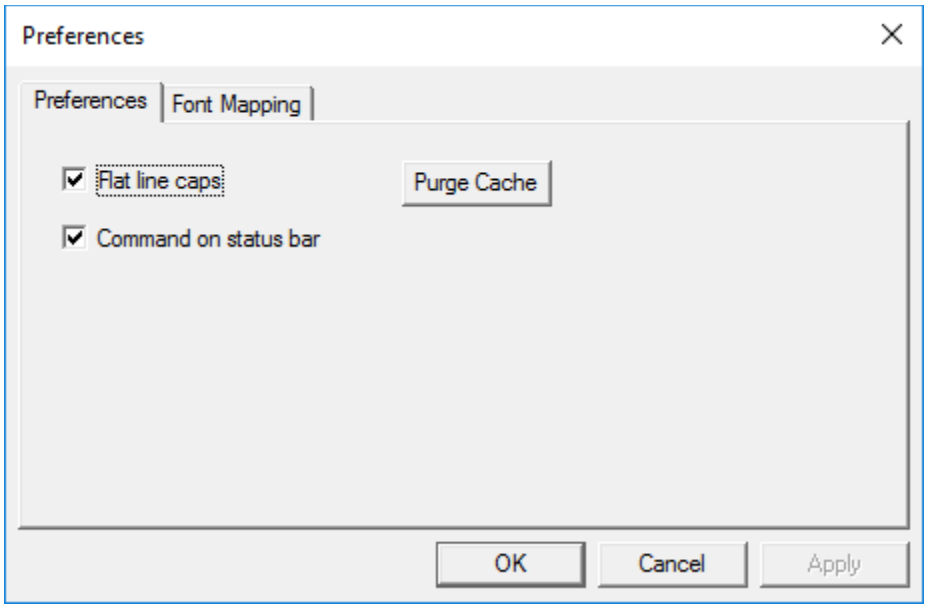
3.3.3.1 Default Menus

The File and Help menus are the main default menus in Browser*:*

The File menu has the following options:

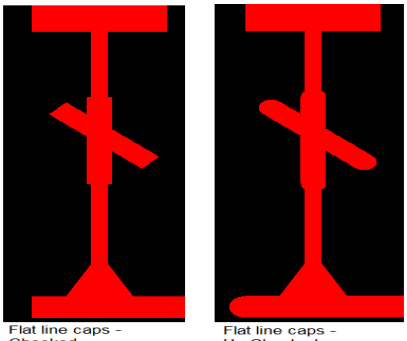
- **Close:** Closes the application viewport.
- **Create SVG:** Executes the Create SVG command, which creates a Scalable Vector Graphics (SVG) file for the display in the active pane and saves it with a filename specified in the Save dialog box.
- **Purge Cache:** Executes the Purge Cache command, which purges the Browser cache.
- **Settings:** The Settings menu has three sub menus. The options in this menu are available when displays are loaded into the viewport/pane.

  Preferences: Sets user preferences. Selecting this option opens the Preferences dialog box.



- The Preferences Tab has the following options:

  Flat Line Caps: This option sets the flat line style to use for drawing lines. The following image provides an example:

  Flat line caps - Checked Flat line caps - Un Checked

  Command On Status Bar: If this option is selected, the command associated with menus and other elements in the display appear in the Viewport Status bar when you hover the mouse pointer over them.

- The Font Mapping Tab helps you map an FG Font family font to a Microsoft Windows font.

Keymap: Displays a dialog box for defining Keymaps. For more information, see section 5.2 The Key Map.

Modes: Opens the Modes Management PERMIT display.

3.3.3.2 Using Drop-Down Menus

To issue a command from a drop-down menu in the main menu bar of a viewport, use the following procedure:

1. Click on the menu label; the menu drops down from the menu bar.
2. Move the pointer to the desired selection; the menu labels will be highlighted in turn as the pointer moves over them.

3. Click the desired selection.
4. (Optional) If the desired selection has a submenu (i.e., it has the ▸ symbol to the far right of the label):
   - Click the menu label; the submenu drops down to either the right or the left of the main menu.
   - Click the desired selection in the submenu.

   The menu disappears and the command associated with the menu option executes.

> **Note:** An ellipsis (...) on a label means that selecting this option will open a dialog box for further operator input, rather than directly issuing a command.

3.3.3.3 Using Pop-Up Menus

By default, clicking the left mouse button will open a pop-up menu. This procedure assumes that the key map has not been modified to change this virtual key.

To use pop-up menus in the active viewport, use the following procedure:

1. Click the object with the associated pop-up menu; the menu pops up under the pointer.
2. Move the pointer to the desired selection; the menu labels will be highlighted in turn as the pointer moves over them.
3. Click the desired selection.
4. (Optional) If the desired selection has a submenu (i.e., it has the ▸ symbol to the far right of the label):
   - Click the menu label; the submenu drops down to either the right or the left of the main menu.
   - Click the desired selection in the submenu.

   The menu disappears and the command associated with the menu option executes.

3.3.3.4 Canceling a Menu

To cancel a menu that has been pulled down or popped up without selecting an option, click in any clear area of the viewport. The menu disappears and no command executes.

3.3.4 Using Toolbars

Browser includes one standard toolbar from which commands can be issued. The display designer can also add customized toolbars to displays. A toolbar contains icons, small buttons linked to commands. To issue a command using the toolbar, click the desired icon.

3.3.4.1 The Display Viewer Default Toolbar

The standard toolbar includes the following icons, from left to right:

- **Back with Display History:** See section 3.5.5.1 Using the Back, Forward, and Previous Buttons.

- **Forward with Display History:** See section 3.5.5.1 Using the Back, Forward, and Previous Buttons.
- **Display History Summary:** See section 3.5.5.2 Using the Display History Lists.
- **Stop the Current Call-Up:** See section 3.5.5.4 Stopping the Current Call-Up.
- **Rooms Directory with Rooms History:** See section 3.6.1 Using the Rooms Directory.
- **Refresh the Current Display:** See section 3.5.6 Updating the Display.
- **Print the Current Display:** See section 4.10 Printing Displays.
- **Print Preview of the Current Display:** See section 4.11 Print Preview.
- **Display Directory:** See section 3.5.1 Using the Display Directory.
- **Open a Trending Frame:** See section 4.9.3 Opening a Trending Frame.
- **Set Modes:** Opens the Modes Management PERMIT display.
- **Sort:** See section 4.6.1 Sorting Data.
- **Filter:** See section 4.6.2 Filtering Data.
- **Purge Cache:** Purge the Internet cache for Browser–related files.
- **Command Window with Command History:** See section 3.3.6 Using the Command Line.
- **Open Frames:** See section 4.9 Viewing Displays in Frames.
- **DPL Disable:** See section 4.12 Disable Dynamic Page Length.

3.3.4.2 Customizing the Default Toolbar

Browser offers the ability to customize the default toolbar. Custom toolbars can be defined on a per-application basis.

For more information, see the *Browser Software Installation and Maintenance Guide* or your system administrator.

3.3.5 Using Directories

Directories allow the user to issue command calling displays or rooms by selecting the desired item from a list. The selected item supplies needed contextual information for the command; for example, the Display directory is organized by application. The actual command issued will depend on the type of directory (see section 3.5.1 Using the Display Directory and section 3.6.1 Using the Rooms Directory).

3.3.6 Using the Command Line

Commands can be directly entered and issued from the viewport command line, and previously issued commands can be recalled and re-issued.

3.3.6.1 Directly Entering and Issuing Commands

To directly enter and issue commands from a command line, use the following procedure:

1. Click in the command window of the viewport for which you want to issue the command. The blinking text-entry cursor appears, indicating that the command window has input focus.
2. Type the command and all associated qualifiers and parameters at the prompt.

   > **Important Note:** Some commands specific to Browser that are issued from the viewport command line must be preceded by the fgcmd qualifier.

   Example: ROOM room_name

   Refer to the *Browser Reference Manual* for information about commands and command syntax.
3. Press Enter.

   The command is logged on the Log tab of the APF console, and the command is executed.

3.3.6.2 Recalling and Reissuing Commands

Issuing a command from the viewport command line allows the recall and re-issue of commands.

> **Note:** Only commands issued from a command line can be recalled using the following procedure. Commands issued by other means (i.e., by menus) cannot be re-issued in this manner.

To recall and re-issue a command, use the following procedure:

1. Click the Arrow button ▾ adjacent to the command line.
2. Select the desired command from the command list, using the scroll bars if necessary.

   > **Note:** Each viewport keeps an individual command history list. For example, a command issued from viewport A will not be available from the viewport B history list.
3. Edit the command as required.
4. Press Enter.

   The command is logged on the Log tab of the APF Console, and the command is executed.

3.3.7 Canceling Synchronized Commands

All commands managed are managed asynchronously by default. That is, they are issued as received without waiting for the completion of the previous command.

Some commands can be issued with a /WAIT qualifier, which will reserve the viewport from which the command was issued until the command completes. Such synchronized commands are indicated by the busy pointer.

Sometimes, such as when errors occur in executing the command, you may want to cancel the synchronized command. To do this, press the Cancel key, which is the pre-defined set of keystrokes that will cancel a synchronized command. The default Cancel key is Ctrl+Y.

Refer to your system manager for the specific Cancel key for your system.

To free a busy viewport:

1. Give input focus to the busy viewport.
2. Press the Cancel key; Ctrl+Y is the default.

   The busy pointer returns to the default pointer, and new commands can now be issued in this viewport.

3.4 Starting Habitat Tasks

The viewport command lines provide a convenient means for starting Habitat application tasks and case manager tasks. The commands accepted by Browser are the Habitat RUN, STOP, SAVE, REMOVE, RETRIEVE, and UPDATE commands. The user can also issue commands that control these tasks by clicking display buttons whose keysets issue the commands.

The APF console tracks task startup and completion on the Log tab, and issues a pop-up message when a task completes.

Refer to the *Browser Reference Manual* for descriptions of Habitat commands.

3.5 Calling Displays

This section describes the methods by which displays in Browser can be called into a viewport. In all of the methods, the display is called into the active viewport by default. By using the /VIEWPORT qualifier, the display can also be called into a specific viewport or into a new dynamic viewport.

3.5.1 Using the Display Directory

To call displays to the workstation using the Display directory (

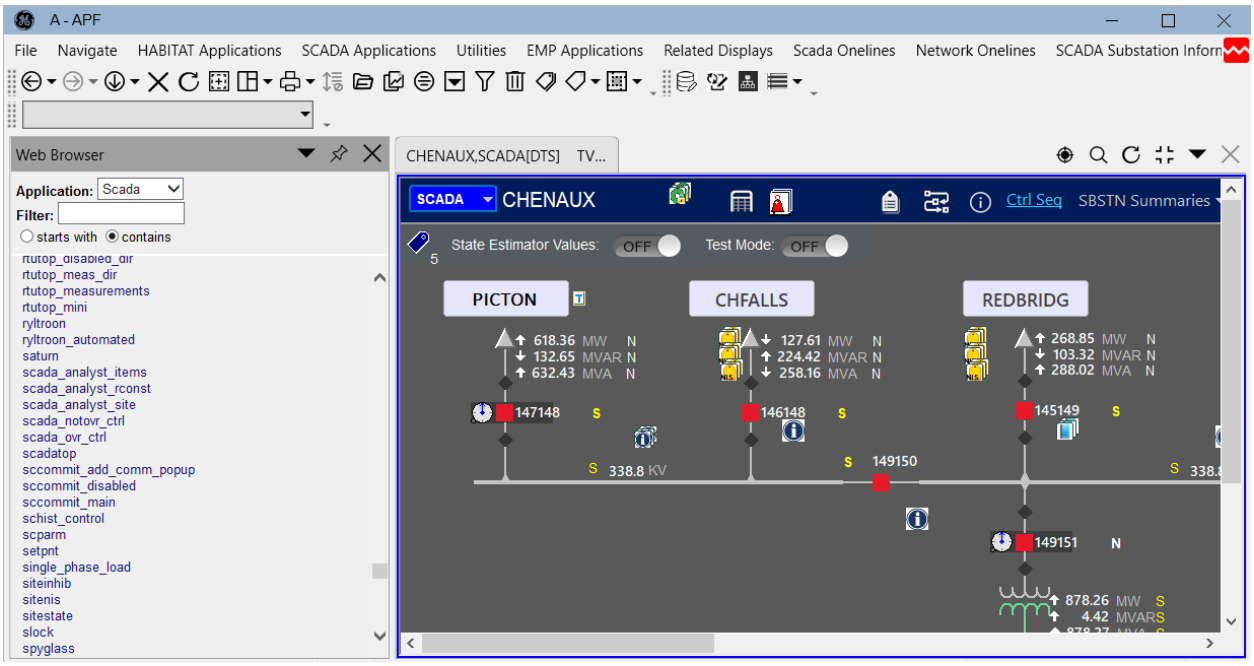


Figure 14), use the following procedure:

1. Give input focus to the viewport into which you want to call the display.
2. Click the Display Directory icon on the standard viewport toolbar. The Display directory will open in a new frame on the left side of the viewport.
3. Select the application for the desired display from the App text box.
4. Click on the desired display from the list. If the display is not visible, use either of the following methods:
   - Scroll through the list using the vertical scroll bars.

   Or:

   - Type all or part of the display name in the Filter text box. This will filter the list to show only those displays matching the entered criteria. For example, typing “S-C-A” into the Filter text box and selecting “starts with” radio button will exclude all displays with names not beginning with those letters.

   The Display command is logged on the Log tab in the APF console, and the display appears in the designated viewport.

3.5.2 Using the Display Command

To call displays to the workstation using the Display command, use the following procedure:

1. Give input focus to the viewport into which you want to call the display. As an option, the viewport may be specified with a command qualifier (see section 3.5.3 Calling the Display into a Specific Viewport).

2. Click in the command line of the viewport.
3. Type the following and press Enter:

   DISPLAY/APP=appname disname

   where *appname* is the application for the display, and *disname* is the name of the display to be called up.

   > **Note:** The /APPLICATION qualifier is not required. The default application is the application of the display in the active viewport. The *disname* can be proxied (%dis%).

   The Display command is logged on the Log tab of the APF console, and the display appears in the designated viewport.

Refer to the *Browser Reference Manual* for information about the valid qualifiers for the Display command.

3.5.3 Calling the Display into a Specific Viewport

To call displays into a specific viewport, use the following procedure:

1. Follow the procedures in either of these sections:
   - 3.5.2 Using the Display Command.
   - 3.5.5.3 Using the Previous/Display Command.
2. Add the viewport qualifier immediately before the display name.

   /VIEWPORT=viewport

   where *viewport* is the name of the specific viewport in which you want the display to appear.

   The Display command is logged on the Log tab of the APF console, and the display appears in the designated viewport. If the specified viewport does not exist, it will be created.

Refer to the *Browser Reference Manual* for information about the valid qualifiers for the Display command.

3.5.4 Display Drag and Drop

The “display drag and drop” feature allows you to select a display call-up button in one viewport, and have the display appear in either a second existing viewport or a new viewport. This feature is available by default.

To use this feature, use the following procedure:

1. Position the pointer on a display call-up button in a viewport and click it until the drag-and-drop indicator appears adjacent to the pointer.
2. Drag the pointer to another viewport. The requested display appears in the second viewport.

   Or:

   Drag the pointer to the Windows desktop. A new viewport opens with the requested display at the location where the drag operation ends.

3.5.5 Calling Up Previously Viewed Displays

You can recall displays to the workstation using toolbar buttons, the Display History lists, or the Previous/Display command.

3.5.5.1 Using the Back, Forward, and Previous Buttons

The standard toolbar includes three navigation arrows (see section 3.3.4.1 The Display Viewer Default Toolbar). Clicking on the desired icon calls up the display.

- Clicking the Back (left-facing) or Forward (right-facing) arrow returns the active frame to the previous or subsequent display, respectively.
- Clicking the Previous (down) arrow returns the active frame to the most recently viewed display, whether it was previous or subsequent to the current display.

3.5.5.2 Using the Display History Lists

Clicking on the small black arrow next to any navigation arrow presents a list of displays accessed during the current session (see section 3.3.4.1 The Display Viewer Default Toolbar). While the history lists adjacent to the Back and Forward arrows track each display in the order viewed (including multiple occurrences of the same display), the history list adjacent to the Previous arrow includes each display viewed during the current session just once, providing a streamlined navigational aid.

3.5.5.3 Using the Previous/Display Command

To recall a display using the Previous/Display command, use the following procedure:

1. Give input focus to the viewport into which you want to call the display. As an option, the viewport may be specified with a command qualifier (see section 3.5.3 Calling the Display into a Specific Viewport).
2. Click in the command window of the viewport, as applicable.
3. Type the following at the prompt and press Enter:

   PREVIOUS/DISPLAY[/BEFORE=*n*]

   where *n* is the *n*th previous display to be called up. The /BEFORE qualifier is not required when you want to call up the most recent previous display.

   The Display command is logged on the Log tab of the APF console, and the display appears in the designated viewport.

Refer to the *Browser Reference Manual* for information about the available qualifiers to the Previous/Display command.

3.5.5.4 Stopping the Current Call-Up

To immediately stop the current display call-up, select the Stop Current Call-Up icon on the standard toolbar.

3.5.6 Updating the Display

Depending upon the design of the display, you may find it necessary to manually request that a display be updated with current data. Displays with critical data may be designed to be continuously refreshed, while others may be designed to be static.

To request current data, click Refresh on the standard toolbar.

3.5.6.1 Refresh On/Off

In displays designed to continuously refresh with updated data, you may at times find it necessary to stop the flow of updated data. To do this, use the Turn Refresh Off button found in each frame in a viewport. This button appears as a circular arrow icon in the upper right corner of the frame.

To turn the refresh off, click the Turn Refresh Off button. The frame will be outlined with a red border, by default, to indicate the refresh status.

When refresh is turned off, the Turn Refresh Off button is replaced by the Turn Refresh On button. This button appears as a red circular arrow.

To turn refresh on, click the Turn Refresh On button.

3.6 Calling Up Rooms

Like displays, rooms can be called to the workstation by using the Rooms directory (see section 3.6.1 Using the Rooms Directory).

> **Note:** Calling a room opens it in place of any room or viewports currently on the screen. Viewports will be opened and closed as required.

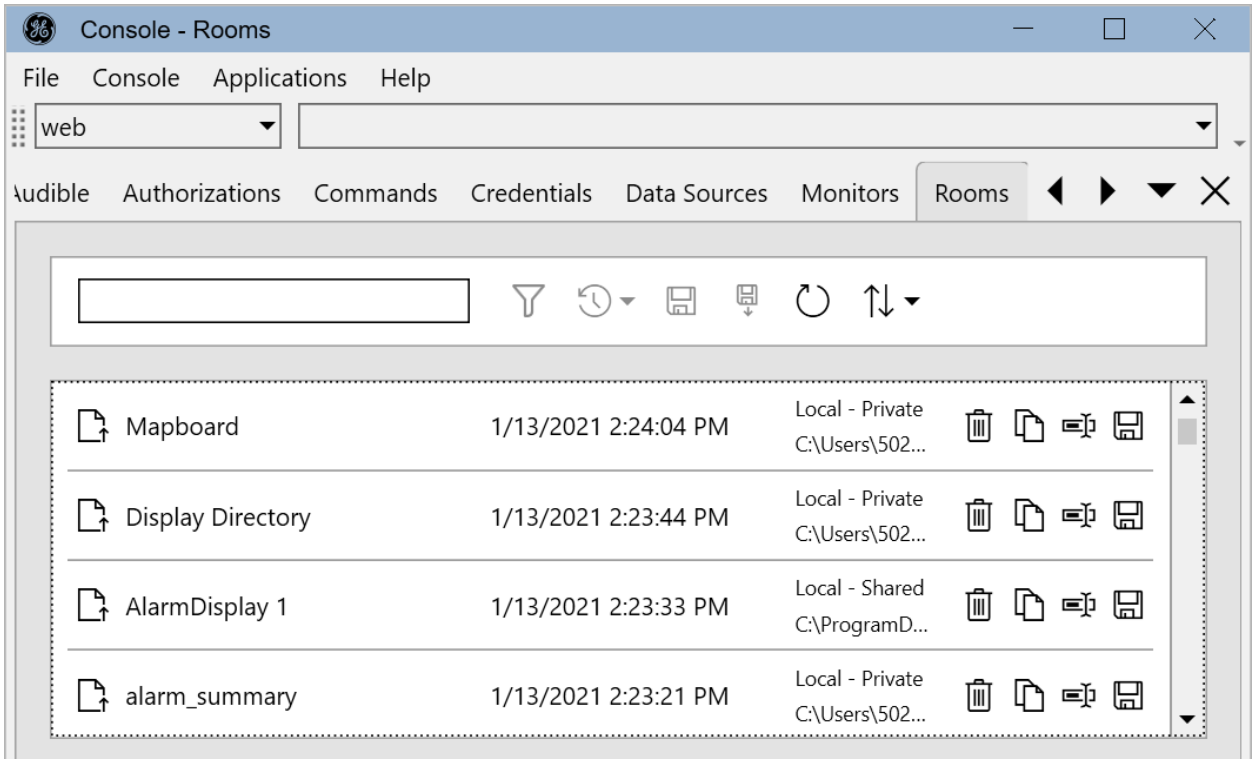
3.6.1 Using the Rooms Directory

The Rooms directory may be accessed in either of two ways:

- Select the Rooms tab in the APF console.
- Select the Rooms icon from the viewport toolbar (see section 3.3.4.1 The Display Viewer Default Toolbar).

From the Rooms tab:

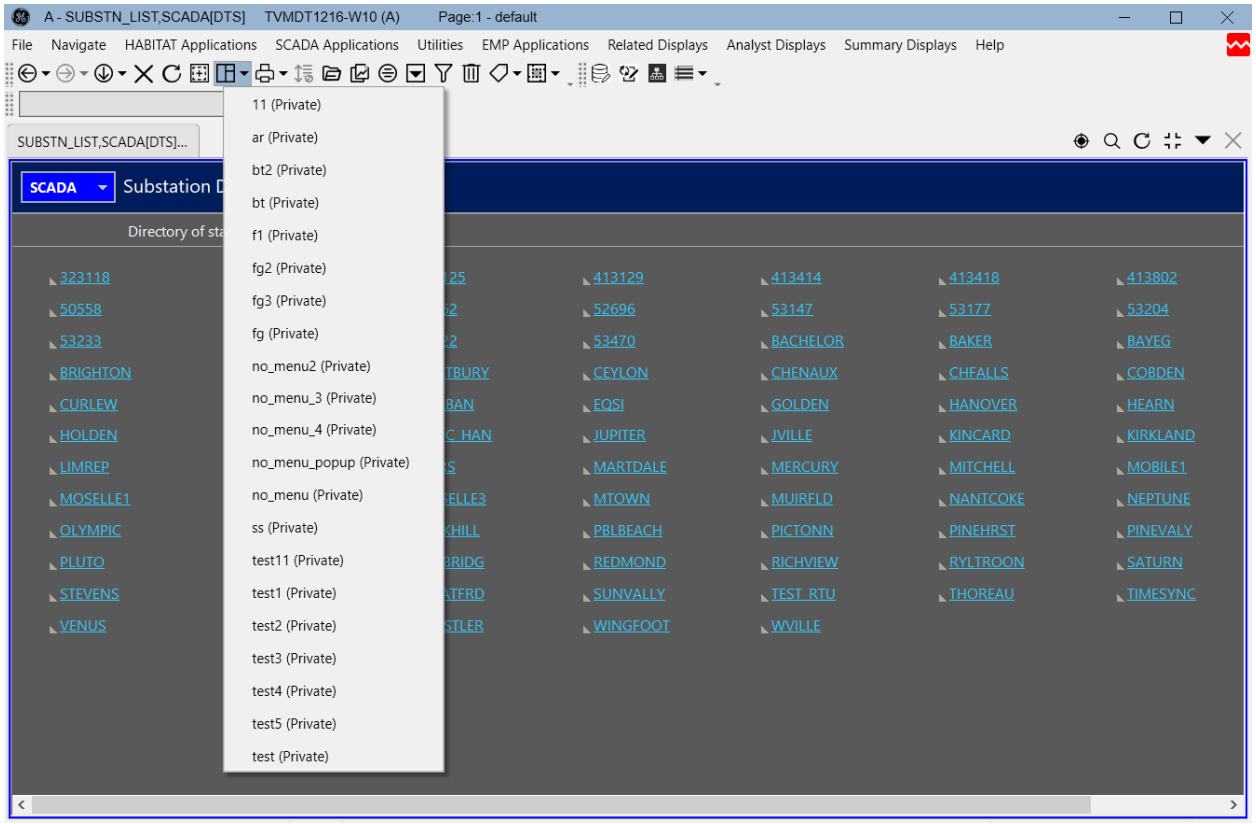
1. Access the Rooms directory.



2. Select the desired room from the list.
3. Click the Load Room command button. The viewports and displays change to reflect the called room.

Using the Rooms icon:

- From the viewport toolbar, click the Room drop-down button and select a room from the list.



3.7 Status Bar

The viewport status bar contains the following information:

Web service connection status icon ( ): This icon appears in the status bar if the connection to the web service is not established.

> **Note:** Browser does not monitor the health of the web services continuously. The status icon is updated only when you interact with web services.
>
> This functionality is not available for the Savecase Service.

3.8 Acknowledging Messages

Informative and decision message boxes require acknowledgment and further action when they appear (see section 2.1.7 Message Boxes).

3.8.1 Acknowledging Informative Messages

To acknowledge informative messages, read the message, and then click OK to acknowledge the message.

The message box closes.

3.8.2 Acknowledging Decision Messages

Decision messages require that you select a course of action for the process or command that initiated the message.

- Continuing as is, according to how the message indicates this will occur (see section 3.8.2.1 Continuing As Is).
- Continuing after providing additional information (see section 3.8.2.2 Continuing after Providing Additional Information).
- Canceling the process/command without completion (see section 3.8.2.3 Canceling the Process/Command).

3.8.2.1 Continuing As Is

To continue the process or command as indicated by the message, use the following procedure:

1. Read the message and ensure that the action that will be performed is what you want.
2. Click OK.

   The message box closes, and the action indicated by the message executes.

3.8.2.2 Continuing after Providing Additional Information

The message box may prompt you to provide additional information or take action. To continue the process or command after providing the required information or taking action, use the following procedure:

1. Carefully read the message to determine what information or action is required, and to ensure that this is the desired option.
2. Provide the information or perform the action(s) indicated by the message.
3. Click the appropriate push button to inform Browser that the information has been supplied or the required action has been taken; this will vary depending on the message and its context.
4. If the message box is still displayed, click OK or Done to close the message box.

   The message box clears the screen, and the process/command completes using the additional information.

3.8.2.3 Canceling the Process/Command

To cancel the process or command without it completing, use the following procedure:

1. Read the message and ensure that you do not want to complete the action indicated by the message.

2. Click Cancel.

   The message box closes, and the process/command stops executing.

3.9 Using Help

Browser provides context-sensitive HTML Help for accessing standard and customized information from Browser displays (see section 3.9.1 Using the Viewport Help Menu).

Help can be received from Browser displays by using the standard viewport Help menu. Help for displays without a menu bar and Message Help for log displays can be obtained using a pre-defined virtual key (see section 5.2 The Key Map). The default virtual key for Message Help is F1.

See your system administrator for additional information about accessing help through virtual keys.

3.9.1 Using the Viewport Help Menu

The following default Help menu options are available.

> **Note:** The default Help menu provided with Browser can be customized for your installation and may appear with different or additional options; contact your system administrator for assistance.

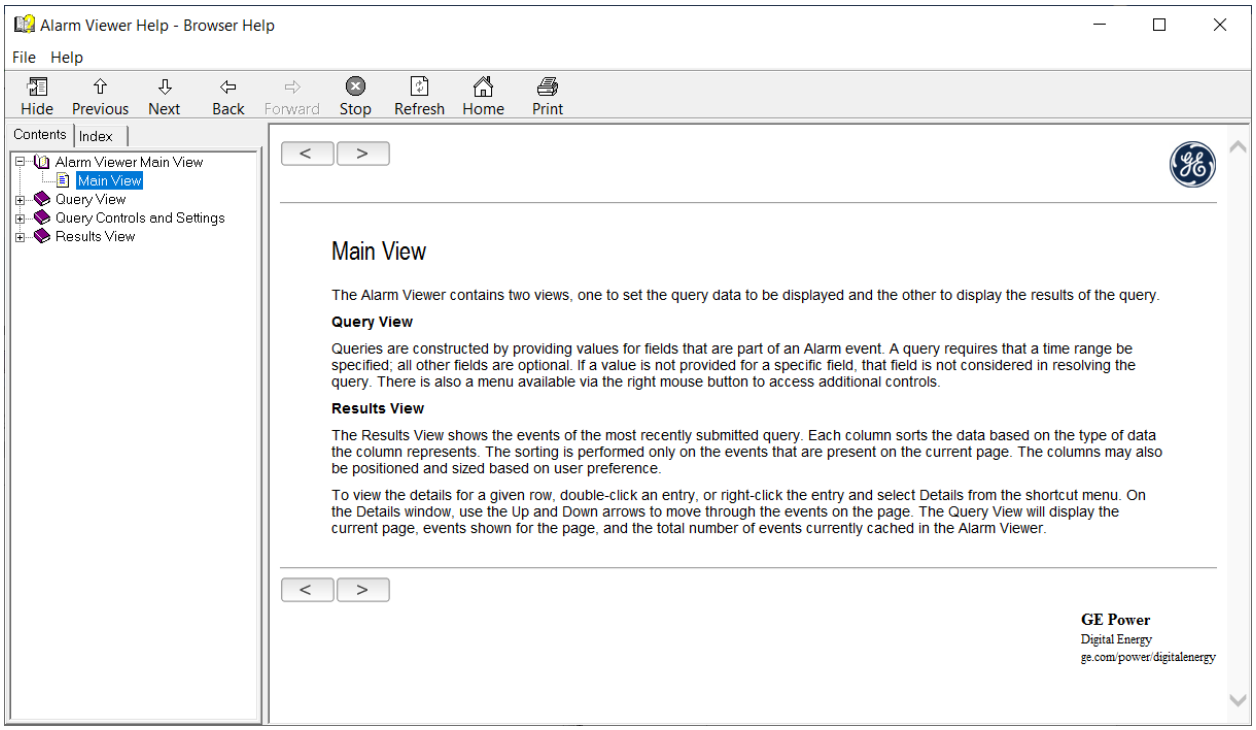
- **On Display:** Provides a description of the current display.
- **On Task:** Opens a list of tasks that can be carried out from the current display. To obtain a detailed description of a particular task, select it from the list.
- **On Application:** Provides information about the features or functionality of the current application.

To use help:

1. Select the Help menu.

   > **Note:** If the current display does not have a Help menu, help may still be available through a virtual key. Contact your system administrator for assistance.

2. Select the type of help required. The online Help window will open.



3. Navigate as desired within the Help Viewer window (see section 3.9.2 Navigating the Help Window).
4. When finished, you can either close the Help window, or minimize it to keep it readily available for future use.

3.9.2 Navigating the Help Window

The Help window is divided into two frames. The search frame, at left, is used to search for Help articles. The display frame, at right, displays Help articles.

Navigation buttons allow fast navigation between help documents. The buttons function as follows:

- **Home:** Returns the display frame to the Help home page. This page provides information about the purpose of the help document as well as the software version, the release date, and copyright information.
- **Back/Forward:** Helps to navigate between the recently viewed document sections.
- **Previous/Next:** Helps to traverse through the table of contents hierarchy:
  - **<<:** Steps backward through the Help articles listed in the Table of Contents.
  - **>>:** Steps forward through the Help articles listed in the Table of Contents.

3.9.3 Finding Information

Help provides two ways to search for information: contents and index.

3.9.3.1 Contents

The Contents tab provides a hierarchical listing, or tree, of topics and articles included in the help document. Topic headings are indicated by either a closed book for a collapsed topic, or an open book for an expanded topic. Help articles are indicated by a document icon.

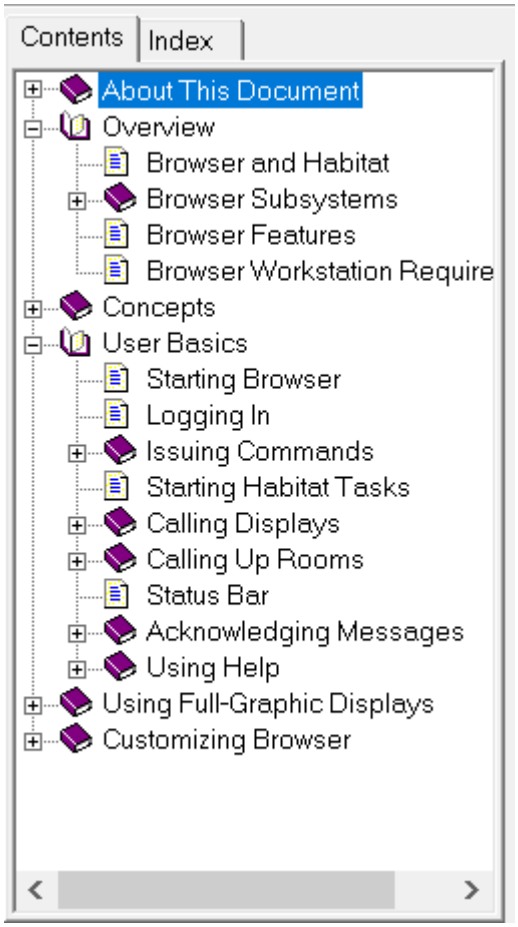


Figure 18. Contents Tab - Expanded View

To locate the desired Help article, use the following procedure:

1. Locate a topic heading related to the information you seek.
2. To expand a topic, click either the topic heading or the plus sign adjacent to the Closed Book icon. The icon will change to an open book, and subtopics will be displayed.
3. Continue navigating through the topic and subtopic headings until you locate the desired information.
4. Click on the name of the desired article. The article will appear in the display frame.

3.9.3.2 Index

The Index tab provides an alphabetical listing of topics that can be browsed or searched.

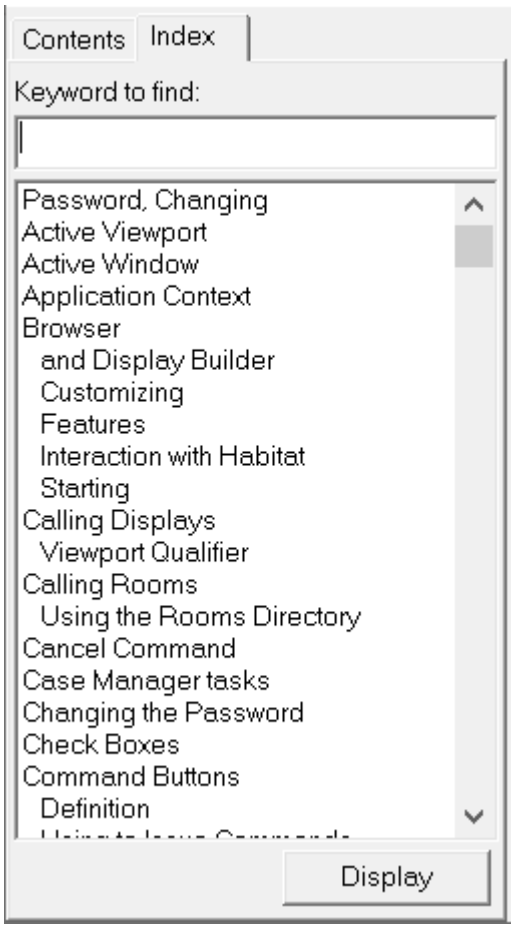


Figure 19. Index Tab - Search Results Displayed

To locate the desired information:

1. Begin typing the topic in the search criteria text box. As you type, different topics are highlighted to match the search criteria.

   At any point, if no topics match your criteria, the highlight box stops moving.

2. When you find a match, click Display. Articles relating to the selected topic are displayed in the search frame.
3. Click the desired article; it will appear in the display frame.
4. If no relevant articles appear, click Index to return to the index view, and try another search.

3.9.4 Obtaining Message Help

Additional information about the messages displayed in any standard Browser log display can be accessed as follows:

1. Call the log display into any standard viewport, and ensure that the viewport has input focus.
2. Position the pointer on the message for which you want more information.

   > **Note:** For messages that extend for more than one line on the display, the pointer must be positioned on the first line of the message.

3. Press the Message Help virtual key (F1 by default) to display additional information about the message or click on the message.

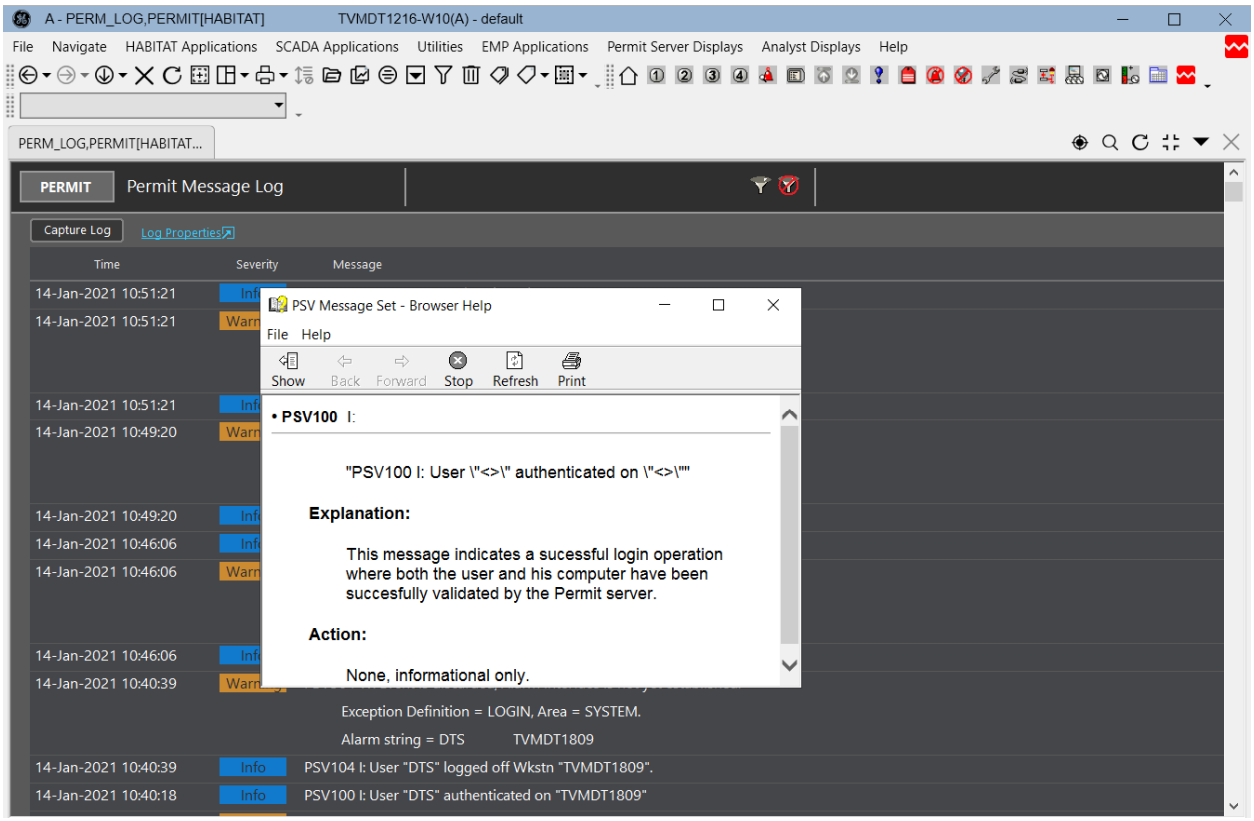


Figure 20. Message Help Example

When finished, you can either close the Help window, or minimize it to keep it readily available for future use, using the appropriate icons in the window title bar.

4. Using Full-Graphic Displays

This chapter provides step-by-step instructions for using Browser displays after they have been called into a viewport.

4.1 Positioning the Display in a Viewport

Several methods are available for positioning the display within the viewport.

4.1.1 Using Scroll Bars

You can scroll through a display using the scroll bars in three ways:

- Scrolling by Unit Steps: This is in small incremental steps called “unit steps,” using the scroll bar stepping arrows (see section 4.1.1.1 Scrolling by Unit Steps).
- Scrolling by Page Steps: This is in large incremental steps called “page steps,” using the scroll region of the scroll bars (see section 4.1.1.2 Scrolling by Page Steps).
- Scrolling by Sliding the scroll box: This is positioning directly to any location in the display using the scroll bar slider (see section 4.1.1.3 Scrolling by Sliding the Scroll Box).

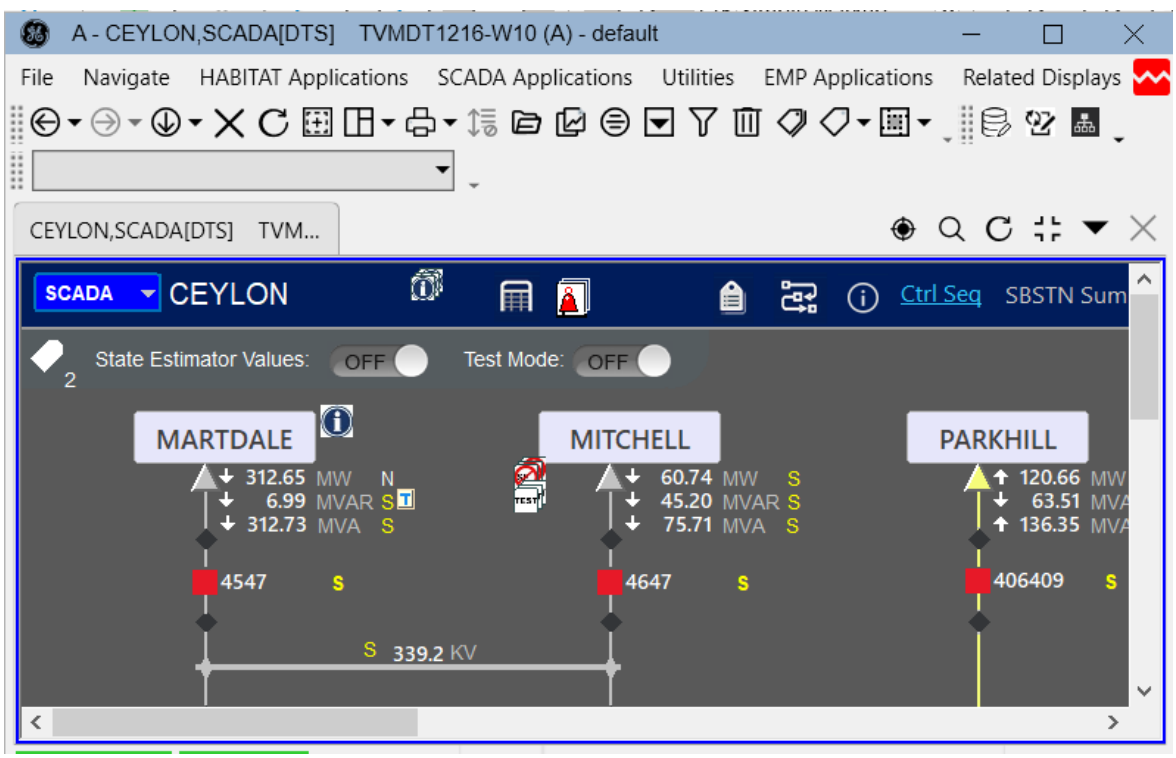


Figure 21. Viewport with Scroll Bars Visible

4.1.1.1 Scrolling by Unit Steps

To scroll through a display in unit steps, either horizontally or vertically, use the following procedure:

1. Position the pointer on the stepping arrow at the end of the horizontal or vertical scroll bar for the direction you want to scroll.
2. Click the left mouse button.

   This will move the viewport a small amount (in increments of a fraction of a viewport) in the desired direction.

   By clicking in rapid succession, or by pressing and holding the left mouse button on the stepping arrow, the viewport will rapidly scroll through the display in the desired direction.

   > **Note:** The size of the unit step is fixed for a given display. Refer to the *Display Builder User Guide* for information about setting the scrolling increments.

3. Stop clicking on the arrow, or release the left mouse button, when the desired position is reached.

4.1.1.2 Scrolling by Page Steps

To scroll through a display in page steps, either horizontally or vertically, use the following procedure:

1. Position the pointer on the scroll region of the scroll bar on the appropriate side of the slider for the dimension and direction you want to scroll.

   Example: To move the viewport up, position the pointer above the slider on the vertical scroll bar.

2. Click the left mouse button, or press and hold it, on the scroll region.

   Each click will move the viewport in increments of a large percentage (up to 100%) of the viewport in the desired direction.

   > **Note:** The size of the page step is fixed for a given display. Refer to the *Display Builder User Guide* for information about setting the page steps.

3. Stop clicking on the scroll bar, or release the left mouse button, when you have positioned the view where you want it.

4.1.1.3 Scrolling by Sliding the Scroll Box

To slide the entire viewport to any new relative position in the display, use the following procedure:

1. Position the pointer on the slider of the scroll bar for the dimension you want to scroll.
2. Press and hold the left mouse button on the slider.
3. Drag the slider in the direction you wish to move the viewport.
4. Stop dragging the slider when you have reached the desired position for the viewport in the display.
5. Release the left mouse button.

4.1.2 Panning Through Displays

The “pan” feature allows you to continuously pan through a display by moving the pointer within the viewport. While scroll bars allow you to move vertically and horizontally through the display, the Pan command is used to toggle the feature on and off.

> **Note:** To effectively use this feature, a virtual key should be defined to execute the Pan On command when the key or mouse button is pressed — key down — and the Pan Off command when the key or mouse button is released — key up (see section 5.2.2 Modifying a Key Map Definition). The default Pan key is the scrolling wheel.

To continually pan through a display in a viewport, use the following procedure:

1. Give input focus to the viewport.
2. Click and hold the scrolling wheel while moving the pointer over the display in the direction you wish to pan.
3. Release the scrolling wheel when the display is in the desired position.

   The viewport is now properly positioned over the display.

4.1.3 Scrolling vs. Panning

Browser will automatically add or remove scroll bars based on the size of the viewport in relation to the size of the display currently visible in the viewport. If the display exceeds the size of the viewport, the horizontal scrollbar, the vertical scrollbar, or both will be added. The scroll bars give you an indication of your relative position on the display; the scroll bars also indicate that you are not viewing the entire display.

Panning allows you to move horizontally and vertically through the display, and it has the added benefit of allowing diagonal movement.

4.1.4 Using the Navigation Window

The navigation window is a separate window that contains a reduced version of the entire display, which can be used to position the display by moving the viewport rectangle.

The viewport rectangle is proportional to the area of the display visible in the viewport. Its position in the navigation window represents the viewport’s relative position over the display, and the viewport’s size and shape.

Three actions can be performed from within the navigation window:

- **Slide the Viewport Over the Display:** By selecting the viewport rectangle using the left mouse button and dragging it to a new position in the navigation window, the viewport will “snap to” the corresponding position over the display (see 4.1.4.2 Sliding the Viewport to a New Position).
- **Pan the Viewport Over the Display:** By selecting the viewport rectangle using Shift + the left mouse button, the viewport will gradually pan over the display as the viewport rectangle is moved in the navigation window (see section 4.1.4.3 Panning the Viewport Over the Display).
- **Zoom a Display:** Pressing Ctrl + clicking the left mouse button within the navigation window will allow you to “rubber-band select” an area within a display. The result is that the display will “zoom to fit” the current viewport size (see section 4.1.4.4 Zooming on Non-Repeating Data).

4.1.4.1 Opening the Navigation Window

To open the navigation window for a given viewport, use the following procedure.

1. Give input focus to the desired viewport or frame within a viewport.
2. Type the following command in the viewport command window, or (if available) press the virtual key:

   OVERVIEW

   The navigation window for the selected viewport pops up on the screen.

> **Note:** To more effectively utilize this feature, a virtual key can be defined to execute the OVERVIEW command (see section 5.2.2 Modifying a Key Map Definition).

4.1.4.2 Sliding the Viewport to a New Position

To slide the viewport to a new position on the display using the navigation window, use the following procedure:

1. Place the pointer on the viewport rectangle in the navigation window.
2. Press and hold the left mouse button.
3. Drag the viewport rectangle to a new position in the navigation window.
4. Release the left mouse button.

   The viewport snaps to the new position.

4.1.4.3 Panning the Viewport Over the Display

To pan the viewport over the display using the navigation window, use the following procedure:

1. Place the pointer on the viewport rectangle in the navigation window.
2. Press and hold Shift + the left mouse button.
3. Drag the viewport rectangle within the navigation window.

   The viewport pans over the display in the direction that the viewport rectangle in the navigation window is moved.
4. Release Shift + the left mouse button when you have panned to the desired location.

4.1.4.4 Zooming on Non-Repeating Data

To “zoom” a simple layer using the navigation window, use the following procedure:

1. Place the pointer in the navigation window at a location that you want to be one corner of the viewport.
2. Press and hold Ctrl + the left mouse button.
3. Drag the pointer to create a new viewport rectangle.
4. Release Ctrl + the left mouse button when the viewport rectangle includes the desired area of the display.

The display zooms to fit the selected area in the current viewport.

4.1.4.5 Closing the Navigation Window

To close the navigation window for a given viewport, click the **X** button in the navigation window title bar.

4.1.5 Positioning to Specific Record Occurrences

Operators can position the viewport around specific record occurrences in the display. This can be done by specifying either:

- The key value in the record type's key field. This is done using the Find command.
- The subscript value of the specific record occurrence. This is done using the Position command.

These methods work for both single occurrences of data (data on simple layers) and tabular data; however, the viewing results will differ as follows:

- For data on simple layers, the specified record will be centered in the viewport.
- For tabular data, the specified record will appear in the upper-left corner of the viewport.

When the display has more than one occurrence of the specified record, Browser will position to the first occurrence. The first occurrence is the one Browser finds as it scans each successive Y coordinate from left to right. Subsequent positioning commands for the same occurrence will position to the next occurrence encountered in the same processing manner.

4.1.5.1 Positioning Using Key Values

To position to a specific record occurrence in a display using the record type's key field, use the following procedure:

1. Give input focus to the viewport with the desired display.
2. Enter the following in the command window of the active viewport:

   FIND record=key[,record=key]

   where *record* is the record type and *key* is the record's key field value. A composite key of record and key values separated by commas may also be specified.

   The view in the viewport changes, positioning to the first occurrence of this record/key combination found on the display.

4.1.5.2 Positioning Using Subscript Values

To position to a specific record occurrence in a display using the subscript value of the record type, use the following procedure:

1. Give input focus to the viewport with the desired display.
2. Enter the following in the command window of the active viewport:

   POSITION record=subscript

   where *record* is the record type and *subscript* is the record occurrence subscript value.

The viewport centers the display view over the occurrence.

4.1.5.3 Positioning to the Next Occurrence

To position to subsequent (if any) locations in the display where the record/key or record/subscript combination may be found, use the following procedure:

1. Select the command window of the active viewport where the Find or Position command was issued.
2. Use the Up Arrow key to bring the previous Find or Position command into the command window.
3. Re-issue the command.

   The viewport positions to the next occurrence of the previously specified record/key or record/subscript combination (if any).

Refer to the *Browser Reference Manual* for more information about the Find and Position commands.

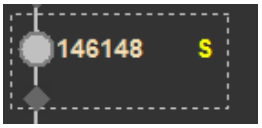
4.1.6 Highlighting all Occurrences of a Specific Record in Simple Displays

LOCATE highlights all the occurrences of the specified record in a simple display. Multiple records can be highlighted at the same time.

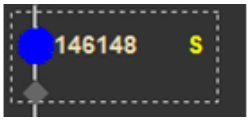
- To show all the highlighted occurrences in the visible area of the viewport, the view is aligned to the LEFT of the screen, by applying a zoom factor if needed.

  If only one placement needs to be highlighted as part of a locate command, that placement is aligned to the CENTER of the viewport and highlighted.

- If the placement being located is not associated with any primitives with “Highlight Me” selected, the highlighting uses a dashed rubber band as shown in the following image.

- If the placement being located is associated with one or more primitives with “Highlight Me” selected, the highlighting is done by changing the color of those primitives using the SELECT-LOCATE Highlight Color.

  You can view this color in the Browser settings, as shown in the Figure 22. This property is read-only in the settings tab. Specify the color value in the SELECTLOCATEHighlightColor property in the configuration file (WebFGConfiguration.defaults.config).

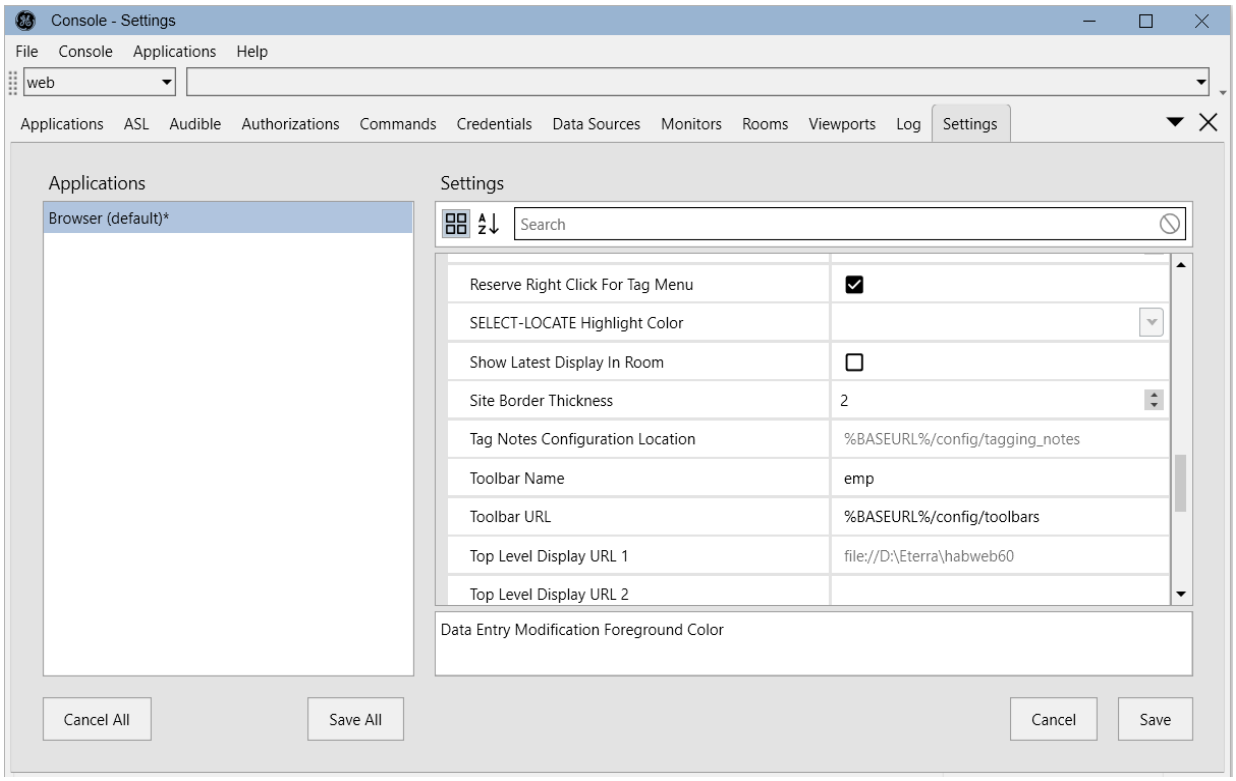


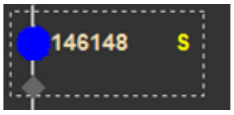
Figure 22. Setting to Change Highlight Color

For more information about the “Highlight Me” property of primitives, refer to the *Display Builder User Guide*. Also refer to the *Browser Reference Manual* for more information about the LOCATE Command.

4.1.6.1 Selecting a Specific Record in Simple Displays for Highlighting

Selecting a specific record using the SELECT command draws a dashed rectangular border around the selected items, if the placement being selected is not associated with any primitives with "Highlight Me" selected.

- If the placement being selected is associated with one or more primitives with “Highlight Me” selected, the highlighting will be done by changing the color of those primitives using the SELECT-LOCATE Highlight Color.

  You can view this color in the Browser settings. This property is read-only in the Settings tab. Specify the color value in the SELECTLOCATEHighlightColor property in the configuration file (WebFGConfiguration.defaults.config).

- While the selection is active on placements having primitives with “Highlight Me” selected, no refresh will take place on those highlighted placements.
- Selection can be cancelled by CANCEL SELECT or by reselecting the placements.

4.1.7 Positioning to Placeholders

To position to a placeholder on a simple layer, enter the following in the command window of the active viewport:

DISPLAY/LAYER=layer_name/PLACEHOLDER=placeholder

where *placeholder* is the name of the placeholder.

The viewport will be positioned with the placeholder approximately in the center.

Refer to the *Browser Reference Manual* for more information about the /PLACEHOLDER qualifier for the Display command.

Consult your system administrator for information about the placeholders defined for a given display.

4.1.8 Positioning to Display Pages

Tabular data can be divided into pages. You can position to the beginning of pages in two ways:

- Positioning to a Specific Page
- Shifting a Specified Number of Pages

> **Note:** These methods apply only to tabular layers in displays defined to have pages. Refer to the *Display Builder User Guide* for more information about tabular data pages.

4.1.8.1 Positioning to a Specific Page

Use the Page command to position to a specific page. To position to a specific page in a tabular layer, use the following procedure:

1. Give input focus to the viewport with the desired display.
2. Enter the following in the command window of the active viewport:

   PAGE number

   where *number* is the exact display page number.

   The viewport positions to the top of the requested page.

Refer to the *Browser Reference Manual* for more information about the Page command.

4.1.8.2 Shifting a Specified Number of Pages

Use the Shift command to shift the display by a number of pages. To shift in a given direction, use either of the following procedures:

Method 1

1. Give input focus to the viewport with the desired display.
2. Enter the following in the command window of the active viewport:

   SHIFT/direction delta

where *direction* is either DOWN (the default) or UP, and *delta* is the number of pages to shift the view. For example:

SHIFT/UP 5

shifts the display up five pages (i.e., from page 10 to page 5).

The viewport positions to the top of the requested page.

Method 2

1. Give input focus to the viewport with the desired display.
2. Enter the following in the command window of the active viewport:

   SHIFT xdelta

   where *x* is either + (DOWN, the default) or – (UP), and *delta* is the number of pages to shift the view. For example:

   SHIFT -5

   shifts the display up five pages (i.e., from page 10 to page 5).

   The viewport positions to the top of the requested page.

Refer to the *Browser Reference Manual* for more information about the Shift command.

4.2 Zooming the Viewport

"Zoom factors" are real number multipliers between 0.01 and 10. Zoom factors greater than one enlarge the display; factors less than one reduce the display.

Zoom factors can change the displayed data relative to either:

- The normalized display size. This is the size of a display when the zoom factor is 1 (100% of its defined size).
- The current display size or the current zoom factor of the display.

The zoom factor of the display can be modified by using commands, menus, the "rubber-band zoom" feature (see section 4.2.3 Using Rubber-Band Zoom), or the navigation window (see section 4.1.3 Scrolling vs. Panning).

An entire display can be set to appear at a given zoom factor by defining a default zoom factor at design time.

On large one-line displays, it might be most effective to use the "zoom around cursor" feature. Place the cursor where you want to enlarge or reduce the display, then hold down the Ctrl key while using the scrolling wheel to zoom in or out (see section 4.2.4 Using Zoom around the Cursor Location).

This section describes methods for zooming a simple layer: by command, by using menu options, and by using rubber-band zoom. In addition, a simple layer can be zoomed using the navigation window (see section 4.1.4.4 Zooming on Non-Repeating Data).

> **Note:** If an embedded user control is present in the display, and the control was configured to 'Fill Entire Display (Zoom Disabled)' when the display was built, zooming and scale to fit functionalities of the display will be disabled in Browser.

4.2.1 Specifying by Command

The Zoom command can be used to modify the size of objects in the display:

- Relative to the current display size
- Relative to the normalized display size (i.e., zoom factor = 1)

Each method is discussed below.

Refer to the *Browser Reference Manual* for more information about the Zoom command.

4.2.1.1 Relative to the Current Display Size

To change the display size relative to its current size, use the following procedure:

1. Give input focus to the viewport with the desired display.
2. Enter the following in the command window of the active viewport:

   ZOOM x

   where *x* is the zoom factor and is a real value between .01 and 10.

   The display increases or decreases in size by the zoom factor specified.

   Examples: If x=1.5, the display will be enlarged to 150% of its current size. If x=0.5, it will be reduced to 50% of its current size.

4.2.1.2 Relative to the Normalized Display Size

The display can be sized relative to the normalized display size (the size of objects when the zoom factor = 1) by specifying either the exact zoom factor or an incremental change in size.

4.2.1.2.1 Specifying the Exact Zoom Factor

To change the size of a display to an exact zoom factor relative to its normalized size, use the following procedure:

1. Give input focus to the viewport with the desired display.
2. Enter the following in the command line of the active viewport:

   ZOOM/TO x

   where *x* is the exact zoom factor and is a real number between .01 and 10.

   The display changes in size to the specified factor of its normalized size, regardless of its current size.

   Examples: If x=1.5, the display will be enlarged to 150% of its normalized size. If x=0.5, it will be reduced to 50% of its normalized size.

4.2.1.2.2 Specifying an Incremental Change in Size

To change the size of a display by some increment of its normalized size, use the following procedure:

1. Give input focus to the viewport with the desired display.

2. Enter the following in the command window of the active viewport:

   ZOOM/BY x

   where *x* is the exact zoom factor and is a real value between 0.1 and 10 (an increase of between 10% and 1000%), or -0.1 and -10 (a decrease of between -10% and -1000%).

   The display changes to a size that represents the percentage increase or decrease in the normalized size, regardless of its current size.

   Examples: A value of 0.1 in the Zoom command enlarges the display to 110% of its normalized size. A value of -0.1 reduces the display to 90% of its normalized size.

4.2.2 Zooming Using Menu Options

Three menu options in the Zoom Display submenu of the viewport's Navigate menu allow you to quickly modify the size of the display.

> **Note:** Not all viewports will have a Navigate menu.

- **Normal:** Changes the size of the display to its normalized size. Equivalent to the command ZOOM/TO 1.
- **Half:** Reduces the size of the display to 50% of its current size. Equivalent to the command ZOOM .5.
- **Double:** Changes the size of the display to 200% (twice as large) of its current size. Equivalent to the command ZOOM 2.

To use any of the menu options, use the following procedure:

1. Select Navigate > Zoom Display from the viewport menu bar.
2. Select the desired option from the submenu.

   The display size changes, according to your choice.

4.2.3 Using Rubber-Band Zoom

Use the rubber-band zoom feature to select a portion of the display to enlarge. This is accomplished by highlighting the desired area, which is then enlarged to fit the viewport.

> **Note:** To effectively use this feature, a virtual key definition for the Zoom/Band command must be defined for action on Key/Button down (see section 5.2.2 Modifying a Key Map Definition). The default key for rubber-band zoom is Shift + the scrolling wheel.

To use rubber-band zoom, use the following procedure:

1. Give input focus to the viewport with the desired display.
2. Position the pointer on the display.
3. Press and hold Shift + the scrolling wheel.
4. Without releasing the virtual key, move the pointer until the area you wish to zoom is included in the highlighted rectangle that is created as you move the pointer.

You have now designated the area to zoom.

5. Release Shift + the scrolling wheel to zoom the display.

   The selected area is sized to fit into the dimensions of the viewport, changing the display's zoom factor as required.

> **Note:** The smallest dimension of the viewport or the largest dimension of the selected area, whichever is the most limiting, determines the actual zoom factor.

4.2.4 Using Zoom around the Cursor Location

Use the "zoom around the cursor location" feature to select a position on the display to zoom around while keeping the location under the cursor fixed.

To use zoom around the cursor location, use the following procedure:

1. Give input focus to the viewport with the desired display.
2. Position the pointer on the desired location within the display.
3. Press and hold Ctrl, and use the scrolling wheel to zoom in or out while keeping the object under the cursor fixed.

4.2.5 Using Scale to Fit

The "scale to fit" feature causes the display to be sized such that the entire display, no matter how large, can be viewed in its entirety in the viewport. The actual zoom factor depends upon the size of the display.

The scale to fit feature is accessed by using frame control buttons (see section 4.9 Viewing Displays in Frames).

4.3 Showing the Current Context

The current context attributes can be displayed using the Show command.

> **Note:** A virtual key definition for the Show Context command is required to display both the viewport and the pointer context (see section 5.2.2 Modifying a Key Map Definition). The default virtual key is F11.

To show all current context attributes, including the current pointer context, use the following procedure:

1. Give input focus to the viewport.
2. Place the pointer on the object whose context you want to show.
3. Press the Show Context virtual key.

   A message box appears with all current pointer, viewport, and station-level context information.
4. Click OK to close the message box when you are finished viewing the context attributes.

   The message box disappears from the screen.

4.4 Displaying Pop-up Pictures

Pop-up pictures are made visible by clicking the object with which the pop-up is associated. The object is often a command button, but it may also be the graphic representation of a system component or other object. When the pointer is positioned over an object with an associated pop-up, the pointer will change from an arrow to a hand.

To display a pop-up, use the following procedure:

1. Select the desired element; when properly positioned on the display element, the cursor will change from the arrow pointer to a hand.

   The pop-up picture containing additional information about the selected element (such as a circuit breaker) is displayed. The pop-up may also contain one or more command buttons. Selecting a button may display additional information, allow you to take an action on the element, or access another pop-up. Taking an action may initiate a confirmation step, depending on specific requirements of the system.

2. To close the pop-up, use either of the following procedures, depending upon the design of the pop-up:

   - Click the Done, OK, or Cancel button normally provided on the pop-up picture.

     Using the **X** in the title bar of the pop-up is not recommended; this could result in important commands related to the pop-up picture not being executed.

   - Click Done or Cancel. The pop-up will close.

4.5 Performing Data Entry Operations

This section provides an overview of data entry operations and instructions for performing the various operations.

4.5.1 Data Entry Overview

In simple terms, a “database” is made up of fields and records. A “field” is a piece of information that cannot reasonably be broken down any further for storage. A “record” is a group of logically related fields. For example, in an address book, the fields *Name*, *Street Address*, *City*, *State*, and *Postal Code* make up the record for each individual.

There are four data entry operations: Delete, Insert, Copy, and Entry (DICE). Delete, Insert, and Copy operations are performed on records, whereas Entry operation is performed on fields.

- **Delete:** This is the process of selecting one or more records, and removing them from an application database. This process removes both the data values in the records and the records themselves.
- **Insert:** This is the process of selecting a record type and (if desired) its children record types, and inserting a blank record of the same type into any legal position in an application database.
- **Copy:** This is the process of selecting one or more existing records from one application database, and copying them to a marked location in the same or another database. This process differs from inserting a record. In the Copy operations, the existing values for the selected fields are also copied; these values can later be modified (if desired). In addition, more than one record and its children can be copied; however, only one record type and its children can be inserted at a time.

- **Entry:** This is the first-time entry or modification of the field values of existing records in application databases.

  One database can include fields of various data types. Examples of data types are *integer*, *boolean*, and *text*. Data entered into a field must correspond to the defined data type of the field. The method of data entry depends on the type of field being displayed and how the field is represented on the display.

  - Toggle and option buttons can be defined to modify boolean or mask fields. Settings (on or off) of the fields are changed and immediately applied by selecting the button or its associated label.
  - Menus can be used to modify any type of field if available. Values are modified and immediately applied by selecting an item from a pop-up menu.
  - Text fields can be defined to allow for direct entry of values in field types such as character, integer, or real. Actual data values are not updated until the Browser Enter command is issued. This command is issued, by default, when the Enter key is pressed.

  Data can be entered into fields on either simple or tabular layers. However, deletion, insertion, and copying of records can be performed only on tabular layers.

There are two additional operations that are performed in conjunction with DICE operations. The first is selecting the field or fields in which to enter data, the record types to insert, or the records to copy or delete (see section 4.5.2 Selecting Fields and Records).

The second operation is marking the location where selected records are to be inserted or copied. This location can be any valid database position in any display, in any viewport on the screen where a tabular layer is shown. There can be only one marked location per Browser session (see section 4.5.3 Marking Fields).

4.5.2 Selecting Fields and Records

Browser provides several methods for selecting fields, depending on the number, location, and relationship of the fields to be selected. Records are selected by selecting a field in the record.

4.5.2.1 Selecting an Individual Field

To select a field for data entry, press Ctrl and click the left mouse button in the desired field.

A blinking text-entry-cursor will appear in the selected field.

4.5.2.2 Selecting Related Fields

To select related fields for data entry, click the left mouse button in any desired field.

A blinking text-entry-cursor will appear in the selected field. In addition, all related fields on the display (i.e., with the same context) will be selected and will appear highlighted on the display.

4.5.2.3 Selecting a Contiguous Group of Fields, Method 1

To select a contiguous group of fields for data entry, use the following procedure:

1. Press Ctrl and click the left mouse button in the first or last desired field. This establishes the anchor location.
2. Press Shift and click the left mouse button in the first or last desired field.

   Example: If you used Ctrl + the left mouse button in the first record, use Shift + the left mouse button in the last record, and vice versa.

   A blinking text-entry-cursor will appear in the second field. All selected fields will be highlighted.

4.5.2.4 Selecting a Contiguous Group of Fields, Method 2

Another method for selecting a group of records is rubber-band select. This method requires a virtual key for the Select/Enter/Band command.

1. Give input focus to the viewport with the fields or records.
2. Place the pointer at the lower right corner of the fields to be selected.
3. Press and hold the Select/Enter/Band virtual key (the scrolling wheel).
4. Move the pointer until all fields to be selected are included in the rubber-band. You have now designated the select area.
5. Release the virtual key to stop the rubber-band selection.
6. Click in one of the fields inside the “rubber-band.”

   A blinking text-entry-cursor will appear in the last field. All fields inside the rubber-band are selected and will be highlighted.

> **Note:** To cancel the rubber-band selection without selecting any fields, press any key that does not have a virtual key definition before releasing the Select/Enter/Band virtual key.

4.5.2.5 Selecting All Enterable Fields in the Viewport

To select all enterable fields in the viewport, use the following procedure:

1. Give input focus to the desired viewport.
2. Enter the following in the command line of the active viewport:

   SELECT/ENTER/VIEW

   All eligible fields in the viewport are selected and will be highlighted.

4.5.2.6 Selecting All Fields in the Display

To select all fields in the display, use the following procedure:

1. Give input focus to the viewport with the desired display.
2. Enter the following in the command window of the active viewport:

   SELECT/ENTER/ALL

All eligible fields in the display are selected and will be highlighted.

4.5.2.7 Selecting a Non-Contiguous Group of Fields

To select a non-contiguous group of fields for data entry, use the following procedure:

1. Press Ctrl and click the left mouse button in any desired field.
2. Without releasing the Ctrl key, click in the next field.
3. Continue selecting fields until all desired fields have been selected.

   A blinking text-entry-cursor will appear in each field as it is selected. When you are finished, all selected fields will be highlighted.

4.5.2.8 Selecting Individual Records for Insert and Delete

Another way to select records is available for insert and delete operations. Records that are enabled for insert and delete operations are identifiable by an adjacent command button:

Figure 23. Insert/Delete Menu Button

Clicking this button pops up a menu that includes the insert and delete options that are available for the selected record.

To cancel field selection and proceed without inserting a record, click at any clear section of the display.

4.5.2.9 De-Selecting Fields

To cancel the selection of all fields for data entry, press the Escape key on your keyboard.

4.5.3 Marking Fields

The default insert and copy location is the current pointer position. By extension this means that, if you select a record to insert or copy without previously marking another location, the record will be inserted or copied immediately below the selected record.

The insert or copy position can also be specified using the Mark command before selecting a record to insert or copy. The /MARK qualifier is then used in conjunction with the Insert or Copy command to locate the new records at the marked position.

When a record is marked using the Mark command, complete context information about the record is maintained in a separate data structure. This information can be used in other Browser commands by using the available Mark proxy strings in the commands.

Refer to the *Browser Reference Manual* for information about the Mark proxy strings.

> **Note:** The following procedure requires a virtual key definition for the Mark command (see section 5.2.2 Modifying a Key Map Definition).

To mark the insert or copy location, use the following procedure:

1. Give input focus to the viewport where the record will be inserted or copied.
2. Place the pointer at the desired insert/copy position on the display.

   > **Note:** Records will be inserted or copied in the first legal position in the database after the record at the pointer position.

3. Press the Mark virtual key.

   The record at the marked location is highlighted with a dashed outline.

   Insertion or copy will be performed at the marker for all Insert or Copy commands that have the /MARK qualifier. Also, the marked record's context can be applied to commands using the Mark proxy strings.

4.5.4 Entering Data

Entering data in fields involves selecting the field, typing in the new value, and committing the changes.

> **Note:** Data entry procedures require a virtual key definition for the Select command (see section 5.2.2 Modifying a Key Map Definition). The default virtual key is the left mouse button.

Fields that are enabled for data entry are usually identifiable by their appearance. For example, on all default displays, data entry–enabled fields appear as recessed text boxes.

4.5.4.1 Moving from Field to Field

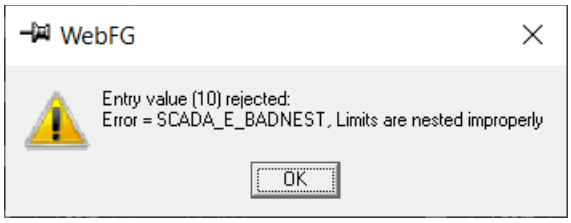
You can move forward through fields selected for data entry by pressing the Tab key, and backward by pressing the Shift+Tab key combination.

4.5.4.2 Entering and Modifying Data Values

To enter data into a field or fields, use the following procedure:

1. Click on the desired field or fields.
2. Enter new data or edit existing data as desired in the field with the text-entry cursor. New and edited values will appear in a different color for easy identification.
3. Cycle through the selected fields using the Tab key.
4. When complete, commit the changes by pressing Enter.

   At this time, the software may perform validation checks to ensure the integrity of the data. If any data fails the validation checks, an informative message box will be displayed.



Failure may be caused by data type incompatibility or incorrect user permissions, among other things.

After the message box has been closed, the rejected data will be highlighted. Any valid changes made prior to committing the data will be saved to the database and the new values displayed. The invalid data must be re-entered and re-committed.

See your system administrator for more information about data types and permissions.

4.5.4.3 Data Entry Keys

Several keys and key combinations provide for efficient data entry.

Table 1. Useful Data Entry Keys

Key or Key Combination	Action
Ctrl+Right Arrow	Moves the text-entry cursor one word to the right.
Ctrl+Left Arrow	Moves the text-entry cursor one word to the left.
Shift+Ctrl+Right Arrow	Highlights the characters to the right of the text-entry cursor, up to the next space character. All highlighted characters will be deleted and replaced by the next keystroke. To highlight more than one word, repeatedly press the arrow key without releasing the Shift and Ctrl keys until all desired words are highlighted.
Shift+Ctrl+Left Arrow	Highlights the characters to the left of the text-entry cursor, up to the next space character. All highlighted characters will be deleted and replaced by the next keystroke. To highlight more than one word, repeatedly press the arrow key without releasing the Shift and Ctrl keys until all desired words are highlighted.
Backspace	Deletes the character immediately to the left of the cursor.
Delete	Deletes the character immediately to the right of the cursor.
Click	Pressing and holding the left mouse button while dragging the cursor across one or more characters highlights the characters. All highlighted characters will be deleted and replaced by the next keystroke.

4.5.4.4 Copying Data Values

Data values can be copied from one record to another with the following procedure:

1. Click in the field containing the data to be copied.
2. Press and hold the left mouse button while dragging the cursor across the data to be copied; this will highlight the data.

3. Press Ctrl+C to copy the data value.
4. Click in the field in which the copied data is to be entered.
5. Press Ctrl+V to paste the data value into the new field.

> **Note:** If the field already contains data, highlight the data to be replaced before pressing Ctrl+V. The highlighted data will be deleted and replaced with the copied data.
>
> In this release, pasting via Ctrl+V or right-click is only supported in text fields.

4.5.4.5 Data Entry using the Date Time Widget

If a selected field is of the type Time/Date and its sub-type is Date, the Date-Time dialog box will appear. The appropriate controls in the dialog box are enabled or disabled according to the sub-type of the selected field.

Table 2. Date Time Dialog Control State

Time/Date Sub-Type	Date Time Dialog Box Controls				Dialog Box Title
	Calendar	Hour	Minute	Second	
Date	Enabled	Disabled	Disabled	Disabled	Select Date Month and Year
Date Time and Zone	Enabled	Enabled	Enabled	Enabled	Select Date and Time
Time and Date	Enabled	Enabled	Enabled	Enabled	Select Date and Time
VMS	Enabled	Enabled	Enabled	Enabled	Select Date and Time
OleDate	Enabled	Enabled	Enabled	Enabled	Select Date and Time
Month	Enabled	Disabled	Disabled	Disabled	Select Month
Day	Enabled	Disabled	Disabled	Disabled	Select Weekday Name
HC	Disabled	Enabled	Disabled	Disabled	Select Hour
HR	Disabled	Enabled	Disabled	Disabled	Select Hour
MN	Disabled	Disabled	Enabled	Disabled	Select Minute
DD	Enabled	Disabled	Disabled	Disabled	Select Date
HM	Disabled	Enabled	Disabled	Disabled	Select Hour
MLTM	Disabled	Enabled	Enabled	Disabled	Select Hour and Minute
Mon	Enabled	Disabled	Disabled	Disabled	Select Month
SC	Disabled	Disabled	Disabled	Enabled	Select Second
Weekday	Enabled	Disabled	Disabled	Disabled	Select Weekday Name
YY	Enabled	Disabled	Disabled	Disabled	Select Year
YYYY	Enabled	Disabled	Disabled	Disabled	Select Year
MM	Enabled	Disabled	Disabled	Disabled	Select Month
DST	Dialog box will not appear				

Time/Date Sub-Type	Date Time Dialog Box Controls				Dialog Box Title
	Calendar	Hour	Minute	Second	
Holiday	Dialog box will not appear				
Am	Dialog box will not appear				
TMZ	Dialog box will not appear				
CTime	Dialog box will not appear				

4.5.5 Inserting Records

Inserting records involves selecting the record type to be inserted, determining the insertion point, and inserting the record.

> **Note:** When records are inserted, they contain no data. To insert a new record that includes data, use the Copy command (see section 4.5.6 Copying Records).

4.5.5.1 Inserting Records at the Current Position

You can insert a single record with the following procedure. Use this method when the record type of the new record corresponds to that of the record at the current pointer position.

1. Click the Insert/Delete Menu button to display the Insert/Delete menu (Figure 23).
2. Make your selection from the menu. Available choices may vary; some menus may offer more than one insert option — for example, if it is possible to insert more than one type of record at the current location.

The new record will be inserted immediately below the selected record and available for data entry.

4.5.5.2 Inserting the Selected Record at the Pointer

> **Note:** This procedure requires a virtual key for the Insert/Select command (see section 5.2.1 Defining a Virtual Key).

To insert a selected record in another location at the current pointer position, use the following procedure.

1. Select the record type you want to insert (see section 4.5.2 Selecting Fields and Records).
2. Place the pointer on the record immediately before the desired insert position. The new record will be inserted in the first valid position after the current pointer location.
3. Press the Insert/Select virtual key.

   The new record will be inserted at the desired location.

4.5.5.3 Inserting the Selected Record at a Marked Location

> **Note:** It is recommended, but not required, that you define a virtual key for the Insert/Select/Mark command (see section 5.2.1 Defining a Virtual Key).

To insert the selected record at a previously marked location, use the following procedure.

1. Select the record type you want to insert (see section 4.5.2 Selecting Fields and Records).
2. Mark the location where you want the new record inserted.

   Either:

   - Press the Insert/Select/Mark virtual key, or
   - Enter the following command in the current viewport command line:

     INSERT/SELECT/MARK

   A new record will be inserted in the first valid position after the marked location.

4.5.5.4 Entering and Modifying Data Values in Inserted Fields

Data values are entered in inserted fields by following standard data entry procedures. See section 4.5.4.2 Entering and Modifying Data Values and section 4.5.4.4 Copying Data Values.

4.5.6 Copying Records

Copying records involves selecting the record type to be copied, determining the location to copy the record, and copying the record.

4.5.6.1 Copying the Record at the Pointer

> **Note:** This procedure requires a virtual key for the Copy command (see section 5.2.1 Defining a Virtual Key).

To copy the record(s) at the current pointer location to the current pointer location, use the following procedure.

1. Place the pointer on the record occurrence you want to copy.
2. Press the Copy virtual key.

   The record at the current pointer location will be copied to the first legal position after the current pointer location.

4.5.6.2 Copying the Selected Record to the Pointer

To copy the selected record or records to the current pointer location, use the following procedure:

1. Select the record or records to copy (see section 4.5.2 Selecting Fields and Records).
2. Place the pointer on the record occurrence after which you want the record(s) copied.
3. Press the Copy/Select virtual key.

The selected record(s) will be copied to the first legal position after the current pointer location. The screen will refresh, and the copied record(s) will show in the display.

4.5.6.3 Copying the Selected Record to the Marked Location

To copy the selected record occurrence(s) to the marker, use the following procedure:

1. Select the record or records you want to copy (see section 4.5.2 Selecting Fields and Records).
2. Mark the record after which you want the record(s) copied.

   Either:

   - Press the Copy/Select/Mark virtual key, or
   - Enter the following in the command line of the current viewport:

     COPY/SELECT/MARK

   The selected record(s) will be copied to the first legal position after the marker location.

4.5.7 Deleting Records

Deleting records involves selecting the record to be deleted, deleting the record, and confirming the deletion.

Records that are enabled for insert and/or delete operations are identifiable by an adjacent command button (Figure 23).

> **Note:** Not all records with an Insert/Delete Menu button will be enabled for deletion. See your system administrator for more information.

4.5.7.1 Deleting a Single Record

To delete a single record, use the following procedure:

1. Click the Insert/Delete Menu button.
2. Select the delete option.

   The selected record is deleted from the database, along with any child records.

4.5.7.2 Deleting the Selected Records

> **Note:** It is recommended, but not required, that a virtual key be defined for the Delete/Select command.

To delete a group of selected records, use the following procedure.

1. Select the records you want to delete (see section 4.5.2 Selecting Fields and Records).
2. Either:

   - Press the Delete/Select virtual key, or
   - Enter the following in the command line of the current viewport:

DELETE/SELECT

The selected records, and all of their child records (if any), are deleted from the database.

4.6 Sorting and Filtering Data

4.6.1 Sorting Data

To sort data, use the following procedure:

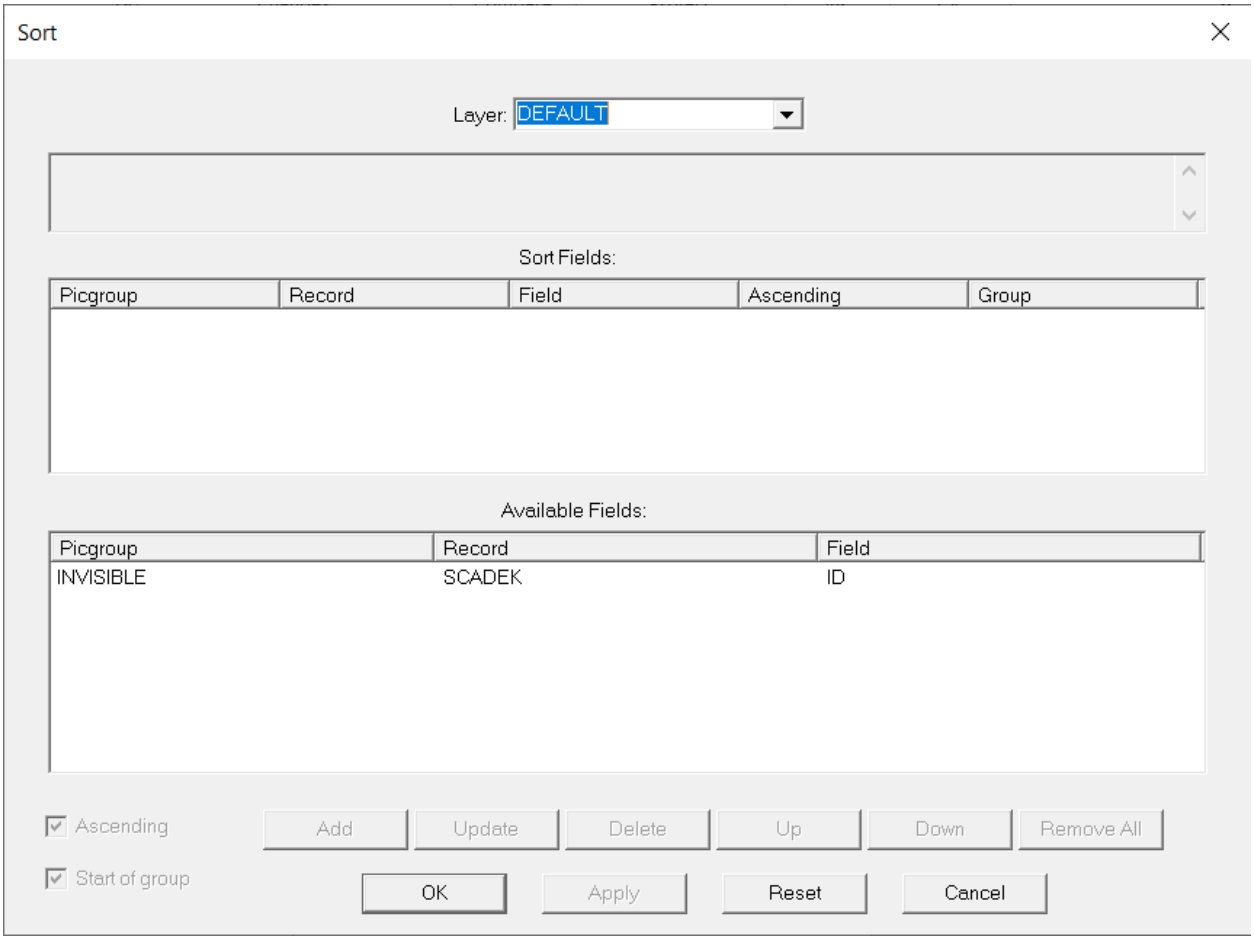


Figure 24. Sort Dialog Box

1. Select the Sort icon from the standard toolbar to open the Sort dialog box (see section 3.3.4.2 Customizing the Default Toolbar).
2. Select the picgroup on which you wish to sort from the Available Fields list box.
3. Click the Add button. The selected picgroup will be added to the Sort Fields list box.
4. Repeat for any additional fields.
5. (Optional) Use the Up and Down buttons to move picgroups up or down in the Sort Fields list. Data will be sorted first on the picgroup at the top of the list, then on each succeeding picgroup.

6. Click Apply to sort the data without closing the Sort dialog box, or click OK to sort the data and close the dialog box.

Note: When connected to Habitat 5.9 or later, only server-side sorting is supported.

4.6.2 Filtering Data

To filter data, use the following procedure:

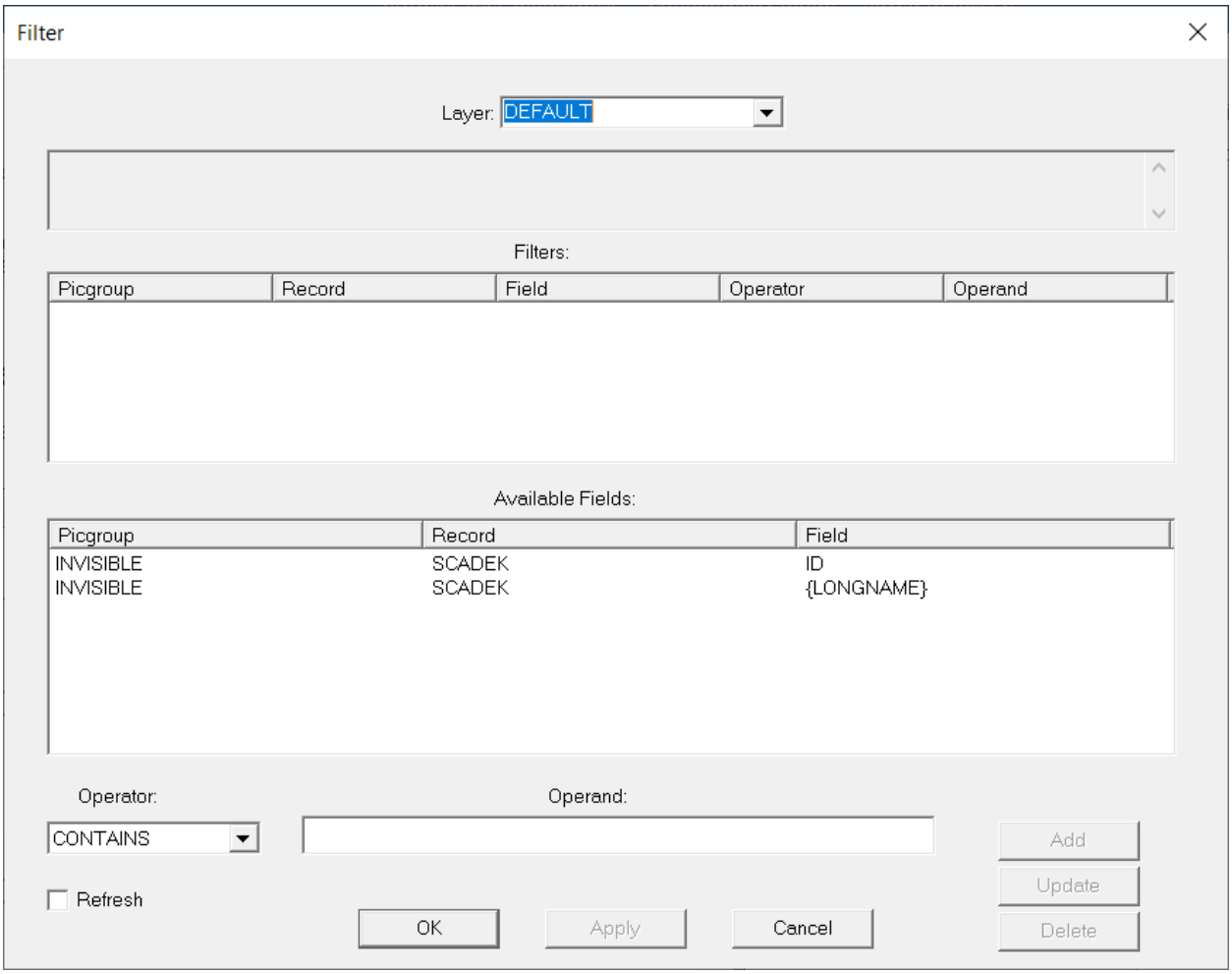


Figure 25. Filter Dialog Box

1. Select the Filter icon to open the Filter dialog box.
2. Select the first picgroup/record/field on which you wish to filter from the Available Fields list box.
3. Select the desired operator from the drop-down list:
   - Contains: A match is returned if the field value contains the criterion value. Both values must be of type CHARACTER.

- EQ: Matches if the field value equals the criterion value. Any data type may be used, but the two values should be of the same type or of compatible types. For instance, INTEGER and REAL may be matched, but not INTEGER and CHARACTER.
- GT: Matches if the field value is greater than the criterion value. Any data type may be used, but the two values should be of the same type or of compatible types.
- GE: Matches if the field value is greater than or equal to the criterion value. Any data type may be used, but the two values should be of the same type or of compatible types.
- IS_TRUE: Matches if the field value evaluates to TRUE. The field must be of LOGICAL or MASK data type.
- IS_FALSE: Matches if the field value evaluates to FALSE. The field must be of LOGICAL or MASK data type.
- LIKE: Matches a CHARACTER field value according to the regular expression specified by the criterion value.
- LT: Matches if the field value is less than the criterion value. Any data type may be used, but the two values should be of the same type or of compatible types.
- LE: Matches if the field value is less than or equal to the criterion value. Any data type may be used, but the two values should be of the same type or of compatible types.
- STARTS WITH: A match is returned if the field value has a proper prefix of the criterion value. Both must be type CHARACTER.
- NOT_LIKE: Matches any/all CHARACTER field value other than the one matching according to the regular expression specified by the criterion value.

4. Enter the filter criteria in the Operand text box.
5. Click Add; the selected picgroup will be added to the Filters list box.
6. Repeat for any additional fields, using the And or Or button.

   Using And requires that results meet all the desired criteria; using Or allows results that meet any of the desired criteria. Browser does not give precedence to either operator, nor does it support nesting of criteria.
7. Click Apply to filter the data without closing the Filter dialog box, or click OK to filter the data and close the dialog box.

4.7 SmartTag UI

You can use the SmartTag UI (user interface) to create, view, and edit either individual tags or multiple tags at the same time.

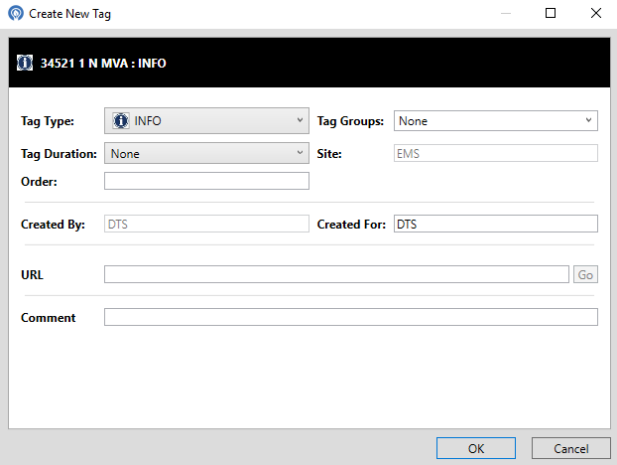


Figure 26. Create New Tag Dialog Box for a Single Tag

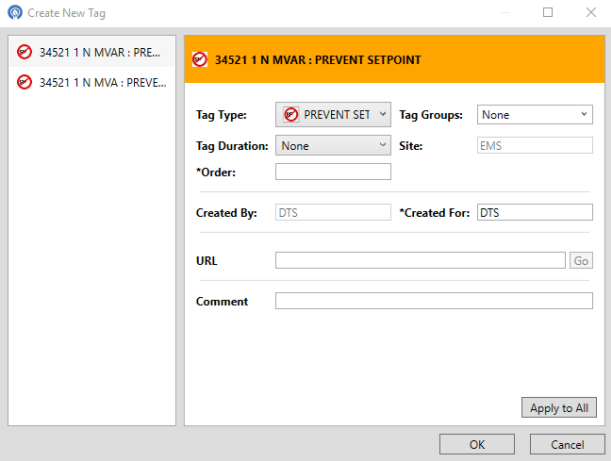


Figure 27. Create New Tag Dialog Box for Multiple Tags

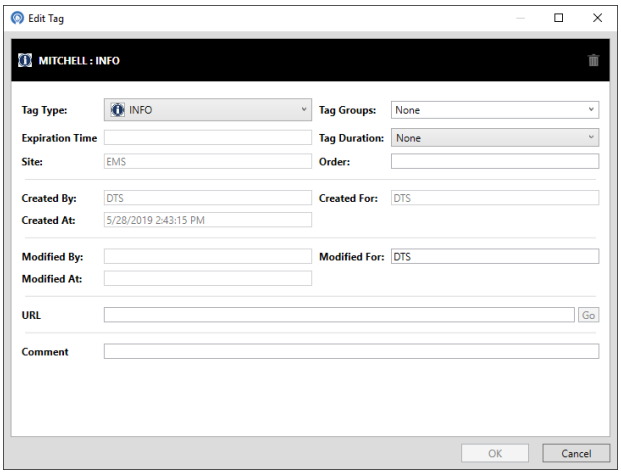


Figure 28. Edit Tag Dialog Box for a Single Tag

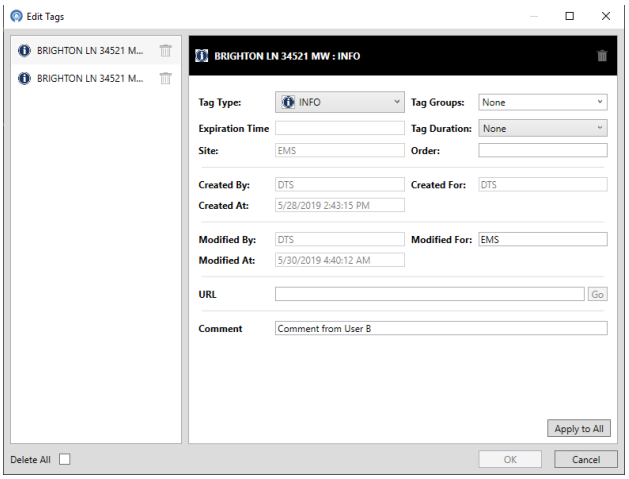


Figure 29. Edit Tag Dialog box for Multiple Tags

4.7.1 Multiple Users Updating Tags at the Same Time

If more than one user updates a tag at the same time, the SmartTag UI shows users a notification of the concurrent changes.

Suppose User A and User B are editing the same tag from two different consoles. The changes made by one user are highlighted with a blue border until that change is committed by the user. For instance, suppose User A is making a change in the Comment field and, in the meantime, User B changes the Comment field of the same tag from a different console and commits the change. In this case, as shown in Figure 30, User A will see the Comment field highlighted with a red border and an exclamation point will appear next to the field.

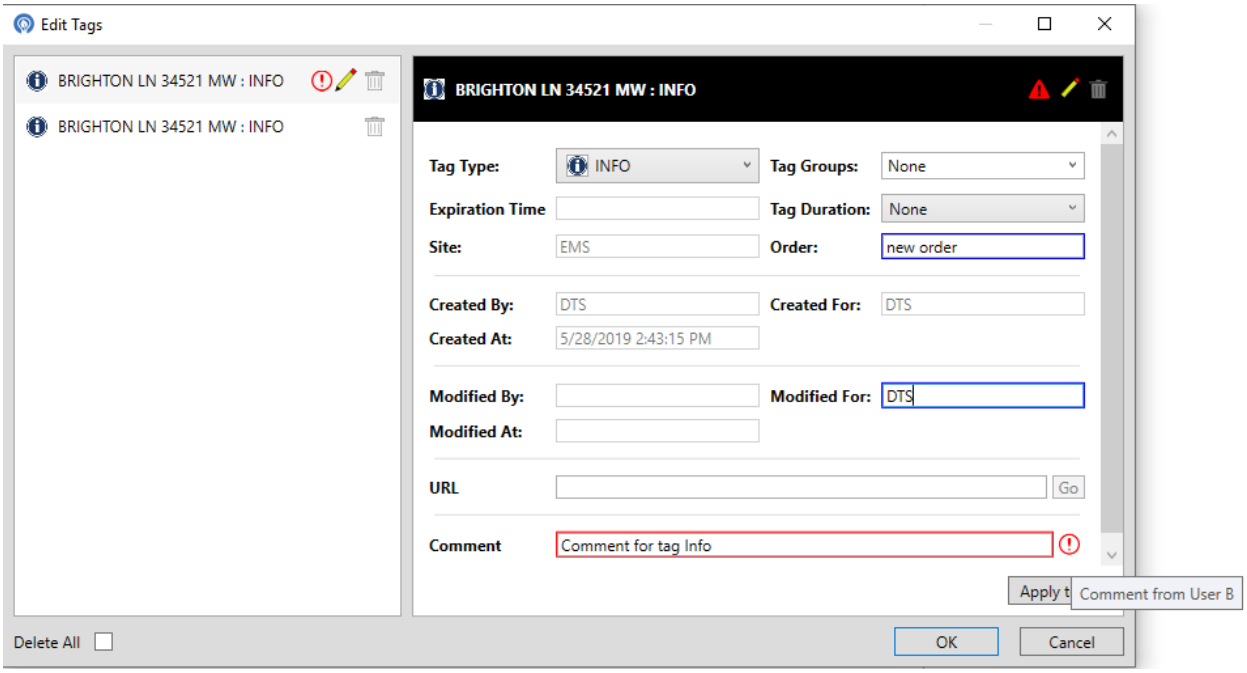


Figure 30. Concurrent Changes Notification in Tag Dialog Box

On hovering the mouse pointer over the exclamation point, User A can view the changes made by User B. After reviewing the changes from User B, User A has three options:

- Accept User B's changes by clicking the exclamation point button at the top of the dialog box.
- Overwrite User B's changes by clicking OK on the dialog box, which commits User A's changes instead.
- Discard all of User A's changes by clicking Cancel on the dialog box.

For more information about the SmartTag UI, refer to the *Browser Software Installation and Maintenance Guide.*

4.8 SmartTag Symbols

You can configure an icon or a color of SmartTags. The icon and color for each TagType are configured in the tag_note_symbol_configuration.xml file. The value of the attribute 'Scheme' of the node 'TagInfos' determines whether an icon or color for tags is shown.

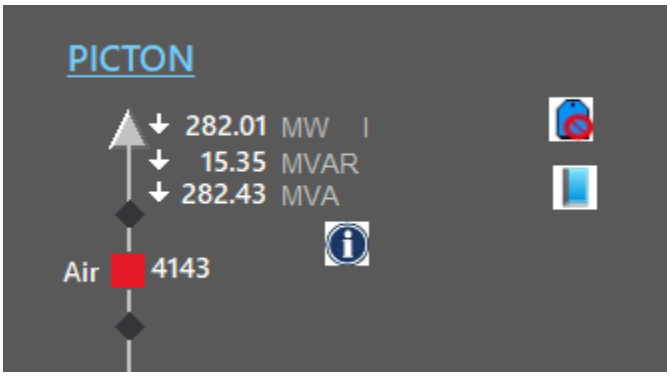
A sample content of the file is given below.

```
<?xml version="1.0" encoding="utf-8"?>
<TagConfig version="1.0">
  <TagInfos Scheme="UseSymbol">
    <TagInfo tagType="DO NOT OPERATE" icon="DoNotOperate.jpg" priority="1"
color="Red" />
    <TagInfo tagType="HOLD CLOSE" icon="HoldClose.jpg" priority="4"
color="Yellow" />
  </TagInfos>
</TagConfig>
```

The possible values of the attribute 'Scheme' are 'UseSymbol', 'UseColor' and 'UseNone'.

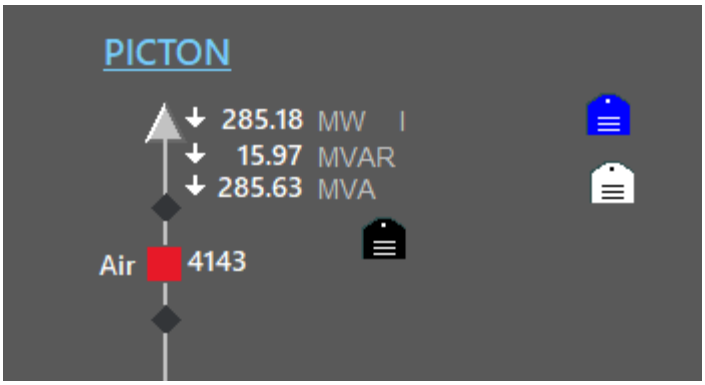
4.8.1 UseSymbol

When Scheme="UseSymbol", the icon corresponding to each tagType is shown in the Browser.



4.8.2 UseColor

When Scheme="UseColor", a rectangle of the color defined for each tagType is shown in the Browser.



4.8.3 UseNone

When Scheme="UseNone", neither the icon nor the color defined for tagTypes is used.

If you are using displays with legacy tagging and you do not want to see SmartTag symbols, you can clear the "Show Tag Symbol" option from the picture placement properties window. Note that clearing the "Show Tag Symbol" option also hides the SmartTag menu and SmartTag UI.

If you do not want to use SmartTag Symbols but still want to have the SmartTag menu and SmartTag dialogs, select the UseNone option.

Note: Selecting the UseNone option will result in the following behavior:

- The TagFilter toolbar menu item option will not be available.
- Browser will not process the FilterTags command.
- The SmartTag Dialog will not use the TagIcon (color/symbol) representations.

4.9 Viewing Displays in Frames

Frames are a means of splitting the viewport so that more than one display, or more than one view of the same display, can be viewed simultaneously in the viewport.

Frames are controlled by a group of five icons contained in a drop-down menu on the standard toolbar (see section 3.3.4.1 The Display Viewer Default Toolbar). These are used to open additional frames and to select a specific frame for a particular action.

Figure 31. Frame Control Tools

Browser supports any number of frames, which can contain any number of views. The frames can be dragged and dropped within a viewport. Normally, frames are docked along one of the four sides of a viewport, but it is also possible to have floating frames, which will appear as a floating window. Views can be dragged from one frame and dropped into another. Drag-and-drop operations are initiated by selecting the border of the frame, and then pressing and holding the left mouse button while moving the pointer to another drop area.

4.9.1 The Target Pane

Within each viewport, there is one and only one target pane. If there are multiple panes, you can choose which pane is the target pane. The target pane is the one that receives all commands, unless the command has a qualifier that directs to a different pane.

To set the target pane, click on the Target button in the mini toolbar above each pane. The Target icon will be green when the pane is the current target and grey when not the target.

Below are the characteristics of the target pane:

- Only one pane can be a target.
- The target pane cannot be closed.
- Menus and toolbars follow the target pane.

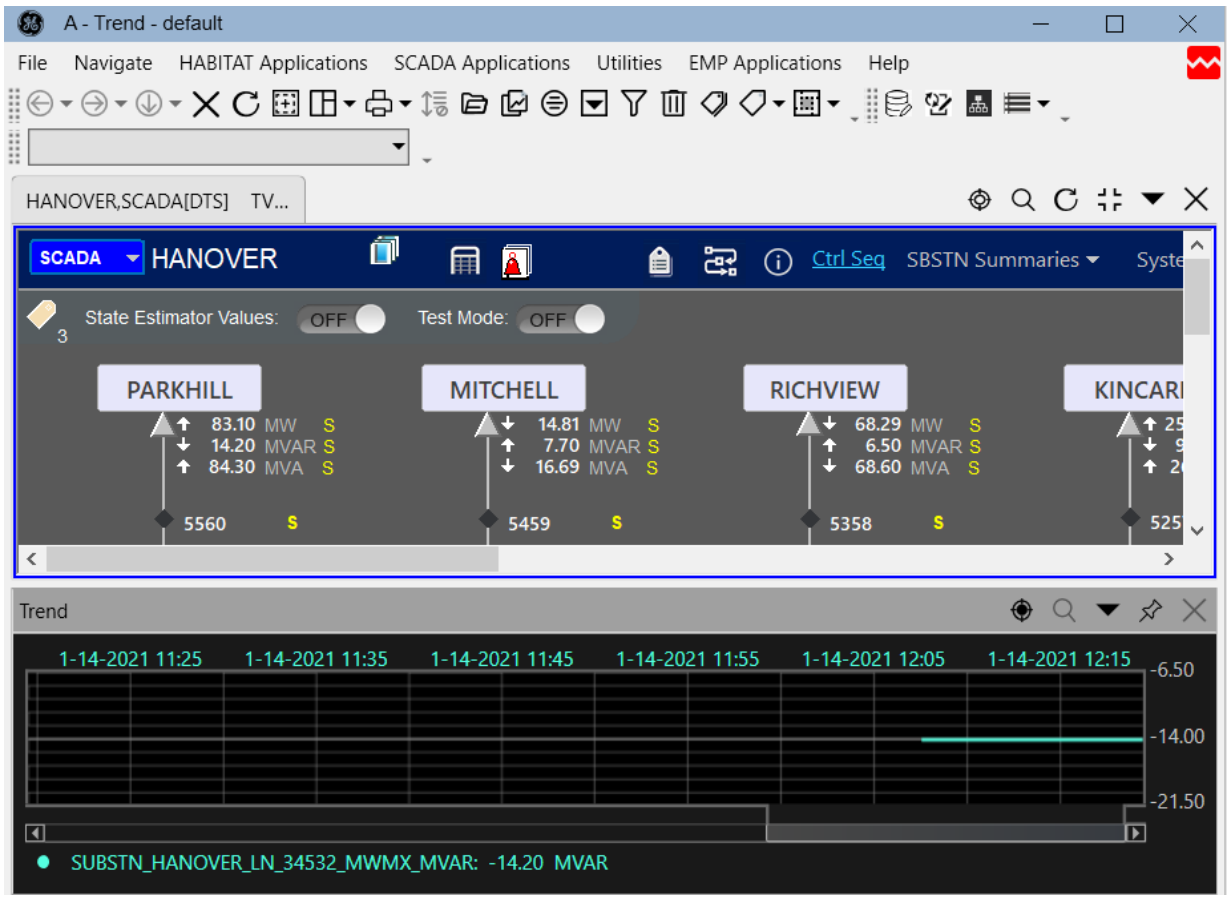


Figure 32. Bottom Pane Set as the Target

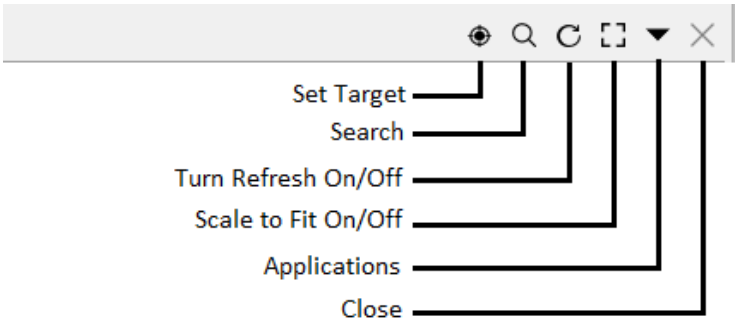
4.9.1.1 Empty Panes

While setting an empty pane to be the target pane, only the standard menus and toolbars will be available. After that, you can call up a display to the empty pane by giving a call-up command.

There are buttons on the Frame Control drop-down menu of the standard toolbar to create empty panes at four positions: Top, Left, Bottom, and Right. A pane created by clicking any of these buttons is automatically set as the target pane.

4.9.1.2 Controlling Displays in the Main Frame

The main frame includes five frame control buttons, located in the upper right corner of the frame.



- **Set Target:** Sets the main frame to be the target pane.
- **Search:** Searches for matching records or text in .fgdisplays. This toolbar button is available only in panes loaded with .fgdisplays. For more information, see section 4.13 Searching in Displays.
- **Turn Refresh On/Off:** Sends a request to the data server to turn refresh on or off for the display in the view. (See section 3.5.6.1 Refresh On/Off.)
- **Turn Scale to Fit On/Off:** Scale to Fit On zooms display in the view so that the entire display is visible. Scale to Fit Off sets the display to its default zoom level. (See section 4.2.5 Using Scale to Fit.)
- **Applications:** For the main frame, this button provides the option to switch between the tabs.
- **Close:** Closes the active view. If the last view within a frame is closed, the entire frame is closed.

4.9.2 Opening a New Frame

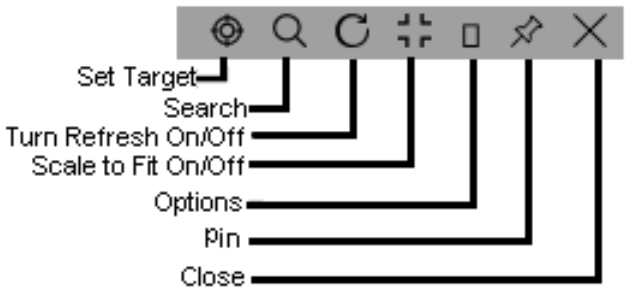
To open a new frame and call a display, use the following procedure:

1. Click the icon representing the location where you want the frame to open.

   The newly opened frame is automatically set as the target pane.
2. Call a display into the frame (see section 3.5 Calling Displays).

4.9.2.1 Controlling Displays in Frames

A number of command options are available to manage the content of a view within a frame.



The following commands are available on the mini-toolbar:

- **Set Target:** Sets the current pane to be the target pane.
- **Search:** Searches for matching records or text in .fgdisplays. This toolbar button is available only in panes loaded with .fgdisplays. For more information, see section 4.13 Searching in Displays.
- **Turn Refresh On/Off:** Sends a request to the data server to turn refresh on or off for the display in the view.
- **Turn Scale to Fit On/Off:** Scale to Fit On zooms the display in the view so that the entire display is visible. Scale to Fit Off set the display to its default zoom level.
- **Options:** This button provides the following options:
  - **Float:** Makes the frame float.
  - **Dock:** Makes the frame dockable.
  - **Dock as Tabbed Document:** Adds the display in this frame as a new tab in the main frame.
  - **Auto Hide:** Autohides the frame. This can also be achieved using the Pin Icon on the mini toolbar for non-main frames.
- **Close:** Closes the active view. If the last view within a frame is closed, the entire frame is closed.

The control buttons are located on the right side of the mini toolbar. Tooltips identify each button. To view tooltips, position your mouse pointer over the button until the tooltip is displayed.

4.9.3 Opening a Trending Frame

To open a dedicated frame for displaying trending information, select the Open Trending Frame icon from the standard toolbar. This opens a new frame at the bottom of the viewport.

4.10 Printing Displays

To print the display in the active frame of the active viewport, click the 'Print the Current Display' icon from the standard toolbar. Clicking this icon prints the entire display directly to the Windows default printer.

> **Note:** This is the only way to print the contents of the Alarm Viewer. The print commands for printing normal displays, Print Page Setup, Print Preview, and Navigate > Print Page do not work for the Alarm Viewer.

To bring up the Windows Print dialog box, activate the pull-down menu item from the Print icon on the standard toolbar and click Print Page Setup. This dialog box allows printing to any allowable printer.

In addition to the Print icon, the Capture and Project commands also allow you to produce hardcopy (printed) output of FG displays using system printing utilities. A common set of command qualifiers provides additional control over the visual characteristics, size, and disposition (i.e., choice of output device) of the printed display.

For further details about the Capture and Project commands, refer to the *Browser Reference Manual*.

4.11 Print Preview

You can preview displays before printing them with Print Preview.

To bring up the Print Preview for a display in the active frame of the active viewport, activate the pull-down menu from the Print icon on the standard toolbar and select Print Preview:

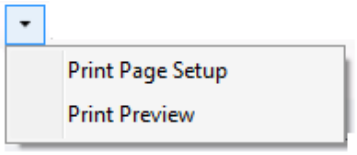


Figure 33. Print Menu

If the viewport contains only the main frame when activating Print Preview, the main frame will change to the Print Preview mode. If there is more than one pane or tab, the active pane will change to Print Preview mode.

A preview toolbar will appear in the Print Preview mode.

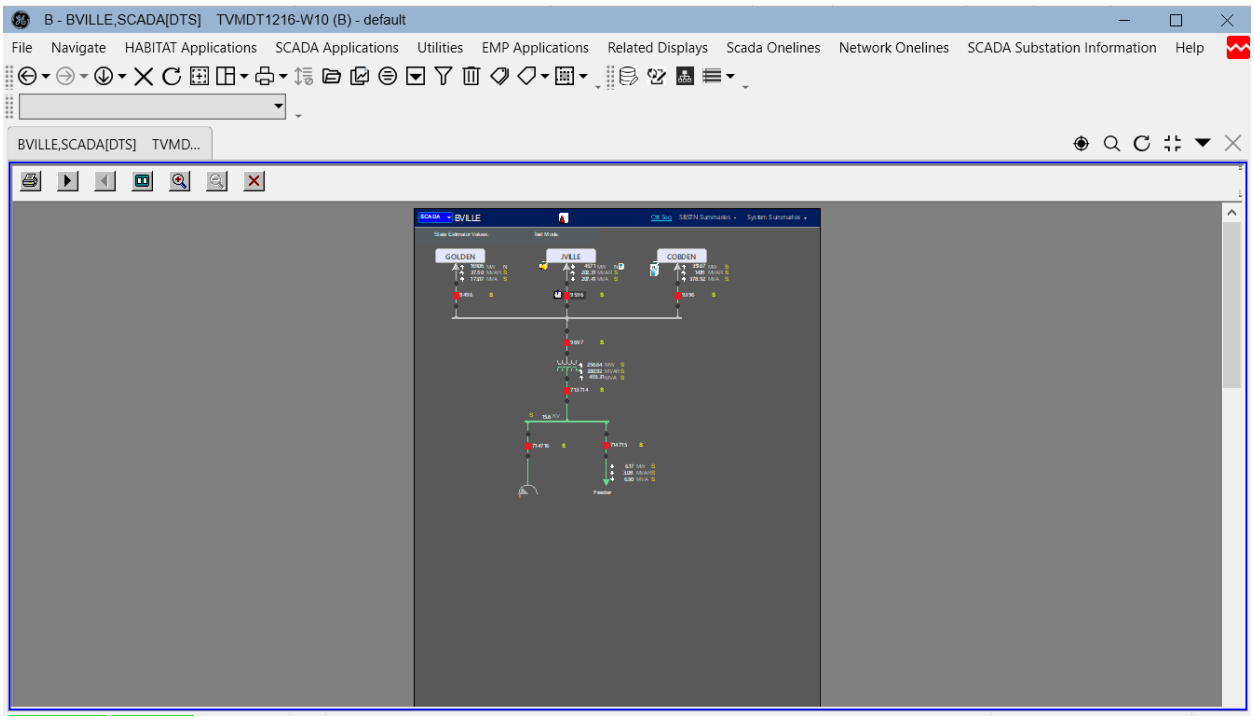
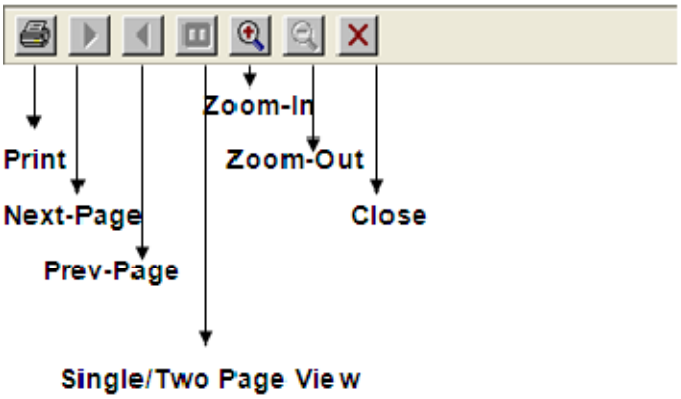


Figure 34. Print Preview of a Typical Viewport

4.11.1 Print Preview Toolbar



A toolbar appears at the top of the print preview, as shown below:

- **Print:** Click the print icon to bring up the Windows Print dialog box.
- **Next/Prev Page:** Click the Next/Prev Page icons to scroll through the pages of the display to be printed.
- **Single/Two-Page View:** Toggle this icon to choose between single-page and two-page views.
- **Zoom-In / Zoom-Out:** Click the Zoom-In/Zoom-Out icons to zoom the print preview in or out.

- **Close:** Click the Close icon to close the preview and return to the normal mode of display.

4.12 Disable Dynamic Page Length

The Disable Dynamic Page Length (DPL) button is for disabling the DPL functionality at runtime for displays that are DPL-enabled.

This toolbar button is available only when a DPL-enabled display is loaded. To disable the DPL functionality of this display, click this button. To reactivate the DPL functionality for this display, click the Refresh button. If different panes contain DPL-enabled and disabled displays, the button state is as per the display loaded in the target pane. After disabling DPL using this button, navigating back and forth enables DPL.

After disabling DPL by using this button, saving the room does not save the button state.

For more information about DPL, refer the Display Tab section of the *Display Builder User Guide*.

4.13 Searching in Displays

You can search for matching records or text in .fgdisplays.

To begin searching, click the Search icon from the view's mini toolbar. (You can also use a shortcut; refer the FAQ section of the *Browser Software Installation and Maintenance Guide*).

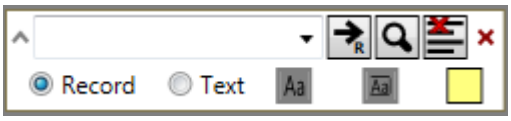
The Search icon appears only if the display loaded in that pane supports searching. (Currently, only fgdisplays support searching). The Search icon will show or hide the search fly-out menu in the corresponding pane.

Two types of searches are supported:

- **Record search** searches for record occurrences in displays.
  - Searches on picture placements, primitives, and symbols with CAMs attached placed directly on the display
  - Searches against the Composite Keys and MRIDs
  - The search is case insensitive
  - Find Next and Find All options are available
- **Text search** searches for plain text in displays.
  - Searches on static text, formatted fields, and conditional text
  - The search can be case sensitive or case insensitive
  - Find Next, Find All, and whole word match options are available

The default search type is the Record search.

To select the search type, click the drop-down button on the left side of the fly-out menu, to show more further options:



Select the option button for the desired search type. The Search Next icon indicates the selected type with a letter in the lower right corner:

- If you select Record search, the icon shows an R: .
- If you select Text search, the Icon shows a T: .

For Record search, the search string is by whole word and is not case sensitive. For Text search, you can select these attributes:

- To set the case sensitivity, use the Match Case icon .
- To seat the word search, use this icon .

While a search is in progress, if you start a new search and change any parameters (search string, search type, match case, or whole word) the current results are cleared and the new search starts from the beginning.

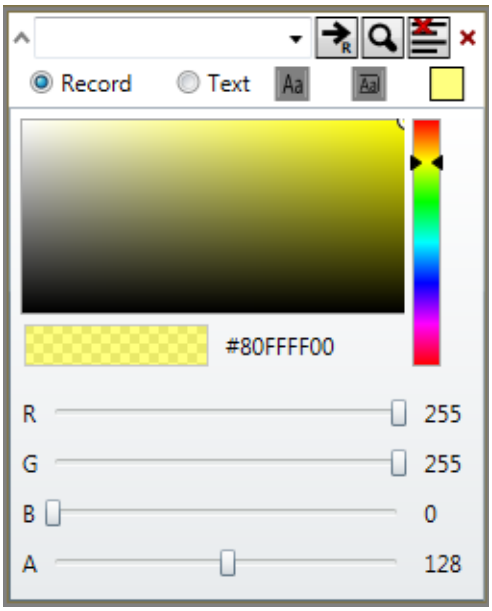
To clear the current search results, click the Clear Search icon .

You can highlight the display elements found by the search all at once or one at a time:

- To highlight all the matching elements, click the Search All icon .
- To highlight the elements one by one, click the Search Next icon (or press Enter with the cursor in the Search text box).

The found matches are highlighted with a colored overlay rectangle. The default highlight color is Yellow.

To choose a highlight different color, click the color selector button: . The fly-out menu expands to show the color parameters:



5. Customizing Browser

This chapter provides instructions for customizing the Browser screen environment. Customizing the screen environment involves customizing its settings and attributes, customizing viewports and rooms, and setting the various aspects of the screen context.

5.1 The Browser Configuration File

All configurable attributes of a Browser session are stored in a configuration text file, which can be modified via the Settings tab of APF Console (Figure 35).

For more information, contact your system administrator, or refer to the *Browser Software Installation and Maintenance Guide*.

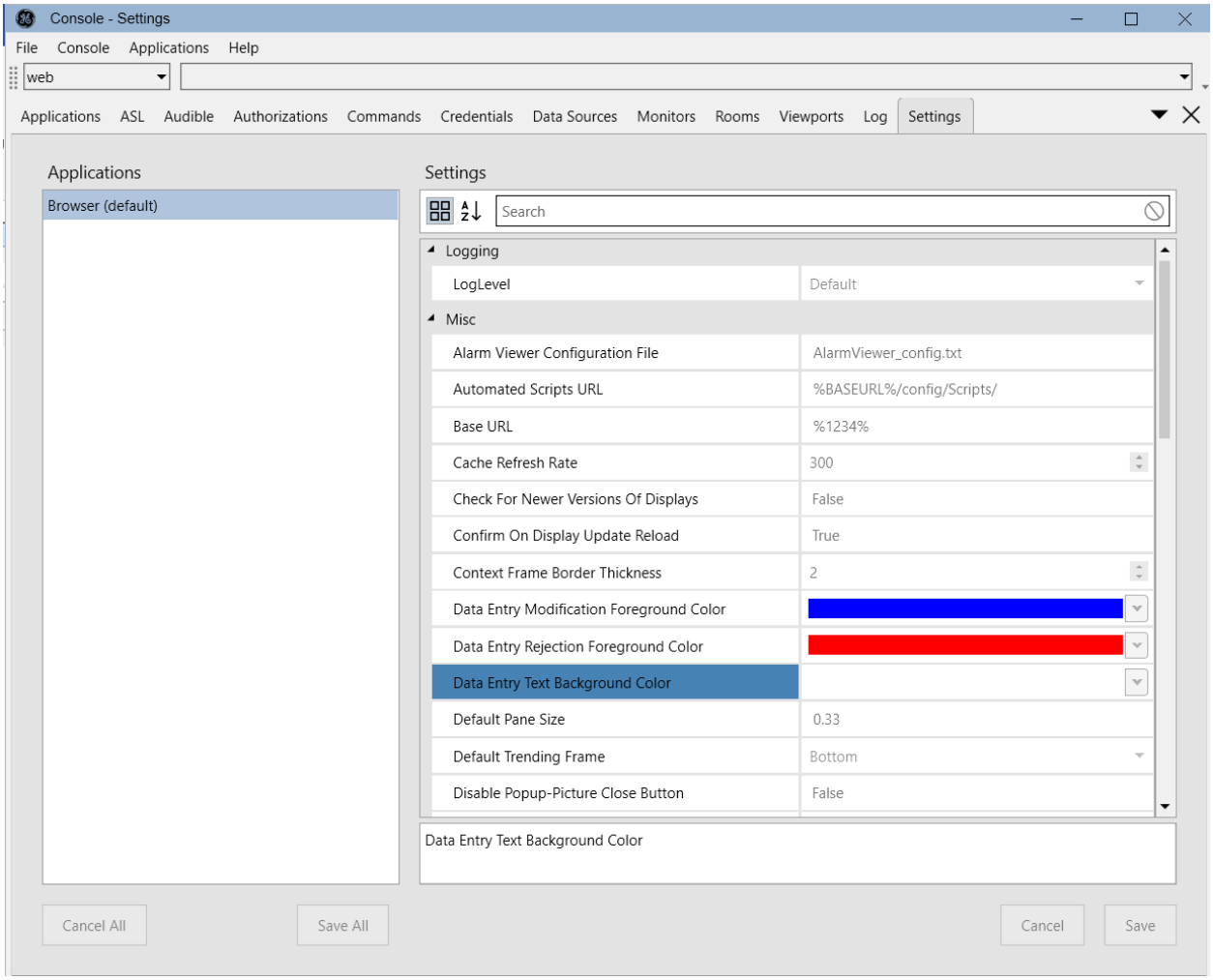


Figure 35. Settings Tab

5.2 The Key Map

A “key map” is a list of those virtual keys that have been defined for the current Browser session.

A virtual key definition consists of four aspects:

- The key, which is the actual keystroke, mouse button action, or series of keystrokes or mouse button actions that activates the virtual key. Examples would be pressing the F6 key, pressing the left mouse button, and the keystroke series (Ctrl+F6).
- The virtual key name. This is an arbitrary string, up to 255 characters long, that names the virtual key. The virtual key name can be mapped to keysets that define buttons on the display.

  Refer to the *Display Builder User Guide* for more information about keysets and defining buttons.
- The Browser command associated with the virtual key name. A virtual key cannot be added with an empty string in the command box. The command can be any Browser command, a series of commands separated by semicolons (;), or a partial command. The command can also include proxy strings.

  Refer to the *Browser Reference Manual* for information about commands and proxy strings.
- Command-line concatenation options. These options allow for text in the command window of the active viewport to be appended to the command defined for the virtual key.

  Defining the virtual key command to accept information from the command window does not mean that such information is required. It means that, if there is anything in the command window, it will be concatenated to the virtual key command.

Virtual key definitions can be applied to only the current session, or they can be saved for future sessions.

When a key on the keyboard or a mouse button is pressed, Browser processes commands in the following hierarchy:

1. The command defined by a keyset. If the pointer is on an item with a keyset attached, Browser will execute the command defined in the keyset and ignore the command defined for the virtual key.
2. The command defined for the virtual key. If the pointer is not positioned on an item with a keyset attached, Browser will execute the command defined for the virtual key, if any.

   If no virtual key name has been defined for the keystroke or mouse button action, the actions will be ignored.

The key map is customized from the Keymap Definition dialog box:

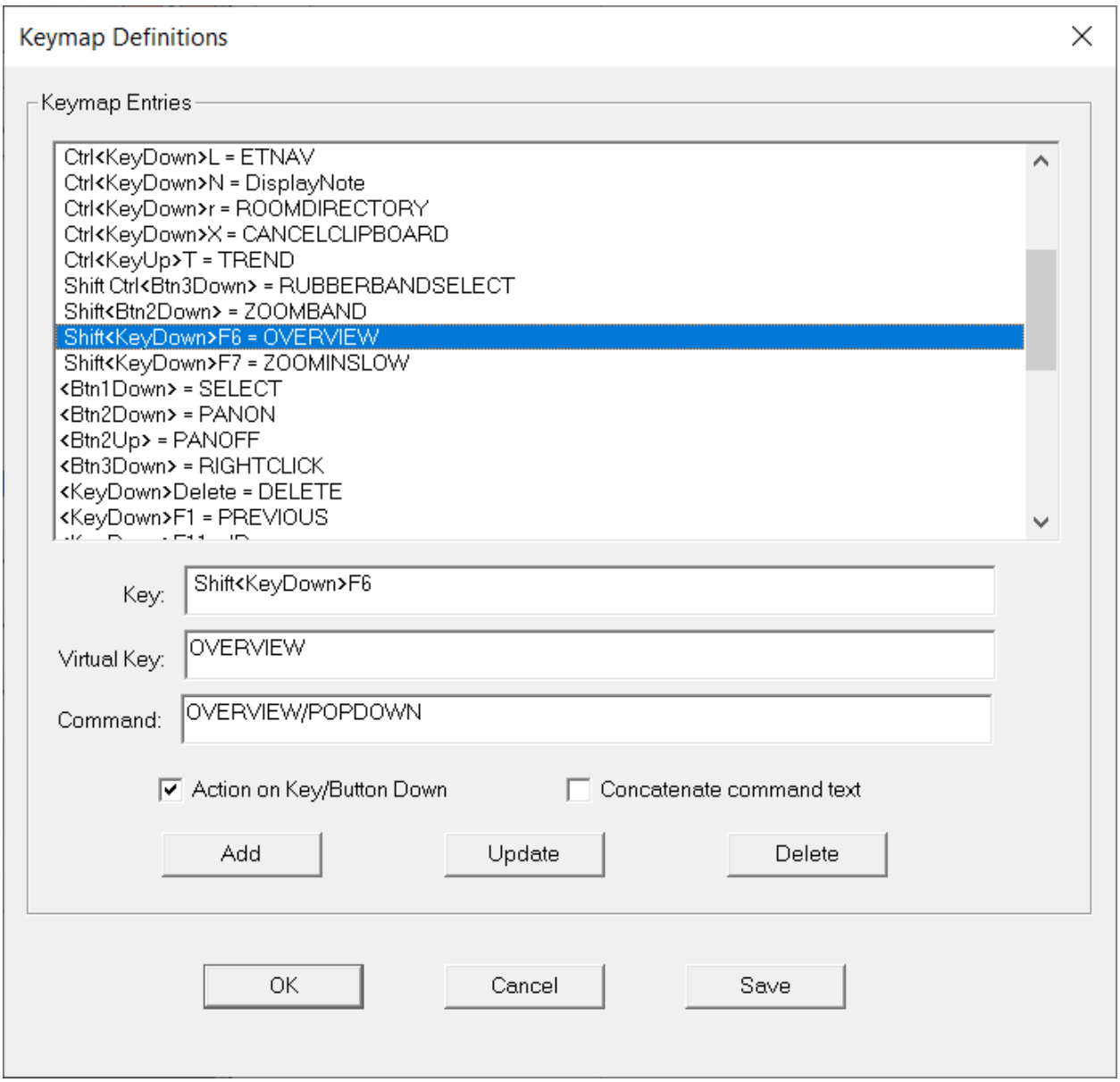


Figure 36. Defining Virtual Keys

5.2.1 Defining a Virtual Key

To define a virtual key, use the following procedure:

1. Select File > Settings > KeyMap from the viewport menu bar. The Key Map Definition dialog box appears (Figure 36).
2. Ensure that the dialog box has input focus and that the text cursor in the Key text box is blinking. If it is not, press Tab until the text-entry cursor in the Key text box blinks.
3. Press the key or series of keystrokes and/or mouse buttons you want for this virtual key definition.

   The Key text box fills in automatically with the desired keystroke(s).

> **Note:** Alpha-numeric keys (ranges 0-9 and A-Z) can only be defined with special keys, that is, Ctrl or an Alt + Ctrl combination. For example, valid alpha-numeric combinations are Ctrl<KeyDown>F or Alt Ctrl<KeyDown>9.
>
> **Note**: The following two keys are assigned to specific functions, and they cannot be used when defining virtual keys:
>
> **F9:** When pressing F9 while the mouse cursor is over a formatted field, it will get the value for the proxy %fv% and copy that value to the toolbox command. This key can be used when combined with special keys.
>
> **F10:** This key is used by the operating system. Pressing F10 will give focus to the active program's menu.

4. Tab to the Virtual Key text box.
5. Type the name of the virtual key in the Virtual Key text box. This can be any character string up to 32 characters.
6. Define a command for this virtual key:
   - Tab to the Command text box.
   - Type the command in the Command text box.
7. (Optional) Toggle the Action on Key/Button Down check box to set whether the command will occur when the key/button is released (key up, the default), or pressed (key down).

   Example: <KeyDown>F6 means the command will execute when the F6 key is pressed. <KeyUp>F6 means the action will take place when the F6 key is pressed and released.
8. (Optional) To allow command-line concatenation to the virtual key command, select the Concatenate Command Text check box. This allows command-line text to be appended to the command.
9. Click Add to make the virtual key available for use in the current session.
10. (Optional) To make the virtual key available for use in future Browser sessions, click Save.
11. Define any other virtual keys, if desired, by repeating steps 2 through 10.
12. Click OK to close the dialog box.

    The new virtual key definition(s) are in effect for the current session and, if saved, for future sessions.

5.2.2 Modifying a Key Map Definition

To modify an existing key map definition, use the following procedure:

1. Select File>Settings > KeyMap from the viewport menu bar to display the Key Map Definition dialog box (Figure 36).
2. Select the virtual key definition you wish to modify from the list box by clicking on it.
3. Make the desired changes to the Key, Virtual Key, and/or Command text boxes, and the Action on Key/Button Down and Concatenate Command Text check boxes.
4. Click the Update button.

5. (Optional) To make the virtual key modifications available for use in future Browser sessions, click Save.
6. Repeat steps 2 through 5 for any other virtual key definitions you wish to modify.
7. Select OK to close the dialog box.

   The changes to the key map are in effect for the current session and, if saved, for future sessions.

5.2.3 Deleting a Key Map Definition

To delete an existing key map definition, use the following procedure:

1. Select File>Settings > Key Map from the viewport menu bar. The Key Map Definition dialog box appears (Figure 36).
2. Select the virtual key definition you wish to delete from the list box by clicking on it.
3. Click Delete.
4. (Optional) To delete the virtual key from future Browser sessions, click Save.
5. Repeat steps 2 through 4 for any other virtual key definitions you wish to delete.
6. Click OK to close the dialog box.

   The changes to the key map are in effect for the current session and, if saved, for future sessions.

5.3 Viewports

The Browser is launched with a default viewport. From one viewport, a new viewports can be created using the "Create New Viewport" icon in the standard toolbar. The state of viewports can be saved, as part of a room (see section 5.4.1 Creating Rooms).

5.3.1 Viewport Attributes

Each viewport has a number of definable attributes:

- **Toolbar:** Controls whether the standard toolbar appears in the viewport.
- **Status Bar:** Controls whether a status bar appears at the bottom of the viewport.
- **Closable:** Controls whether a viewport can be closed by clicking the **X** in the upper-right corner of a viewport.
- **Sizable:** Controls whether a viewport can be resized.

Select or clear the Toolbar check box to specify the default value of the Toolbar attribute.

Select or clear the Show Status Bar check box to specify the default value of the Status Bar attribute.

In addition, the display called into the viewport will modify the viewport attributes, according to its definition. The attributes defined in the display will always override those set through either of the above check boxes. Refer to the *Display Builder User Guide* for information about customizing the viewport from the display definition.

5.3.2 Creating Viewports using /VIEWPORT Qualifier

The /VIEWPORT qualifier for the Display or Previous/Display command can be used to create dynamic viewports, 'on the fly'. A dynamic viewport can be given a specific name, or the default name can be applied (see section 3.5.3 Calling the Display into a Specific Viewport).

Refer to the *Browser Reference Manual* for more information about the /VIEWPORT qualifier for the Display command.

5.4 Rooms

Rooms are created using the Rooms tab of the APF console.

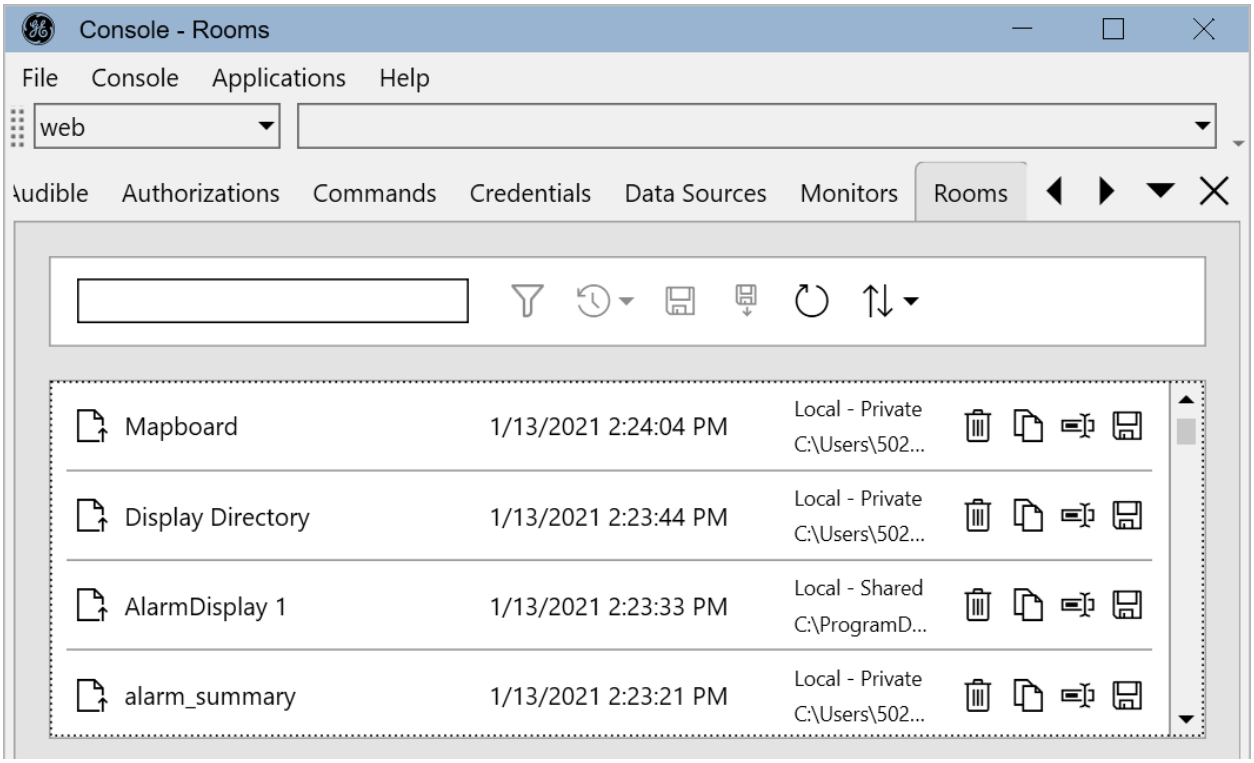


Figure 37. APF Console, Rooms Tab

5.4.1 Creating Rooms

To create a room, use the following procedure:

1. Create the viewports you want in the room (see section 5.3 Viewports).

   > **Note:** In a multi-headed configuration, the viewports can be placed on any of the workstation's screens.

2. Call the displays required for the room into the viewports (see section 3.5 Calling Displays).
3. Position and size the viewports on the screen where you want them to appear when the room is called, including minimized as a task bar icon.

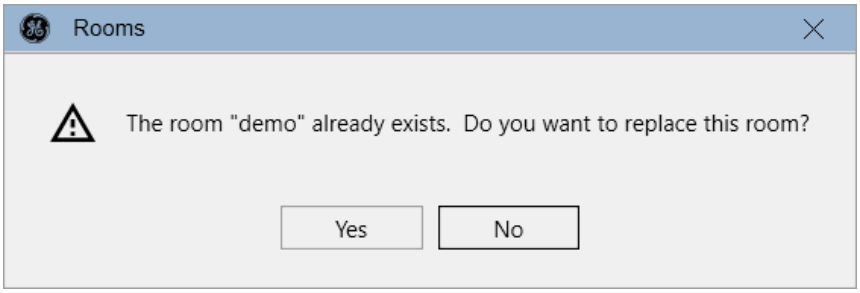
4. Enter a name for the room in the text box in the toolbar of the Rooms tab.
5. Click one of the Save buttons.

   The new room will appear in the list of available rooms, and it will be available for future sessions.

5.4.2 Modifying Rooms

To modify a previously created room, use the following procedure:

1. Call up the room you want to modify.
2. Make any modifications to the position of the viewports, to the number of viewports, or to the displays in the viewports, as desired.
3. Click Save. The following message box appears:



4. Click Yes to save the changes to the room, or No to cancel the changes.

   The room is now saved with the new configuration.

5.5 Monitor Configurations

Browser is designed to handle many multi-monitor configurations correctly using default settings. If required, modifications can easily be made via the Monitors tab in the APF console.

When correctly configured, you can expect your multi-monitor configuration to perform as follows:

- New viewports will be sized to the smaller of:
  - The size of the display
  - The size of one full monitor

  This means that viewports will not span monitors.
- Pop-ups and message boxes will not span monitors.
- Drag-and-drop viewports will be placed on the monitor where the left mouse button is released.
- Pop-up messages will be centered over the viewport from which the originating command is issued.
- A monitor or monitors may be designated as restricted; dynamic viewports will be prohibited on these monitors. When a viewport is opened from a restricted monitor, it will be placed on another monitor, rotating from left to right.

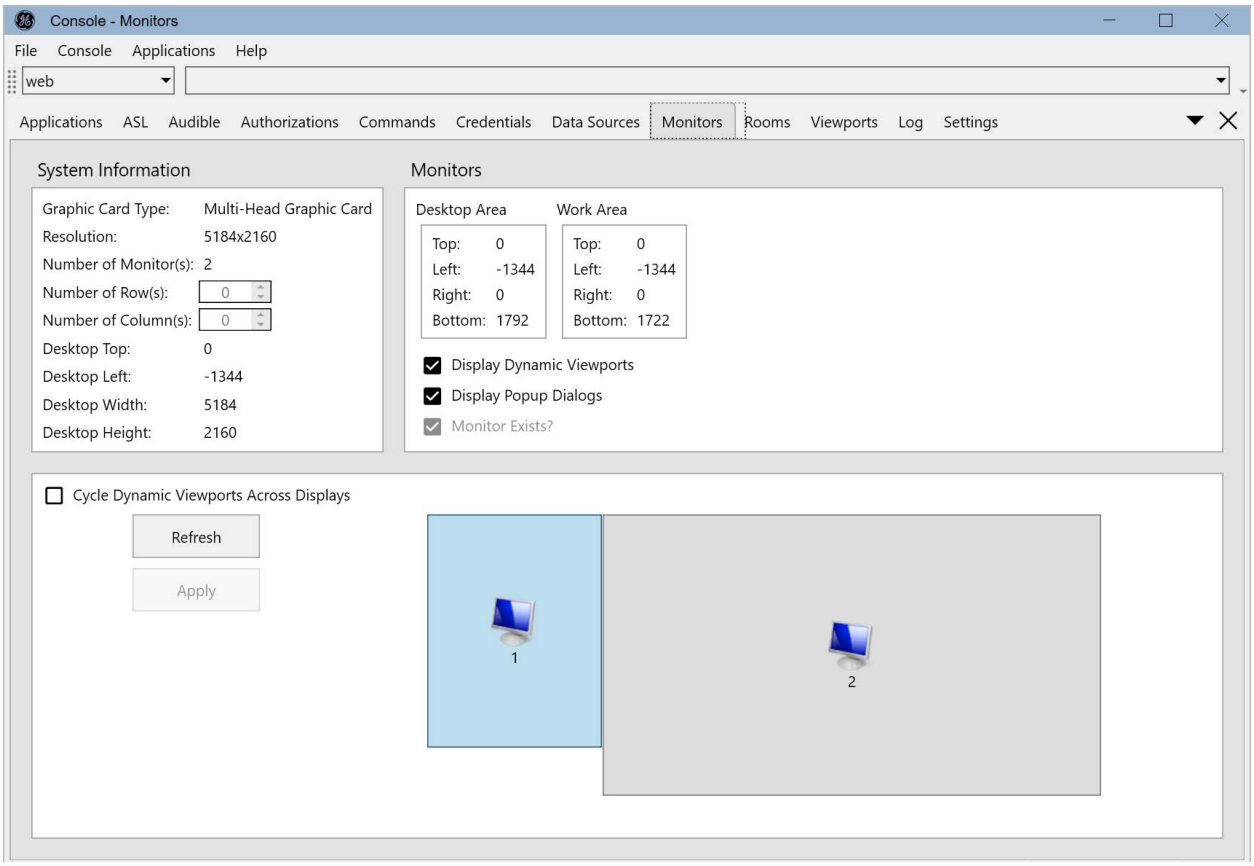


Figure 38. APF Console, Monitors Tab

Refer to the *Browser Software Installation and Maintenance Guide* for more information.

5.6 Data Sources

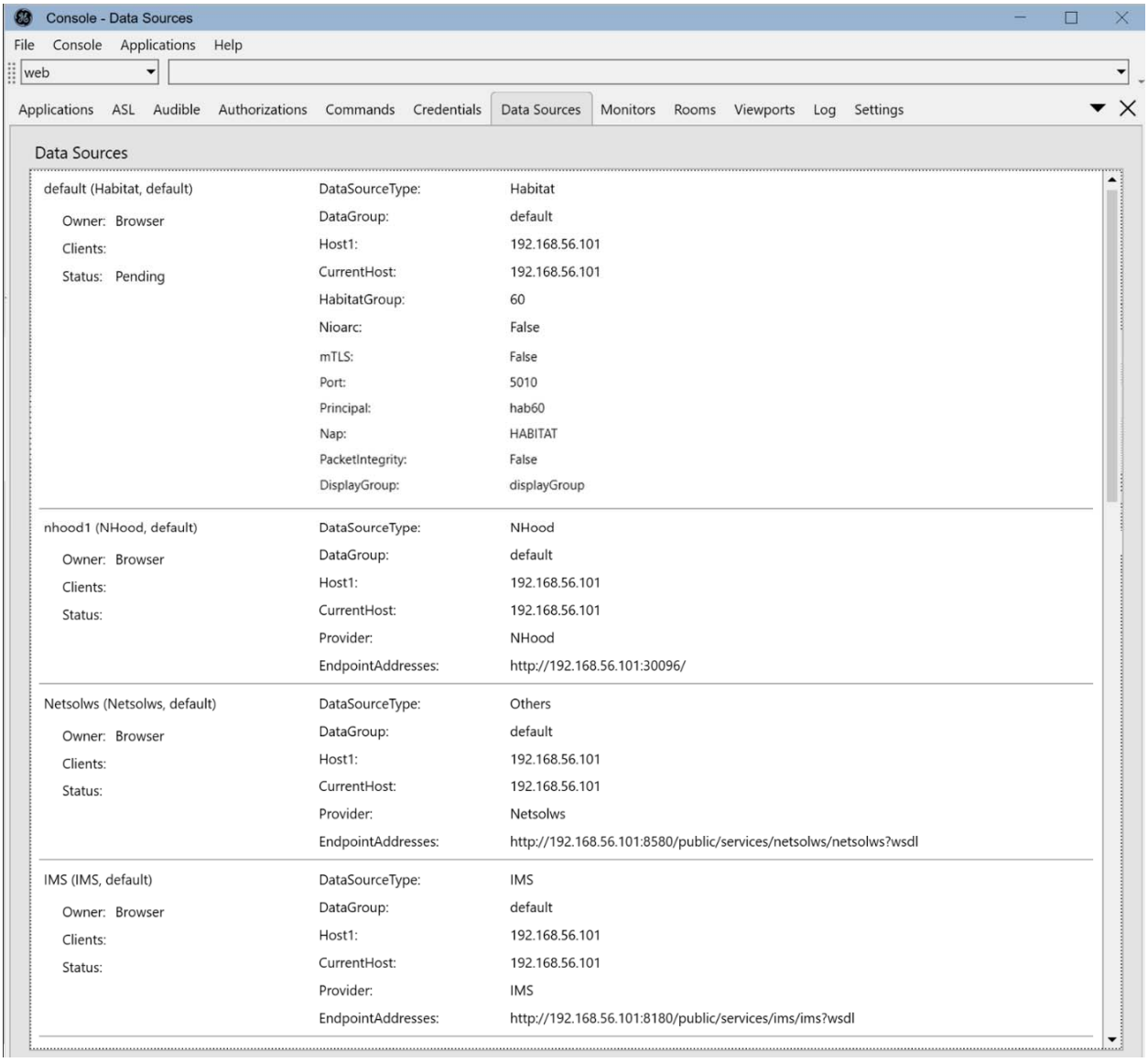


Figure 39. APF Console, Data Sources Tab

The Data Sources tab of the APF console is the central location where all the data sources used by Browser are displayed:

- The data source name.
- The Owner field shows the name of the application that created this data source.
- The Clients field lists different types of displays. For this release, the Clients field is marked as “Browser”.
- The Status field shows the current connection state.

- The Configuration Properties column lists detailed information about the connection, such as host, port, username etc.

The Data Sources tab is read-only. Data sources appear when the connections are made, and remain in the view until the associated application is shut down or that APF client is closed.

5.7 Changing Users

Previously, the user switching was done by Browser. Version 4.0 onwards, this is done using Switch User command under the APF Console. Once the command is triggered, the user change is applied to all applications. The Browser application first attempts to use the current user; if the credential type is not usable, or the authentication fails, a credentials dialog is displayed.

However, if the PERMIT credentials are correct but the authentication still fails, this may be the indication of a problem with the active modes of the user, especially if the original or the new user is an ADMIN user. For more information about changing users, refer to the *PERMIT User Guide*.

6. mTLS Enabled Communication

Browser supports mTLS enable communication with Browser Data Server. By default, the communication between Browser and Browser Data Server is non-mTLS.

For enabling mTLS support in Browser Data Server, refer to the section “Encrypted Port Configuration” in the *Habitat Software Installation and Maintenance Guide*.

The Browser configuration to enable mTLS based communication is defined in two locations.

- You must configure mTLS="true" in DataSourcesConfiguration.xml (for the attribute mTLS in DatasourcesConfiguration.xml, Browser considers the configuration of the “default” webfg DataSource in the “default” DataGroup only; it does not consider the configuration of other DataSources).

  For example:

  ```
  <webfg displaygroup="DefaultDisplayGroup" provider="RdOleDB" id="default"
  principal="hab60" nioarc="false" mTLS="true" nap="HABITAT" group="60"
  port=”xxxx”>
  ```

- The **netio_well_known_ports.dat** file is delivered with Browser in the Browser Install directory.

  > **Note:** This file containts default entries and should be updated with a new entry with corresponding NAP of the server the client is trying to connect.

  Example:

  ```
  TYPE:NETIO_SSL_CONNECT    NAP:HABITAT    SSL:MTLS
  TYPE:NETIO_SSL_CONNECT    NAP:DBSEQ      SSL:MTLS
  TYPE:NETIO_SSL_CONNECT    NAP:EMS        SSL:MTLS
  ```

In addition to this, the user has to generate and configure the certificates on server and client machines (for generation and installation of the certificates, refer to the section “SSL/mTLS Certificate Configuration” in the *Habitat Netio User Guide*).

> **Note:** The following environment variables will help with getting log on mTLS connection details.
>
> NIOUTIL_WKPORT_DEBUG=VERBOSE
>
> NIONTS_SOCKBERK_DEBUG=ALL

> **Important:** NIOARC with mTLS (both values set to true) is not a supported/tested configuration.

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